

BRAMBLELOOP FINAL MASTER UPGRADE

v0.1 - Etsy Search, Listing Conversion & Search-Visibility Intelligence

STATUS: STARTED. Living final-upgrade file. Append future final audits before the final handoff.

Objective: turn Etsy's own current search guidance into executable Brambleloop requirements. The core model is two-stage: **query matching** determines eligibility; **ranking** orders eligible listings. Search success therefore requires truthful relevance, click-worthy creative, conversion, shop/listing quality, and customer-service trust - not keyword stuffing.

Audited source set: Etsy's Ultimate Guide to Etsy Search; Keywords 101; Add Attributes to Help Increase Your Shop's Visibility. Because the Ultimate Guide explicitly delegates core mechanics to linked official guides, this audit also followed Etsy's How Search Works, Well-Crafted Listing, Search Visibility, Listings That Convert, and Customer Service search-ranking guidance.

1. DEEP AUDIT - OPERATING FINDINGS

Two separate search problems: Brambleloop must model query matching and ranking separately. A single SEO score can hide whether a listing fails to match or matches but loses the ranking/conversion battle.

Holistic relevance: Titles are only one input. Etsy describes title, tags, attributes, descriptions, categories, first photo, reviews and other shop/listing information as part of matching/ranking.

Structured data is search inventory: Specific categories and truthful attributes create matching/filter opportunities. Missing a relevant attribute can exclude a listing from filtered results.

Use all 13 tags intelligently: Etsy recommends all 13 tags, varied multi-word phrases, long-tail terms, and avoiding repetition of tags/categories/attributes.

Readable titles, not keyword soup: Lead with what the item is for mobile/search scanning. Etsy says title phrase position does not itself change ranking, so front-loading is a buyer-comprehension rule, not a ranking myth.

Descriptions matter: Relevant phrases belong naturally in early description copy. Copying the title or dumping keywords is discouraged.

Hero image is search infrastructure: The first photo earns the click. Etsy advises clear high-resolution imagery and warns against confusing first-image collages/text-heavy thumbnails.

Conversion feeds ranking: Clicks, favorites and purchases contribute to engagement. Search optimization must therefore include CTR and conversion experiments.

New-listing boost is a learning window: A new listing gets a small temporary boost so Etsy can learn shopper response. Treat this as an evidence-collection window, not proof of demand.

Trust affects visibility: Shop icon, listing completeness, multiple strong images, policies, reviews, message response and case rate matter to buyer confidence and search performance.

Search Visibility is first-party evidence: Etsy's Shop Manager Search Visibility recommendations should be ingested when browser access exists; do not approximate first-party warnings with internal guesses.

Search rules change: Etsy explicitly says search changes. Brambleloop needs dated policy snapshots and automatic re-audits.

2. FINAL-UPGRADE REQUIREMENTS

New namespace: **F-001+**. These are additive to Build 2's 1-320 registry.

ID	Requirement	Executable definition
F-001	Two-Stage Search Model	Separate query-match eligibility from ranking/conversion readiness.

ID	Requirement	Executable definition
F-002	Query Coverage Matrix	Maintain target query families and the exact listing fields supplying each match.
F-003	Ranking Readiness Profile	Track listing quality, click readiness, conversion readiness, shop quality and service quality independently.
F-004	Holistic Search Gate	No search certification from title/tags alone; category, attributes, description and first-image readiness must pass.
F-005	Deepest Truthful Category	Choose the most specific valid Etsy category; refuse unnecessary broad categorization.
F-006	Category Inheritance Awareness	Account for parent-category coverage created by specific subcategories.
F-007	Attribute Completeness	Require every relevant truthful category-specific attribute.
F-008	Attribute Truth Gate	Reject irrelevant holiday, occasion, material, color, recipient, style or other attributes.
F-009	Filter-Surface Audit	Show which Etsy filters a listing can truthfully enter and which are lost through missing data.
F-010	Attribute Drift Monitor	Periodically re-check category/attribute schemas for platform changes.
F-011	13-Tag Gate	Use all 13 tag slots unless a documented platform limitation prevents it.
F-012	Multi-Word Tag Optimizer	Prefer useful multi-word phrases over weak single-word fragmentation.
F-013	Tag Diversity Gate	Detect semantic duplication across tags and force distinct intent coverage.
F-014	Structured-Field Deduplication	Do not waste standalone tags repeating exact category/attribute phrases already supplied.
F-015	Long-Tail Portfolio	Mix core item terms with evidence-supported specific purchase-intent phrases.
F-016	Regional Phrase Intelligence	Capture meaningful regional phrases Etsy does not normalize; avoid redundant spelling variants it already handles.
F-017	No Typo SEO	Prohibit deliberate misspellings.
F-018	Language Discipline	Use shop language canonically; add supported translations only when quality is sufficient.
F-019	Plural/Root Waste Detector	Do not burn tag slots on singular/plural variants Etsy already root-matches.
F-020	Keyword Provenance	Every target phrase carries evidence provenance and freshness.
F-021	Title Clarity Gate	Titles must identify the item clearly and remain buyer-readable.
F-022	Front-Scan Optimization	Put item-defining language and strongest differentiator early for mobile/search comprehension.
F-023	Title Length Discipline	Treat Etsy's concise-title guidance (roughly under 15 words) as a readability target, not a ranking law.
F-024	Subjective-Word Relocation	Move empty subjective claims out of scarce title space.
F-025	Description Opening Optimizer	Naturally place important relevant phrases in the first sentences without copying the title.
F-026	Human Description Gate	Reject keyword-dump descriptions.
F-027	First-Image Search Gate	Hero must clearly show what is sold and survive realistic Etsy thumbnail scale.
F-028	Hero Collage/Text Restraint	Avoid confusing collages/text overlays in the first image unless measured evidence justifies an exception.
F-029	Image Resolution Gate	Target strong Etsy-ready resolution, preferably ~2000px+ where practical; never fall below current visibility guidance.
F-030	Gallery Information Architecture	Use later frames for angles, detail, texture, scale, lifestyle/use, sizing, inclusions and category-specific proof.
F-031	Thumbnail Simulator	Render heroes at realistic mobile/search sizes and reject unreadable/ambiguous creative.
F-032	CTR Experiment Registry	Measure impressions-to-clicks and isolate hero/title experiments.
F-033	Conversion Experiment Registry	Measure click-to-favorite/purchase and isolate listing changes.
F-034	New-Listing Learning Window	Schedule evidence capture during Etsy's temporary new-listing boost.
F-035	Engagement Feedback Loop	Feed click/favorite/purchase evidence back into creative, copy, price and product decisions.

ID	Requirement	Executable definition
F-036	Query-Specific Performance	Where data permits, evaluate performance by query family, not only aggregate conversion.
F-037	Search Visibility Ingestion	Browser-ingest Etsy's first-party Search Visibility recommendations with date, listing, reason and resolution.
F-038	First-Party Priority	Direct Etsy recommendations outrank speculative third-party SEO rules when applicable.
F-039	Search Policy Snapshot	Version official Etsy guidance and trigger re-audit when it changes.
F-040	Shop Quality Gate	Icon, About/brand presentation and core trust elements must be complete before launch.
F-041	Listing Quality Gate	Require multiple strong images, clear information and applicable policy/digital-product expectations.
F-042	Customer-Service Search Monitor	Track review rating, first-message response and Etsy-counted case rate.
F-043	48-Hour Response Protection	Alert before Etsy's initial-message response standard is breached.
F-044	Review Quality Loop	Route recurring review confusion into listing, pattern QA and support improvements.
F-045	Case-Rate Incident	Any Etsy-counted case triggers root-cause and corrective action.
F-046	No Paid-Rank Myth	Never model Etsy Ads or Etsy Plus as organic ranking boosts.
F-047	Star Seller Framing	Treat Star Seller as trust/conversion support, not a direct organic ranking factor.
F-048	Search Launch Checklist	No listing publishes until relevance, structured data, hero, gallery, conversion, policy and truth gates pass.
F-049	Search Re-Certification	Material listing/product changes invalidate prior search certification.
F-050	Query Cannibalization Audit	Detect portfolio listings targeting materially identical intent without strategic reason.
F-051	Inventory Diversification	Expand legitimate query coverage through relevant product/keyword diversity, not low-quality SKU spam.
F-052	Occasion Truth Model	Use holiday/occasion attributes only when the product is actually for that occasion; use tags for broader gift suitability.
F-053	Recipient/Gift Intent	Use truthful recipient and gift-intent phrases where evidence supports them.
F-054	Solution-Oriented Intent	Model the shopper problem solved and test relevant solution language.
F-055	Style/Aesthetic Intent	Use accurate evidence-supported aesthetic language, not trend stuffing.
F-056	Size/Scale Intent	Capture size/shape intent where meaningful.
F-057	Materials/Technique Intent	Capture buyer-relevant crochet technique/material differentiators truthfully.
F-058	Search Evidence Dashboard	Expose match coverage, rank readiness, visibility warnings, CTR, conversion and experiments.
F-059	SEO Self-Flattery Refusal	Internal checks cannot prove optimization; post-launch real Etsy evidence must take over.
F-060	Search Supremacy Gate	Before scaling, require truthful query coverage, competitive thumbnail quality, conversion readiness and no unresolved first-party warning.

3. REQUIRED ARCHITECTURE

SearchPolicySnapshot: Dated official-rule registry with source, retrieval date, extracted rule, applicability and supersession.

SearchIntentGraph: Product -> query families -> category/attributes/tags/title/description coverage, with provenance.

EtsySearchCertificate: Versioned certificate binding listing content hash to search/truth/creative/conversion readiness.

SearchExperiment: One-variable experiments with baseline, treatment, window, evidence volume, verdict and rollback.

SearchVisibilityFinding: First-party Etsy warning/recommendation captured from Shop Manager with lifecycle state.

SearchPerformanceFact: Observed impressions, clicks, favorites, orders and conversion where accessible; never fabricate missing metrics.

ServiceQualityFact: Reviews, response SLA and case-rate evidence used as operational/search-health signals.

Policy Change Watch: Regularly re-fetch official Etsy guidance and invalidate affected assumptions when source text materially changes.

4. RELEASE / ACCEPTANCE TESTS

- Listing with 13 tags but missing a relevant attribute must fail search certification.
- Listing with an irrelevant Christmas attribute must fail even if predicted impressions rise.
- Title keyword stuffing must lose to a clearer title when query coverage is otherwise preserved.
- Changing a hero image must invalidate click-readiness certification and re-run thumbnail QA.
- Changing category must re-resolve valid attributes and invalidate stale tags/coverage assumptions.
- Search Visibility warning cannot be marked resolved until the Etsy surface no longer reports it or a documented exception is approved.
- Ads spend cannot be credited as an organic-ranking improvement.
- A new listing's temporary boost cannot be treated as demand proof until engagement/conversion evidence matures.
- Zero impressions/clicks must remain zero/unknown; never impute neutral performance.
- Search guidance older than the active policy snapshot must not silently override newer official Etsy guidance.

5. PRIORITY FOR BRAMBLELOOP

P0 before launch: F-001-015, F-020-031, F-039-043, F-048-049, F-052, F-058-060.

P1 immediately after real traffic: F-032-038, F-044-045, F-050-051, F-053-057, because these require real search/customer evidence to become meaningful.

Key implication for the current build: MJs intelligence tells Brambleloop what markets and creative standards exist. This final Search OS tells Brambleloop how a strong product becomes discoverable, clickable and rank-worthy on Etsy without lying to the algorithm or the buyer.

6. SOURCE REGISTER

Primary official Etsy Seller Handbook sources audited for v0.1:

The Ultimate Guide to Etsy Search - <https://www.etsy.com/seller-handbook/article/ultimate-guide-to-etsy-search/366469415790>

Keywords 101: Everything You Need to Know - <https://www.etsy.com/seller-handbook/article/382774281517>

Add Attributes to Help Increase Your Shop's Visibility - <https://www.etsy.com/seller-handbook/article/604203126614>

How Etsy Search Works - Official Etsy Seller Handbook, linked from the Ultimate Guide

The Anatomy of a Well-Crafted Etsy Listing - Official Etsy Seller Handbook, linked from the Ultimate Guide

The New Etsy Search Visibility Page - Official Etsy Seller Handbook, linked from the Ultimate Guide

Creating Listings That Convert - Official Etsy Seller Handbook, linked from the Ultimate Guide

How Great Customer Service Can Improve Your Search Ranking - Official Etsy Seller Handbook, linked from the Ultimate Guide

Version note: v0.1 starts the Final Master Upgrade. Do not hand this to Claude as a replacement for Build 2 while Build 2 is still executing. Continue appending final audits, then deliver the consolidated final-upgrade file at the chosen checkpoint.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.2 Addendum - Benchmark Truth, Validation & Launch Evidence

Purpose: strengthen Build 2's existing purchased-pattern Teardown Laboratory and resolve the mismatch between the owner's intended benchmark workflow and any launch gate that assumes the owner personally crochets a sample. This addendum does not weaken pattern truth. It replaces a person-specific action with an evidence hierarchy and risk-based validation system.

7. PURCHASED BENCHMARK PROGRAM - FINAL HARDENING

ID	Requirement	Executable definition
F-061	Benchmark Purchase Selector	Before the owner buys files, automatically recommend approximately 10 high-information MJsOffTheHookDesigns patterns spanning the product classes Brambleloop intends to sell. Optimize for information diversity, not ten near-duplicates.
F-062	Benchmark Coverage Matrix	The purchased set should deliberately cover multiple construction/experience types where available: garment/sizing, blanket/motif, hat/wearable, stocking/seasonal, ornament/quick make, bag/accessory, dimensional/soft sculpture, home/kitchen and bundle/ebook/premium delivery.
F-063	Purchase Manifest	For each purchased product store canonical listing identity, shop, purchase date, price/sale state, category, why selected, promised deliverables, files received, original filenames and version/hash.
F-064	Quarantine Enforcement	Purchased competitor pattern text, charts, photos and videos remain inaccessible to canonical Brambleloop pattern generation. Analysis produces abstract quality/features/metrics, not reusable competitor instructions.
F-065	Promise-vs-Delivery Audit	Compare listing promises with actual purchased deliverables: file count, page depth, sizing, materials, gauge, charts, photos, video, navigation, printing, support, bonuses and update/version information.
F-066	Best-of-Breed Standard Builder	Construct the Brambleloop Premium Product Standard from the strongest mechanisms across the purchased set rather than copying one seller wholesale.
F-067	Category-Matched Challenge	A Brambleloop release must be compared with the most relevant purchased benchmark for that product class on clarity, completeness, usability, presentation, value, support and confidence.
F-068	No Fake Win	If the benchmark is not genuinely comparable in product form, complexity or customer promise, the comparison is invalid rather than a Brambleloop victory.
F-069	Benchmark Refresh Trigger	Acquire additional benchmark files only when they answer a distinct unknown: new category, unusually strong competitor format, changed customer experience, or major market shift.
F-070	Benchmark Spend Governance	Track benchmark purchase cost and information value; owner approval required above existing authorized research spend.

8. PHYSICAL-VALIDATION GATE - RECONCILIATION

Decision: remove any requirement that specifically requires the OWNER to crochet a sample. The business may use independent testers/contract crocheters when physical evidence is necessary. Purchased competitor files are valuable benchmarks but are **not** evidence that a newly generated Brambleloop pattern physically works. Benchmark quality and pattern-validity evidence remain separate.

ID	Requirement	Executable definition
F-071	Owner Is Not the Tester	No launch requirement may require the owner personally to crochet a product. Physical validation, when required, may be delegated to an independent tester/contract crocheter.
F-072	Validation Evidence Hierarchy	Maintain distinct evidence classes: deterministic compile/reverse-compile/twin; benchmark teardown; model/visual review; independent human pattern review; partial physical swatch/component test; full physical make. Never substitute one class while claiming another.

ID	Requirement	Executable definition
F-073	Risk-Class Validation Matrix	Define minimum evidence by product risk/novelty. Simple constructions with fully proven primitives may qualify for stronger automated validation; fitted garments, complex grading, closed 3-D forms, novel assemblies, high-yardage claims and new construction primitives require stronger independent/physical evidence.
F-074	New Primitive Physical Calibration	Any new stitch/construction primitive whose real-world gauge, drape, assembly or yardage behavior materially affects claims must be physically calibrated before unrestricted use.
F-075	Yardage Calibration Without Owner Labor	Where yardage accuracy cannot be established to the required confidence from validated historical calibration, commission a tester to swatch or make the relevant product/component. Record actual yarn usage, gauge, dimensions, time and deviations.
F-076	Independent Tester Protocol	Tester receives the customer-facing pattern version, not internal CIR hints. Record start/end version, skill level, yarn substitution, gauge, completion status, confusion points, corrections, actual dimensions, yardage and make time.
F-077	Tester Independence	The same agent/system that generated a pattern cannot mark a human/physical test passed without external evidence. Tester results are immutable source evidence; interpretation may be revised, source observations may not.
F-078	Pattern Version Binding	Physical/test evidence applies only to the exact certified pattern version and declared compatible variants. Material pattern changes invalidate affected evidence and trigger re-test/re-certification.
F-079	No Benchmark-as-Physical-Proof	A purchased MJ's or other competitor pattern may prove customer-experience standards and market expectations, but it cannot prove that Brambleloop's independently generated stitch counts, yardage, sizing or assembly physically work.
F-080	No Universal Full-Make Requirement	Do not force a full physical make of every low-risk product if deterministic evidence plus calibrated primitives meets the defined confidence threshold. Spend human testing where uncertainty is material.
F-081	High-Risk Full-Make Gate	Class C/high-risk products - especially fitted garments, novel grading, complex dimensional forms or new structural systems - remain blocked from live sale until required physical evidence exists.
F-082	Calibration Library	Store physical-test results as reusable calibrated evidence for yarn families, gauge ranges, stitch primitives, construction methods and make-time distributions, with explicit applicability bounds.
F-083	Make-Time Distribution Calibration	Replace single assumed maker speed with observed distributions by product class/skill level as tester/customer evidence accumulates; seasonal runway uses uncertainty bounds.
F-084	Yardage Confidence Interval	Report yardage as validated estimate plus safety allowance/confidence; never present false precision derived solely from a digital twin.
F-085	Tester Marketplace Workflow	Build an owner-light workflow to recruit/assign compensated independent crochet testers, deliver files, collect structured results, track deadlines and close validation gates. Owner approval governs spend, not manual testing labor.
F-086	Launch Gate Reconciliation	Replace any Build-2 launch condition phrased as 'owner must crochet/sample' with the risk-based evidence requirement above. Do not delete the underlying truth objective.
F-087	Opening Catalogue Risk Mix	The initial live catalogue should favor products whose construction is already strongly validated while higher-risk products finish testing. This prevents one difficult garment from holding the entire store hostage.
F-088	Post-Launch Reality Feedback	Support tickets, refunds, reviews, customer project photos and version-specific completion evidence update confidence. Any serious pattern defect automatically blocks affected version and descendants.
F-089	Physical Truth Dashboard	Expose products requiring physical evidence, tester assigned, pattern version, test stage, due date, findings, yardage/gauge deltas and whether launch is blocked.
F-090	No Convenience Downgrade	The system may reduce testing only because evidence justifies it, never because launch timing is inconvenient.

9. LAUNCH CONSEQUENCE

The benchmark purchase and physical-validation systems answer different questions. **Benchmark teardown asks: is Brambleloop's customer experience competitive?** Physical/test evidence asks: **does Brambleloop's own pattern actually produce what it claims?** The final launch architecture must satisfy both at the appropriate risk level.

Owner workload target: the owner chooses/approves spend and reviews commercial output. The system selects benchmark purchases, manages testers, gathers evidence and closes gates. The owner is not converted into Brambleloop's crochet production department.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.3 Addendum - AI Resilience, Spend Truth & Legacy Catalogue

Trigger: Build 2 found one load-bearing AI provider, a historical model-spend accounting gap, and eleven legacy products whose declared gauge conflicts with the stated yarn band. Capture these for final hardening without interrupting the current Build-2 executor.

10. AI PROVIDER RESILIENCE

ID	Requirement	Executable definition
F-091	Provider Abstraction Boundary	Route judgement/vision/creative work through a provider-neutral gateway; business agents must not depend directly on one vendor SDK.
F-092	AI Dependency Register	Measure provider concentration by calls, spend, critical workflows and requirements lost during outage.
F-093	Existential Dependency Classification	A single provider carrying all load-bearing judgement work is explicitly an existential dependency until tested fallback exists.
F-094	Second-Provider Trigger	Do not add a second paid provider for symmetry; trigger diversification when outage, price/terms, volume, revenue dependence or downtime cost justifies it.
F-095	Fallback Capability Contract	Define required structured output, vision, context, quality, latency and cost before a fallback can satisfy a gate.
F-096	Provider Compatibility Pack	Maintain provider-neutral golden tasks for extraction, vision, creative jury, benchmark comparison and improvement analysis.
F-097	Shadow Failover Drill	Once a fallback exists, periodically route safe sampled work through it and compare quality/cost/latency.
F-098	Outage Degradation Plan	Deterministic work continues during outage; AI-dependent tasks park explicitly and no weaker substitute silently passes a high-quality gate.
F-099	Provider Change Watch	Version provider pricing/model availability and re-run routing economics on material change.
F-100	Owner-Gated Diversification Spend	Funding/connecting another provider requires owner approval backed by measured resilience value.

11. MODEL-SPEND ACCOUNTING TRUTH

ID	Requirement	Executable definition
F-101	Canonical Billable Taxonomy	All billable model calls enter one canonical accounting taxonomy; aliases such as model/llm cannot bypass ceilings.
F-102	Pre-Call Reservation	Reserve a conservative upper-bound cost before every call; refuse unpriced/unbounded models.
F-103	Post-Call Reconciliation	Reconcile reserved cost to actual measured/provider usage and retain estimate plus settled amount.
F-104	No Missing-Price Zero	Unknown model/pricing or missing ledger data is UNKNOWN/UNMEASURED, never CA\$0.
F-105	Historical Spend Honesty	Dashboard separates measured spend, conservative estimate/upper bound and historical unknown; never rewrite unknown history as measured.
F-106	Provider Billing Reconciliation	Where accessible, compare internal ledger to provider usage/billing and incident material discrepancies.
F-107	Spend by Value Unit	Attribute spend by model, provider, agent, department, requirement, product, benchmark and experiment.
F-108	Real-Gateway Budget Test	Spend-limit tests must exercise the same gateway/accounting path production uses.
F-109	Budget Breach Persistence	A denied over-budget call leaves durable blocked/paused evidence even if surrounding work rolls back.
F-110	Quality Escalation	If budget constrains valuable creative/vision work, request more budget with measured burn/value rather than silently degrading quality.

12. LEGACY CATALOGUE GAUGE / RELEASE DISPOSITION

ID	Requirement	Executable definition
F-111	Legacy Product Quarantine	Affected Build-1 products are pre-calibration and ineligible for live publication until re-engineered or independently validated.
F-112	No Warning-as-Truth Substitute	A 'swatch first' warning cannot compensate for a known implausible gauge/yardage assumption.
F-113	Commercial Salvage Test	Before rescue, re-run current creative, MJs-market, seasonal, novelty and portfolio gates; weak legacy products retire.
F-114	Re-Engineer Worthy Legacy Products	Worthy concepts go through current yarn-band gauge derivation, compiler, twin, reverse compiler and release chain.
F-115	Legacy Artifact Invalidation	Re-engineering invalidates old PDFs, charts, imagery, listings, SEO, support knowledge and certificates.
F-116	Gauge Evidence Gate	New authoring derives initial gauge from declared yarn/material evidence or calibrated physical evidence, not arbitrary constants.
F-117	Twin Plausibility Gate	Compilation alone cannot certify; finished dimensions must remain inside declared tolerance.
F-118	Risk-Based Physical Escalation	If calibrated evidence cannot support gauge, drape, fit, assembly or yardage, route to independent tester workflow F-071-F-090.
F-119	Legacy Launch Count Zero Until Cleared	Old certification does not count as current launch readiness; affected legacy products count zero until cleared.
F-120	Quality Over Salvage	Do not delay launch to rehabilitate all eleven; launch strongest newly engineered/validated products and salvage only those that earn it.

13. FINAL ACCEPTANCE ADDITIONS

- Simulate primary AI provider unavailable: deterministic work continues, AI work parks explicitly, and no fake fallback success is recorded.
- Inject an unpriced model call: it must be refused or conservatively bounded; never recorded as zero-cost.
- Reconcile internal model ledger against provider usage over a bounded window and explain material differences.
- Attempt to release an uncleared legacy product: release must fail.
- Re-engineer one commercially worthy legacy concept and prove downstream invalidation/rebuild.
- Feed a compile-valid but dimensionally wrong CIR to the twin: certification must fail on size drift.

Final-upgrade registry now runs F-001 through F-120.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.4 Addendum - Operational Assurance, Evidence Integrity & Build Hygiene

Trigger: late Build-2 work exposed recurring classes that deserve permanent controls: green tests without outcome proof, vacuous zero-exposure metrics, stale registry states, expected refusals polluting dead-letter health, test-runner disk leakage, misleading commit-signing configuration, incomplete provenance, and production defects invisible to local tests.

14. CLAIMS MUST BE PROVED, NOT INFERRED

ID	Requirement	Executable definition
F-121	Claim-to-Evidence Contract	Every material status/dashboard claim names the direct evidence proving it; configuration, code existence or absence of errors is not outcome proof.
F-122	Positive-Evidence Requirement	Zero exposure/zero denominator returns UNMEASURED or NOT-YET-OBSERVED, never pass, clean, 0% conversion/refund, or failed.
F-123	Vacuous-Test Detector	Reject tests that pass over empty eligible sets unless emptiness itself is the behavior under test.
F-124	Postcondition Verification	After operational changes verify the claimed result directly: cleanup happened, restore digest matched, job completed, artifact rebuilt, credential served a request.
F-125	Production Disagreement Escalation	When production disagrees with tests/modules, treat disagreement as a defect until reconciled.
F-126	Superseded Evidence Retention	Keep invalid/confounded experiments marked superseded with reason.
F-127	Unknown Is First-Class	UNKNOWN, UNMEASURED, NOT-YET-WALKED and NOT-APPLICABLE remain distinct from zero/false/fail/pass.
F-128	No Self-Grading Labels	Labels like verified/live/browser_traversal cannot prove themselves; source capability is independently established.

15. REGISTRY / EXECUTOR TRUTH

ID	Requirement	Executable definition
F-129	Registry Re-Audit	Periodically reconcile requirement states against actual code, tests and production evidence.
F-130	Executable-vs-Ready Clarity	Report executable remaining, READY, RUNNING, dependency-waiting, PARKED, owner-gated and data-gated separately.
F-131	Gate Semantic Accuracy	Gate the capability actually required, not proxies such as env-var presence.
F-132	Automatic Gate Reconciliation	Successful/failed capability evidence automatically un-parks/re-parks dependents.
F-133	No Orphaned Partial	Every partial names the exact missing behavior/evidence and what can close it.
F-134	Never-Idle Proof	READY work with no completion inside the window raises a stall; fully parked work does not.
F-135	Lease-Recovery Drill	Kill a worker mid-requirement and prove lease expiry safely returns work without duplicate consequential effects.
F-136	Executor Completion Gate	Completion is machine-derived from requirement reconciliation, not a session summary.

16. OPERATIONAL SIGNAL HYGIENE

ID	Requirement	Executable definition
F-137	Expected Refusal Separation	Shadow/policy/capability refusals are separate from unexpected dead letters.
F-138	No Retry Storm	Retry only when conditions changed/backoff allows or remediation can plausibly fix the failure.
F-139	Incident Aggregation	Homogeneous large backlogs create one actionable incident with count/scope while retaining item evidence.
F-140	Signal Actionability	A permanently noisy/false health signal is itself a defect.
F-141	Useful-Work Health	Healthy means useful progress where work exists, not merely HTTP 200/heartbeat.
F-142	Split-Process Safe Health	Distributed liveness uses shared/durable evidence, not process-local globals.

ID	Requirement	Executable definition
F-143	Live Console Truth	Console distinguishes healthy, correctly idle, stalled, owner-blocked and degraded.
F-144	Production Readback Gate	Major operational features require deployed readback/verification after local tests.

17. TEST / TEMP / STORAGE HYGIENE

ID	Requirement	Executable definition
F-145	Single Test Scratch Root	Full-suite runs use scoped TMPDIR with cleanup on success/failure/INT/TERM.
F-146	Single Cleanup Trap	One cleanup function/trap; lint/test against a second EXIT trap silently replacing it.
F-147	Leak Postcondition	Green tests do not prove cleanup; verify leftover scratch count/size after run.
F-148	Direct-Test Hygiene	Direct/single-suite runs receive the same scratch isolation.
F-149	Disk Pressure Monitor	Monitor scratch/cache growth, disk utilization and abnormal file-count growth before ENOSPC.
F-150	Failure Classification Before Revert	Classify ENOSPC/permissions/network/provider failures before reverting code.
F-151	Safe Cleanup Scope	Cleanup only Brambleloop-owned paths/prefixes; never broad-wipe shared /tmp.
F-152	Scratch Non-Authority	Recovery/audit evidence cannot exist only in ephemeral scratch.

18. SOURCE-CONTROL / SUPPLY-CHAIN ASSURANCE

ID	Requirement	Executable definition
F-153	Commit Signing Truth Audit	If signing is claimed, prove signatures are generated and independently verifiable; %G?=N or empty key means unavailable.
F-154	No Security Theatre	Configured-but-nonfunctional security controls are reported unavailable.
F-155	Durable Signing Helper	Signing helper/key cannot depend on ephemeral /tmp.
F-156	Signing Recovery Procedure	Document/test restoring signing in a new environment without committing private keys.
F-157	Unsigned-Commit Visibility	Intentional unsigned commits are explicit in release evidence when signing is required.
F-158	Dependency Integrity Audit	Final release verifies dependency locks/hashes where supported and records material supply-chain changes.
F-159	Secret Scan Gate	Scan repo/logs/fixtures/evidence/generated artifacts for credential patterns before release.
F-160	Credential Rotation Register	Secrets passed through chat/manual handoff remain rotation-required until replaced; values themselves are never stored.

19. PROVENANCE GRADUATION & REBUILD SAFETY

ID	Requirement	Executable definition
F-161	Provenance Coverage Metric	Report provenance coverage numerator/denominator by artifact class; absence is unproven, not fresh.
F-162	Graduated Enforcement	During migration count missing provenance; after declared class graduation, absence becomes release-blocking.
F-163	Topological Rebuild	Rebuild descendants in dependency order so children never regenerate from stale parents.
F-164	Cycle Refusal	Dependency cycles are structural defects and refused/escalated.
F-165	Rebuild Coverage Disclosure	Show graph coverage beside rebuild set; zero rebuild with incomplete graph cannot imply freshness.
F-166	Rebuild Effect Proof	Verify fingerprints/certificates and descendant freshness after rebuild.
F-167	Legacy Provenance Backfill	Backfill launch-relevant legacy artifacts; retire obsolete low-value ones.
F-168	Publication Halting Scope	Block affected publication scope rather than whole company unless evidence shows systemic corruption.

20. FINAL BUILD / RELEASE ASSURANCE

ID	Requirement	Executable definition
F-169	Clean-Tree Full Suite	Final handoff requires full suite against exact committed tree intended for deploy.
F-170	Suite Result Binding	Test totals/suite counts/SHA come from runner/git, not manually carried values.
F-171	Production Smoke Matrix	Exercise critical APIs, cadences, queues, provider probes, continuity and search intelligence after final deploy.
F-172	Shadow-Mode Publication Proof	Prove consequential actions blocked before phase change while expected refusals remain non-defects.
F-173	Owner-Gate Inventory	One authoritative launch owner-action list: capability, cost, time, credentials/legal/physical requirement and consequence.
F-174	Data-Gate Inventory	List requirements waiting on customers/orders/traffic/reviews and prevent pre-launch false completion.
F-175	Final Unknowns Register	Launch report explicitly names assumptions/unknowns; zero unknowns is not a goal without evidence.
F-176	Rollback Baseline	Create recoverable pre-launch baseline with code SHA, schema, config names, coverage map, tests and restore proof.
F-177	Final Master Reconciliation	Map F-001+ to Build-2 capabilities first; reuse existing work and build only real gaps.
F-178	Final Master Completion Standard	Complete only when applicable items are implemented/tested/production-verified or genuinely gated.

21. ACCEPTANCE ADDITIONS

Final tests include: empty experiment returns UNMEASURED; Shadow refusal is not an unexpected dead letter; killed requirement worker recovers after lease expiry; scratch root is removed after success and forced termination; ENOSPC is classified as infrastructure failure; signing is cryptographically verified if claimed; incomplete provenance is shown beside stale counts; upstream fingerprint change rebuilds descendants topologically.

Final-upgrade registry now runs F-001 through F-178.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.5 Addendum - Dashboard Truth, Owner UX & Commercial State

Source: live Brambleloop dashboard screenshots supplied 2026-09-21. These requirements preserve what the screenshots actually showed: stale owner-action surfaces, ambiguous spend language, a 134 dead-letter headline whose entries are expected Shadow refusals, legacy products still labelled certified despite current creative failure, a 0.00 CA\$5K probability used where evidence is absent, and mobile status buried under long tables. This addendum does not assume those defects remain after later deploys; it makes them regression classes.

22. DASHBOARD / OWNER-STATE HARDENING

ID	Requirement	Executable definition
F-179	Single Owner-Action Source	Dashboard owner-action surfaces must render from one canonical action/gate state. A stale legacy action table cannot coexist with a corrected queue and ask for already-completed work.
F-180	Owner Action Lifecycle	Owner actions have explicit open, satisfied, superseded, withdrawn, parked and expired states with timestamps/reasons. Satisfied or superseded actions disappear from the active queue without erasing history.
F-181	Action-to-Gate Binding	Every active owner action identifies the exact gate/requirements it can unlock. An action that no longer unblocks anything is not active.
F-182	Stale-Action Deployment Check	After deployment, compare active owner actions with live gate evidence and fail the dashboard verification if already-satisfied actions remain presented as required.
F-183	Spend Vocabulary Unification	Replace ambiguous 'No spend limits configured' wording when a company/model ceiling or agent caps exist. Distinguish global AI ceiling, agent daily caps, infra ceiling, research approvals, ad authority and actual MTD spend.
F-184	Spend Limit Precedence	Dashboard shows which ceiling is binding and how nested limits interact; an agent cap cannot visually imply the company has no limit and vice versa.
F-185	Expected Dead-Letter Presentation	Top-level DEAD LETTERS must split expected refusals from unexpected defects. A headline '134 dead letters' without '134 expected / 0 defects' is operationally misleading.
F-186	Certified-vs-Launch-Cleared Inventory	Top-level Certified Releases and Products must distinguish legacy certification from current launch-standard clearance. Legacy products failing current creative/gauge standards cannot visually read as launch-ready inventory.
F-187	Creative Bottleneck Prominence	When 0 products survive the current creative gate, surface that near the launch summary and opening-catalogue state; do not bury the principal commercial blocker below infrastructure metrics.
F-188	Creative Metric Cohort Label	Creative Standard states which cohort/version it measures (legacy 11 vs new tournament outputs) so old failures do not masquerade as the current generator's performance, and new success cannot erase legacy evidence.
F-189	CA\$5K Probability Semantics	A CA\$5K model with insufficient evidence must display 'UNMEASURED / insufficient commercial evidence' rather than 0.00 probability. Zero is a quantitative claim, not a refusal.
F-190	Evidence-Gate UI	Probability/forecast cards show the missing evidence classes and observed share beside the status, making the difference between 'cannot estimate' and 'estimated near zero' explicit.
F-191	MJs Scan Consistency	All MJs dashboard counters must derive from the same canonical observation state and explain distinctions such as listings known, listings audited, images pending and gallery backlog. Conflicting counts require reconciliation.
F-192	Scan Health Semantics	If MJs observation is actively processing sanctioned evidence, 'scan health: unverified' must explain the unverified dimension rather than implying the whole scanner is unhealthy.
F-193	Gallery Cost Forecast	Show observed cost/image, remaining images, projected drain cost, allocation consumed and expected completion time together; forecasts must update from measured token usage.
F-194	Seasonal Legacy Separation	Seasonal at-risk/missed counts identify whether items are legacy catalogue, current launch candidates or future programme work. Obsolete legacy Halloween products cannot dominate current planning without context.
F-195	Incident Freshness/Actionability	Open incidents include last-confirmed time, current applicability and owning remediation/gate. Policy incidents that cannot be read automatically must not remain indefinitely without an explicit owner/source path.

ID	Requirement	Executable definition
F-196	Rendered-Pages Gate Reclassification	Rendered-page/browser requirements that are intentionally unavailable because sanctioned access does not exist are labelled external-capability unavailable, not ordinary owner negligence. Do not repeatedly ask the owner to buy an ineffective browser.
F-197	Tester Path in Owner UI	Physical proof owner action offers the independent tester route as the normal path and never defaults to asking the owner to crochet. It links to tester roster/status once Final Master validation architecture lands.
F-198	Owned-Surface Definition	Owned-surfaces gate names the specific acceptable surfaces and why they are required. Etsy shop existence must not be conflated with an owned editorial/distribution surface if the requirement means site/Pinterest/etc.
F-199	Launch Readiness Current-Evidence Audit	Each launch-readiness 'ready' row is periodically re-proved against current artifacts/evidence. Legacy ready flags cannot survive a new stricter standard without re-evaluation.
F-200	Dashboard Mobile Priority Order	On mobile, order the page by owner decision value: health/phase -> true active owner actions -> launch blockers -> product/creative truth -> spend -> operations detail/audit trail. Avoid forcing the owner through long low-value tables to find blockers.
F-201	Dashboard Compact Drilldown	Large tables (audit trail, incidents, agents, learning changes) default to concise summaries with expandable/drilldown detail so mobile status remains readable without losing evidence.
F-202	Last-Deployed SHA and Verification	Dashboard header shows deployed commit SHA, deployment time and last successful production verification, preventing confusion between newly pushed fixes and an older rendered dashboard.
F-203	Data Timestamp Consistency	Every major card/table exposes observation/as-of time where freshness matters; mixed snapshots cannot silently appear as one current state.
F-204	Owner Queue Empty Guard	An empty active owner queue closes nothing by itself; assessment health and gate evidence must prove the empty state, preserving the guard Claude just implemented.
F-205	Dashboard Contradiction Test	Automated acceptance test loads the assembled dashboard model and rejects contradictions such as model ceiling present + 'no spend limits', expected refusals + defect-like headline, or satisfied shop + active 'open shop' action.
F-206	Commercial Truth Above Vanity Counts	Top-level counts such as content pieces, listing images and certified releases are secondary to launch-cleared products, creative survivors, benchmark status and real commercial evidence. Do not let output volume imply business readiness.

23. ACCEPTANCE ADDITIONS

- Render the dashboard from a fixture where the Etsy shop/app/model key are already proven; none may appear in the active owner queue.
- With CA\$100 company AI ceiling and agent caps configured, no surface may state 'No spend limits configured' without qualification.
- With 134 expected Shadow refusals and zero unexpected defects, the top-level health presentation must communicate both facts together.
- With zero commercial history, the CA\$5K card must be UNMEASURED/INSUFFICIENT EVIDENCE, not 0.00 probability.
- With legacy products certified under an older standard but 0/11 surviving the current creative gate, no top-level count may imply those products are launch-cleared.
- Mobile viewport acceptance test must expose current owner actions and principal launch/product blockers before long audit/agent tables.
- Header/deployment fixture must prove the dashboard identifies the exact deployed SHA and verification timestamp.

Final-upgrade registry now runs F-001 through F-206.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.6 Addendum - Canonical Model Whole-Person Identity

Observed failure class: the live reference-conditioned trial preserved a strongly similar face across generated images, but the reference and generated subject showed materially different chest/upper-torso proportions. The prior statement 'identity lock held' therefore used an identity definition that was too narrow. This addendum makes face identity necessary but not sufficient.

24. WHOLE-PERSON IDENTITY HARDENING

ID	Requirement	Executable definition
F-207	Whole-Person Identity, Not Face Identity	A canonical Brambleloop model passes identity only when facial identity AND whole-person morphology remain consistent. A convincing face match cannot satisfy the gate by itself.
F-208	Independent Morphology Hard Floor	Whole-person morphology is a separate hard floor from facial identity. Evaluate stature/build, shoulder width, torso length, waist/hip relationship, and chest/bust proportions within pose- and garment-aware tolerances. Material drift on any defining dimension fails the identity trial even when the face matches.
F-209	Chest/Bust Continuity	For adult garment imagery, chest/bust volume and torso proportions are explicit identity features. Clothing may obscure or redistribute apparent contour, but the generator may not materially resize the person's anatomy between reference and output. The reference/output pair observed on 2026-09-20 is a regression fixture: the face is similar while chest morphology materially differs, therefore the trial must fail whole-person identity.
F-210	Garment-Aware Morphology Judge	Identity judging must distinguish anatomy drift from silhouette changes caused by garment structure, padding, ease, pose, camera perspective and layering. When clothing prevents a defensible anatomy comparison, return UNMEASURABLE for that dimension rather than pass.
F-211	Reference + Candidate Required	Every identity judge receives the canonical reference and generated candidate together. A one-image 'same person?' judgment is invalid and cannot write a score.
F-212	Reference Conditioning Proven End-to-End	The generation request must prove the canonical reference was actually transmitted through the provider-specific adapter. Prompt text claiming 'same person as reference' is not evidence of conditioning.
F-213	Identity Trial Validity	A trial with missing reference bytes, unsupported provider conditioning, expired reference, missing judge pair, or unmeasurable morphology is INVALID/UNMEASURED - never a zero score and never evidence against a provider.
F-214	Cross-Season Identity Pack	The canonical model is tested across at least three materially different contexts (for example fitted garment, loose/layered garment, and accessory/outerwear) so consistency is not inferred from two near-identical portraits.
F-215	Garment Fit Requires Stable Body	No garment-fit or sizing score is valid unless the model's relevant body morphology is within identity tolerance. If the body changes, the system cannot attribute fit differences to the garment.
F-216	Provider Benchmark Re-run Rule	Any provider scores produced by the broken identity protocol are superseded for identity-related dimensions. A corrected benchmark reruns affected trials; invalid historical scores remain in provenance but cannot contribute to winner selection.
F-217	Identity Subscores Exposed	Benchmark output reports face identity, whole-person morphology, and overall identity separately. A high face score may not hide a morphology failure inside an average.
F-218	Canonical Model Drift Sentinel	Customer-facing generations are compared with the canonical identity pack before release. Repeated near-threshold morphology drift raises an incident even if individual images narrowly pass, preventing gradual body drift across a season.
F-219	Human Review Escalation Band	Borderline whole-person identity results enter a review band rather than being auto-passed. The system may auto-refuse obvious drift; it must not auto-approve ambiguous anatomy from obscured clothing or pose.
F-220	Identity Benchmark Regression Fixture	Keep a sanitized regression representation of the 2026-09-20 failure class: same-looking face, materially different chest/torso morphology. Tests must prove the face-only implementation passes the face component but the whole-person gate refuses the asset.

25. REQUIRED ACCEPTANCE TESTS

- A candidate with a strong facial match but materially changed chest/torso morphology **MUST** fail whole-person identity.

- A loose or structured garment that prevents a defensible morphology comparison MUST yield UNMEASURABLE for the obscured dimensions, not PASS.
- The judge MUST receive both reference and candidate images. Removing either image invalidates the trial.
- The provider adapter MUST demonstrate that reference bytes/asset were sent, not merely that the prompt mentioned a reference.
- Garment-fit scoring MUST refuse to run when relevant body morphology is outside tolerance.
- Historical identity scores from the broken one-image/no-reference protocol MUST be excluded from provider selection.
- Identity reporting MUST expose face and morphology separately; averaging cannot rescue a hard-floor failure.

Final-upgrade registry now runs F-001 through F-220.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.7 Addendum - Natural Listing Photography & Photographic Realism

Owner-authoritative visual standard: customer-facing listing imagery must look like believable real photography, not obvious AI imagery. MJsOffTheHookDesigns may be used as benchmark evidence for the general photographic qualities and gallery structure that make strong crochet listings feel authentic, while competitor images remain analysis-only and may not be copied or reproduced. Product truth and photographic realism are independent release gates.

26. NATURAL LISTING IMAGE HARDENING

ID	Requirement	Executable definition
F-221	Photographic Realism Hard Floor	Every customer-facing listing image must independently pass a photographic-realism gate. The target is a believable high-quality photograph of a real crochet product in a plausible environment, not merely attractive generated art. Obvious synthetic/AI appearance blocks release.
F-222	Natural, Not Over-Polished	Do not optimize every frame toward sterile perfection. Prefer believable natural light, skin texture, hair, fabric behaviour, folds, ordinary poses, realistic homes/outdoor settings and controlled imperfection. Plastic skin, excessive beauty retouching, implausibly pristine scenes and generic AI-editorial/Pinterest perfection are release defects.
F-223	Generative-Tell Rejection	Reject or regenerate images with impossible anatomy, malformed hands, inconsistent shadows/reflections, artificial depth of field, repeated or melted textures, incoherent props, fake-looking yarn, impossible stitch geometry, unexplained text/logos/brands, or other material generative artifacts.
F-224	MJs Photographic Benchmark	Use sanctioned MJsOffTheHookDesigns listing evidence as a benchmark for general photographic principles: lighting, framing, product scale, environment, styling, image sequence, detail coverage, model-vs-product balance and perceived naturalness. Learn mechanisms and standards only; do not reproduce their specific photos, compositions or protected creative expression.
F-225	Gallery-as-Real-Photo-Shoot	A listing gallery must read as one coherent, believable photo shoot rather than unrelated generated hero images. Where relevant include hero, useful alternate angles, scale/context, construction/detail close-ups and worn/fit views. Maintain consistent product identity, materials, colour and construction across frames.
F-226	Canonical Model Naturalism	The canonical Brambleloop model must remain identity- and morphology-locked while still looking like a real person photographed on different days. Permit natural variation in pose, expression, hair and setting; reject copy-paste sameness, over-smoothed skin and artificial glamour that makes the model look synthetic.
F-227	Product Truth Outranks Beauty	Crochet fidelity outranks photographic beauty. The pictured object must agree with the certified pattern/CIR/digital twin on visually material construction: stitch structure, motif, repeat geometry, shaping, seams/joins, edging, dimensions, colour placement, drape and garment fit. A beautiful photograph of the wrong object fails.
F-228	Motif & Construction Fidelity Gate	Where the certified pattern specifies a motif, repeat or visually material construction feature, imagery must faithfully depict it. The observed checkerboard-versus-diamond-lattice failure is a permanent regression fixture: matching category, colour, dimensions and lighting cannot rescue incorrect motif geometry.
F-229	Independent Dual Verdict	Before release, independently judge (A) PRODUCT TRUTH - does the image depict what the certified pattern actually makes? and (B) PHOTOGRAPHIC REALISM - would the frame plausibly exist as an ordinary high-quality photograph without obvious generative artifacts? Both must pass. Never average them into one score.
F-230	Unreadable Is Not Pass	If stitch structure, motif, fit, anatomy or another required visual fact cannot be reliably assessed because of framing, distance, clothing, blur, pose or occlusion, return UNREADABLE/UNMEASURABLE for that dimension. Customer-facing release cannot treat absence of evidence as a pass.
F-231	Benchmark-Derived Photo Grammar	After the MJs teardown is legally purchased and ingested, quantify high-level gallery grammar such as image count/types, hero composition, close-up frequency, background/environment patterns, model-versus-product-only mix, lighting, framing and within-listing consistency. Store derived abstractions with provenance; competitor images remain source artifacts only.
F-232	Generated-Content Disclosure Integrity	Generated-image disclosure requirements remain authoritative. Photographic realism must never be achieved by concealing a material fact, fabricating proof, or representing generated imagery as real-world evidence when platform rules or Brambleloop policy require disclosure.

27. REQUIRED ACCEPTANCE TESTS

- A visually attractive image with obvious synthetic/AI artifacts MUST fail photographic realism even if product truth passes.
- A photorealistic image depicting the wrong motif, stitch structure or construction MUST fail product truth even if photographic realism passes.
- The checkerboard-versus-diamond-lattice regression fixture MUST continue to fail.
- An unreadable motif/detail or obscured body/fit dimension MUST be UNREADABLE/UNMEASURABLE, never PASS.
- A model-bearing gallery MUST preserve canonical identity and morphology without forcing identical pose/expression/hair across every frame.
- A gallery fixture composed of individually plausible but visually unrelated frames MUST fail gallery coherence.
- MJIs-derived benchmark data MAY inform general photographic standards and gallery grammar; no acceptance test may require copying a competitor image or composition.
- Competitor source images MUST remain quarantined from Brambleloop product generation and marketing outputs; only provenance-backed abstractions may cross the boundary.
- Generated-content disclosure MUST remain correct even when the image is sufficiently realistic that a viewer could mistake it for a photograph.

Final-upgrade registry now runs F-001 through F-232.

Version note: v0.7 appends the owner-authoritative natural-listing-image standard to v0.6. All earlier requirements remain in force unless explicitly superseded.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.8 Addendum - Etsy Launch, Storefront, Search, Ads & Visibility OS

P0 / launch-critical. A great pattern that nobody sees is not a commercial product. This addendum makes storefront completion, Etsy search visibility, click-through, conversion, advertising, attribution and launch learning first-class systems. Official Etsy guidance audited 2026-09-21 states that search uses query matching and ranking; relevant titles/tags/descriptions/categories/attributes, shop/listing quality, customer service and listing engagement matter. Etsy Ads buy designated paid exposure and do not directly improve organic placement.

28. ETSY STOREFRONT / SHOP QUALITY

ID	Requirement	Executable definition
F-233	Storefront Completion Gate	No paid traffic or public launch until icon, banner/cover where available, shop title/tagline, announcement, About, policies/FAQ, seller/public identity, sections, featured/ordered listings and coherent branding are complete. Blank/default surfaces block launch.
F-234	Shop Trust Architecture	The shop clearly explains what Brambleloop sells, what the buyer receives, digital-delivery expectations, skill expectations, support path and policies without forcing buyers to infer them.
F-235	About / Expertise / Process	Create a truthful, substantive About section and appropriate images/video showing Brambleloop's real process/quality system. Never invent years of experience, physical-making claims or credentials.
F-236	Shop SEO Surface	Tagline and shop description use natural buyer language to explain the category/value proposition and support external search understanding without keyword stuffing.
F-237	Section Architecture	Sections reflect buyer browsing logic such as occasion, use/recipient and product family rather than opaque internal department names.
F-238	Opening Grid Merchandising	Deliberately merchandise the first visible grid: strongest products first, seasonal opportunity visible, visual repetition controlled, price/value ladder apparent, and no weak legacy filler.
F-239	Mobile Storefront QA	Before launch verify icon, banner crop, announcement opening, sections, first grid and thumbnails on mobile.
F-240	Public Seller Identity Integrity	Approved public shop identity may differ from legal/tax presentation only where Etsy permits; legal, payout and tax identity remain truthful.
F-241	Seasonal Storefront Takeover	Seasonal banner/announcement/featured-product changes have start/revert rules and never erase permanent Brambleloop identity.

29. ETSY SEARCH VISIBILITY

ID	Requirement	Executable definition
F-242	Official Search Policy Snapshot	Maintain dated official Etsy search/search-visibility guidance. Material changes invalidate affected assumptions/certificates.
F-243	Query-Matching Completeness	Each listing uses the most specific accurate category, relevant attributes, all 13 useful tags, clear title, useful description and truthful metadata.
F-244	Search Intent Graph	Map each product to buyer query families and corresponding category/attributes/title/tag/description coverage with provenance; avoid needless catalogue cannibalization.
F-245	Buyer-Friendly Title Gate	Titles clearly identify the item and meaningful differentiators while preserving relevant query coverage without stuffing or repetitive phrasing.
F-246	Tag Diversity Gate	Use varied relevant tags, not near-duplicates. Seasonal terms must be truthful. Refresh from observed search terms/current evidence rather than fabricated demand.
F-247	Attribute / Category Integrity	Categories and attributes are maximally specific and accurate; traffic opportunity never justifies irrelevant metadata.
F-248	Search Visibility Intake	Capture accessible Etsy Search Visibility recommendations with date, listing, shop/listing/service category, remediation and lifecycle. Code changes alone cannot mark them resolved.
F-249	Personalized-Ranking Humility	Manual searches of our own listings are not ranking proof. Use Shop Stats/Search Visibility/attribution traffic because Etsy results are personalized/context-specific.
F-250	Recency Non-Gaming	New/renewed listing boosts are temporary learning opportunities, not a renewal-spam strategy.
F-251	Search Language Integrity	Titles/tags follow shop-language rules; translations are deliberate and not used as impression bait.

30. CLICK-THROUGH / CONVERSION

ID	Requirement	Executable definition
F-252	First-Photo Click Gate	First image clearly shows the product, survives thumbnail/mobile crop, is high-resolution/well-lit, avoids confusing collage treatment and maximizes truthful click appeal.
F-253	Natural Photo + Product Truth	Final Master F-221-F-232 applies to every gallery: natural photographic realism and exact pattern/product fidelity are independent hard gates.

ID	Requirement	Executable definition
F-254	Gallery Information Architecture	Gallery frames have jobs: hero, useful alternate angle, stitch/motif detail, scale/context, construction detail, colour context and fit/worn views where applicable.
F-255	Description Conversion Architecture	Descriptions lead with what is sold and why it matters, then skill level, result, sizing, materials, yarn/hook, gauge, inclusions, delivery, support/terms and decision-critical facts.
F-256	Digital-Pattern Expectation Truth	Listings cannot imply a finished physical item when selling a digital pattern. Make the offer understandable before purchase without destroying visual appeal.
F-257	Price / Value Coherence	Price, bundle, sale state and presentation tell one value story; fake anchors/permanent discounts/confusing jumps are refused.
F-258	Conversion Evidence Loop	Measure impressions -> clicks -> favorites -> orders where accessible. Diagnose visibility, click and conversion as different failure stages.
F-259	Neutral New-Listing Prior	New listings begin UNMEASURED, not successful/failed. Update from exposure and engagement evidence.
F-260	Listing Experiment Discipline	Version controlled tests on meaningful variables preserve baseline/treatment windows and refuse conclusions without exposure.
F-261	Underperformer Remediation	No impressions -> relevance/query; impressions/no clicks -> hero/title/value; clicks/no sale -> product/value/trust/price; post-sale friction -> product/customer experience.

31. ETSY ADS - PAID VISIBILITY

ID	Requirement	Executable definition
F-262	Paid vs Organic Separation	Etsy Ads and organic search are measured separately. Ads do not directly improve organic placement outside designated ad spaces.
F-263	Ads Launch Readiness Gate	Never buy traffic to unfinished storefronts or uncertified listings. Ads require storefront, listing truth, search, imagery, offer, policy and attribution readiness.
F-264	Initial Etsy Ads Learning Test	Design launch testing consistent with current Etsy guidance: sufficient budget/duration to learn. Current Etsy guidance recommends at least US\$3-5/day initially and, when practical, promoting all active listings to gather engagement data.
F-265	Per-Listing Strategy Threshold	Recognize Etsy's current US\$25/day maximum-budget threshold for per-listing strategies; below it do not pretend visibility/efficient/lower-click-cost controls are available.
F-266	Learning-Window Discipline	Avoid daily strategy thrash. Preserve meaningful optimization windows unless truth, safety, runaway spend or obviously invalid traffic requires intervention.
F-267	Ad Budget Owner Authority	Advertising is consequential spend. New/increased ad ceilings require owner approval and are enforced before spend.
F-268	Ad Allocation Evolution	Broad launch learning may narrow after evidence. Never cut unmeasured listings merely because null sorts below measured winners.
F-269	Ad Funnel Attribution	Track spend, impressions, clicks, CPC, orders/revenue where accessible, conversion, contribution and evidence volume per advertised listing.
F-270	Evidence-Aware Stop / Scale	Pause/continue/scale rules require adequate exposure. Spend with clicks/no conversion triggers diagnosis before more budget.
F-271	Creative Before Spend Escalation	Impressions/weak clicks -> improve hero/title/value; healthy clicks/weak conversion -> diagnose product/listing/trust/price. Do not solve every problem with more budget.
F-272	Ads Cannot Rescue Bad Products	Ads never override Creative Standard, Pattern Truth, Asset Truth or customer-experience gates.
F-273	Offsite Ads Accounting	Track Etsy Offsite Ads separately; account for attributed fees/economics without claiming control over which listings Etsy promotes.
F-274	Paid Experiment Ledger	Every campaign/change records objective, authorization, listings, budget, settings, window, evidence, verdict and rollback/next action.

32. LAUNCH / SEASONAL ORCHESTRATION

ID	Requirement	Executable definition
F-275	Launch Demand Capture Plan	Before phase change link opening products to search intents, seasonal runway, storefront placement, organic readiness, Etsy Ads plan and measurement checkpoints.
F-276	Christmas Search Compression	Prepare Christmas query/title/tag/attribute/gallery/storefront/ad work early enough to collect engagement before peak buying; urgency cannot bypass truth/quality.
F-277	Portfolio Query Coverage	Opening catalogue covers multiple relevant intents/product families rather than many listings fighting for one query.
F-278	Hero Thumbnail Tournament	Challenge first-photo candidates at Etsy-like thumbnail/mobile size; select for truthful clarity and click appeal, not full-resolution beauty.

ID	Requirement	Executable definition
F-279	Storefront-to-Listing Continuity	Banner, announcement, sections and featured products reinforce the same seasonal/value story as search/ad landing listings.
F-280	Launch Search Visibility Watch	During launch week inspect accessible Search Visibility issues frequently enough to react quickly without evading unsanctioned surfaces.
F-281	First-30-Day Learning Plan	Predefine day 1/3/7/14/30 reviews of impressions, clicks, favorites, sales, conversion, search terms, ads, support friction and contribution.
F-282	No Zero-Sales Panic	Short early zero-sales windows require exposure diagnosis first; adequate qualified exposure with persistent no-conversion cannot be excused indefinitely as 'new shop'.
F-283	First-Sale Attribution	Record first-sale source honestly (organic Etsy, Etsy Ads, Offsite Ads, owned/direct, unknown). One sale proves possibility, not scale.
F-284	Review / Service Flywheel	Prioritize flawless download, rapid truthful support and defect prevention. Reviews/response/case health feed trust/search; never buy or gate reviews.
F-285	Seasonal Keyword Refresh	Before major seasons refresh query/title/tag strategy from current Etsy guidance, Search Visibility, observed terms and market evidence.
F-286	Launch Success Multi-Signal	Launch success requires visibility + clicks + honest conversion + product truth + customer experience + unit economics, not merely 'live' or 'ads running'.

33. OWNER VISIBILITY COMMAND CENTER

ID	Requirement	Executable definition
F-287	Visibility Command Center	Dashboard shows storefront completion, Search Visibility issues, search readiness, search terms, impressions, clicks, conversion, ads, Offsite Ads, seasonal readiness and real owner actions.
F-288	Actionable Recommendations	Every search/ads recommendation names evidence, affected listings, expected mechanism, cost, confidence and rollback. 'Improve SEO' is not an action.
F-289	Traffic Source Separation	UI separates Etsy organic, Etsy Ads, Offsite Ads and owned/direct traffic; blended totals never hide components.
F-290	Owner Ads Approval Cards	Ad authorization cards state daily/monthly ceiling, listings, objective, evidence, expected learning period, worst-case spend and consequence of waiting.
F-291	Search / Ads Policy Watch	Re-fetch official Etsy search/ads guidance and invalidate assumptions when thresholds, controls or guidance change.
F-292	No Vanity Optimization	Do not optimize raw impressions/clicks/favorites/content count/ad spend in isolation; prioritize qualified traffic, conversion, contribution, truth and customer experience.

34. FINAL ETSY LAUNCH ACCEPTANCE

ID	Requirement	Executable definition
F-293	Storefront Visual Inspection	Review complete BrambleloopStudio storefront at desktop/mobile before launch; no blank/default surfaces, stale identity, weak opening grid or seasonal mismatch.
F-294	Search Certificate Required	Opening listings require current search/truth/creative/conversion certification bound to exact content/image fingerprint; material edits invalidate affected checks.
F-295	Ads Dry Run	Simulate campaign plan before spend and prove uncertified listings, stale assets and over-budget actions cannot enter.
F-296	Launch-Week Ad Envelope	Owner explicitly approves launch-week Etsy Ads ceiling separately from AI/infrastructure budgets.
F-297	Measurement Before Optimization	Prove accessible organic/paid evidence can be recorded before enabling automated optimization.
F-298	No Visibility Shortcut	No misleading keyword, irrelevant attribute, renewal trick, clickbait image or ad tactic may bypass truth, IP/policy, product accuracy or customer experience.
F-299	Controlled Phase Change	Shadow -> staging -> live publication/ads is explicit owner-approved transition with rollback.
F-300	Launch Readiness Verdict	Final challenge answers separately: Can buyers find it? Will they click? Does it convert honestly? Does the pattern make what is shown? Is the shop trustworthy? Are ads measurable/bounded? Any failed hard gate blocks launch.

35. REQUIRED ACCEPTANCE TESTS

- A complete product with no shop icon/About/policies/sections MUST fail storefront launch readiness.
- A listing with strong ads but weak organic evidence MUST NOT be reported as organically improved.
- A listing with impressions but weak clicks MUST route to click/hero/title diagnosis before budget escalation.
- A listing with healthy clicks but no sales MUST route to product/value/trust/price diagnosis rather than SEO by default.

- A new listing with inadequate exposure MUST remain UNMEASURED, not weak.
- A misleading seasonal tag/attribute MUST fail even if it increases impressions.
- The first-photo candidate MUST pass Etsy-like thumbnail/mobile review and Final Master photographic/product-truth gates.
- Ads cannot activate for uncertified listings or exceed owner-authorized ad ceilings.
- Organic Etsy, Etsy Ads, Offsite Ads and owned/direct traffic MUST remain separately attributable.
- Launch cannot pass while measurement ingestion is incapable of observing the signals the optimization loop claims to optimize.

Final-upgrade registry now runs F-001 through F-300.

Official Etsy sources audited for this addendum: How Etsy Search Works (Aug. 26, 2025); The Ultimate Guide to Etsy Search; Checklist: Optimize Your Shop for Etsy Search; Creating Listings That Convert; The Anatomy of a Well-Crafted Etsy Listing; The New Etsy Search Visibility Page; How Great Customer Service Can Improve Your Search Ranking; Getting Started With Etsy Ads (June 1, 2026); How to Set Up and Manage an Etsy Ads Campaign; 7 Common Questions About Etsy Ads; How Etsy's Offsite Ads Work; Search Engine Optimization for Shop and Listing Pages.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.9 Addendum - Production Economics, API Sustainability & Cost Governance

P0 / launch-critical sustainability standard. The build has demonstrated that a system can be technically correct while economically wasteful. Brambleloop must be capable of using premium models when quality requires them without repeatedly purchasing the same evidence, retrying settled failures, or hiding provider spend behind estimates. The governing rule remains: **QUALITY FIRST. COST SECOND. WASTE NEVER.** Cost optimization may remove waste; it may not weaken Pattern Truth, Product Truth, photographic realism, identity, customer experience or other hard quality floors.

36. PROVIDER COST TRUTH & RECONCILIATION

ID	Requirement	Executable definition
F-301	Provider-Observed Spend Reconciliation	Internal spend is not authoritative by itself. Reconcile Brambleloop's ledger against actual provider-observed usage/cost for Anthropic, OpenAI, Google, BFL and future paid providers. Unexplained variance is a trust blocker and must be surfaced, not silently absorbed.
F-302	Credits Are Not Budgets	Provider prepaid balance, credit purchase, account limit and Brambleloop spending authorization are four separate facts. Available provider credit never grants permission to spend beyond Brambleloop's owner-approved ceiling.
F-303	Development vs Production Economics	Tag spend as development/build, benchmark/research, one-time product creation, listing/creative production, recurring production, support/operations, advertising-analysis, or other named purpose. One-time build burn must never be presented as steady-state operating cost.
F-304	Per-Operation Cost Attribution	Every paid call records provider, model, job/run, agent, product/listing where applicable, purpose, input/output usage or image count, estimated cost, actual/observed cost when available, timestamp and evidence produced.
F-305	Unattributed Spend Is a Defect	Every provider dollar must be attributable or explicitly placed in an unattributed bucket. Attributed + unattributed must reconcile to provider-observed spend. Unattributed spend above a defined tolerance raises an incident.
F-306	No Duplicate Paid Evidence	Certified facts and valid observations are durable evidence. Never repay an LLM/vision model to judge identical evidence merely because a cadence runs again. Content/evidence fingerprints must make duplicate paid work idempotent.
F-307	Paid-Job Idempotency	Retries, deploys, worker restarts and concurrent claims cannot cause the same paid operation to execute twice without an explicit new evidence/version key. Reservation, execution and durable completion are transactionally linked where practical.
F-308	Bounded Retry Economics	Every paid job has a bounded retry policy differentiated by failure type. Authentication/funding/configuration/content refusals do not enter generic transient retry loops. Retry storms are a financial-integrity incident.
F-309	Systematic-Failure Circuit Breaker	When repeated attempts fail the same hard floor across products/methods/providers strongly enough to indicate a systematic block, stop paid retries, name method_blocked_on, preserve evidence and require a changed method/provider/owner decision before spending again.
F-310	New Evidence Requirement	A recurring paid cadence must produce new evidence or measurable state progress. Repeated cost without new durable evidence is detected as stalled/waste and automatically stops or escalates.
F-311	Cache Certified Evidence	Pattern certificates, image judgments, identity verdicts, policy snapshots, benchmark observations and other stable evidence are cached/versioned and reused until an upstream fingerprint changes. Cache invalidation follows provenance, not elapsed time alone.
F-312	Deterministic Before Generative	Use deterministic code, database queries, local computation or cached evidence when they can answer the question with equivalent correctness. Paid model calls are reserved for judgment/generation tasks that genuinely require them.
F-313	Quality-Proven Model Routing	Route each task to the least expensive model/provider that has demonstrated equivalent performance on that task's required quality floors. Cost alone can never justify a downgrade. Quality-first routing decisions are evidence-backed and reversible.
F-314	Deep-Tier Escalation Policy	Expensive/deep models are escalation resources, not defaults. Define when Sonnet/Opus/deep judging is warranted, when a standard/cheap tier is sufficient, and when deterministic checks eliminate model use entirely.
F-315	Prompt/Input Token Discipline	Measure and control context size. Do not repeatedly send large unchanged histories, full catalogues or evidence bundles when a fingerprint, summary, retrieval slice or structured record suffices. Token reduction may not remove evidence required for correctness.
F-316	Output Token Discipline	Model outputs use the minimum structured schema needed for the decision. Long narrative reasoning is not purchased where a compact verdict/evidence schema is sufficient. Truncation risk must still be prevented.
F-317	Batching With Isolation	Batch model work only when it reduces cost without contaminating evidence between products/items. A batch failure cannot silently invalidate or duplicate-charge unrelated completed items.
F-318	Backlog Must Actually Drain	Any paid backlog processor reports unique pending before, unique completed, unique pending after and spend. Completion must durably reduce backlog. A cadence repeatedly processing the same items while backlog remains flat is a financial-integrity failure.

ID	Requirement	Executable definition
F-319	Spend-to-Progress Ratio	Track cost per new durable evidence unit and cost per requirement/product/listing advanced. Large spend with little new evidence triggers investigation even when the monthly ceiling has not been reached.
F-320	Per-Product Creation Cost	For every sellable pattern, compute total one-time creation cost across concept/research, pattern engineering, certification, imagery, listing creation and necessary judging. Preserve provider/model breakdown and distinguish sunk development infrastructure.
F-321	Per-Listing Maintenance Cost	Track recurring AI/API cost attributable to each live listing: observation, revalidation, seasonal refresh, support intelligence and other maintenance. Stable listings should approach very low recurring inference cost.
F-322	Cost Per Usable Asset	For generated imagery record attempts, failures, accepted assets, generation cost, judging cost and cost per completely usable asset/gallery. A low first-pass rate cannot be hidden behind one successful render.
F-323	Cost Per Sale / Contribution	Where attribution permits, calculate revenue minus Etsy fees, advertising, refunds, variable AI/API cost and other variable costs. AI cost is part of contribution economics, not an infrastructure footnote.
F-324	Break-Even Sales Requirement	For each product estimate how many sales are required to recover one-time product creation cost, then distinguish that from recurring margin. High-quality one-time creation spend can be rational if the reusable digital product has strong lifetime economics.
F-325	Steady-State Sustainability Forecast	Forecast monthly AI/API cost at realistic catalogue sizes, sales volumes, support loads and observation cadences. Include low/base/high cases and name assumptions. Refuse a precise forecast when operational data is insufficient.
F-326	Provider Spend Anomaly Detection	Detect abnormal daily/hourly burn relative to measured baseline and workload. Alert on sudden token/image spikes, repeated identical purposes, unexpected expensive-model routing, backlog loops and provider/internal ledger divergence.
F-327	Hard Spend Reservations	Consequential paid jobs reserve expected budget before execution and settle actual cost afterward. A job cannot begin when its reservation would exceed the applicable owner-authorized ceiling.
F-328	No Spend To Look Busy	Autonomous agents/cadences may not consume paid inference merely because they are scheduled or READY. Every paid call must answer a specific unresolved question, produce new evidence, create a required asset, or perform necessary live work.
F-329	Sustainability Launch Gate	Before launch, demonstrate estimated steady-state Brambleloop economics are commercially sustainable at opening prices. If expected recurring AI/API cost threatens product contribution or requires constant owner top-ups unrelated to sales, launch is blocked until corrected.
F-330	Post-Launch Cost Learning	After launch review actual unit economics at day 1/7/14/30 and thereafter. Optimize waste aggressively while preserving Pattern Truth, Product Truth, photographic realism, customer experience and other quality floors. Never cheapen the product to make a cost dashboard green.

37. REQUIRED ECONOMICS / WASTE ACCEPTANCE TESTS

- Provider-observed Anthropic/OpenAI/etc. spend and internal ledger totals must reconcile within an explicit tolerance; unexplained variance raises an incident.
- Re-running a cadence over unchanged certified evidence must create zero duplicate paid judgments.
- A worker restart during a paid job must not cause a second billable execution after durable completion.
- A funding/configuration refusal must not enter a generic paid retry loop.
- A systematic repeated hard-floor failure must open a circuit breaker before the remaining monthly budget is consumed.
- A stable live listing with no changed upstream evidence must not repeatedly incur full re-certification cost.
- Changing an upstream fingerprint must invalidate only the dependent evidence that genuinely requires recomputation.
- An expensive model may not replace a cheaper proven model by default, and a cheaper model may not replace a quality-proven model without equivalent-floor evidence.
- Per-product economics must distinguish one-time creation cost from recurring listing cost.
- A product with positive gross sales but negative contribution after Etsy/ads/AI variable cost must be surfaced as economically unsustainable.
- Launch readiness must include a steady-state API-cost forecast and owner-visible worst-case monthly burn.
- Provider credit availability must never override an owner-approved Brambleloop spending ceiling.

Observed build lesson incorporated: large provider token usage during development is not automatically a business-model failure; the launch requirement is that Brambleloop can explain what the spend bought, eliminate repeated/unproductive inference, and demonstrate sustainable steady-state unit economics.

Final-upgrade registry now runs F-001 through F-330.

Version note: v0.9 appends production-economics and API-sustainability requirements to v0.8. All earlier requirements remain in force unless explicitly superseded.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.10 Addendum - Durable Job Observation, Waiter Safety & Progress Truth

P0 / launch-critical reliability standard. This addendum incorporates the production lesson in which the full test suite completed successfully in roughly seven minutes (3,127 passing / 0 failing), while four observer shells continued waiting for roughly 100 minutes because each used a process-signature search that matched its own command line. Additional waiters increased the match count and were incorrectly read as workload progress. Brambleloop must never infer job liveness or progress from an observer that can observe itself.

Governing rule: OBSERVE THE WORK, NOT THE OBSERVER.

38. DURABLE JOB OBSERVATION & WAITER SAFETY

ID	Requirement	Executable definition
F-331	Authoritative Job Identity	Every long-running operation receives a durable unique job/run ID. Consumers observe that ID and its persisted state/result rather than rediscovering work by fuzzy process-name matching.
F-332	No Self-Matching Liveness Probe	A liveness/progress probe must be structurally incapable of matching its own observer/waiter process. Process-signature searches such as pgrep/grep are prohibited as authoritative evidence unless self-exclusion and target identity are proven.
F-333	One Operation, One State	Multiple agents, dashboards or waiters observing the same operation must subscribe/read the same authoritative state. They may not spawn duplicate underlying operations or independent chains of waiters merely to learn completion.
F-334	Waiter Deduplication	At most one active waiter/monitor is created for a given job when a waiter is required. Duplicate wait requests attach to the existing monitor or read durable state; duplicate waiters are detected and terminated safely.
F-335	Completion Is Durable	Completion, failure, cancellation, timeout and result/evidence are persisted independently of the observer session. A completed job cannot remain 'running' merely because an observer missed the terminal event.
F-336	Progress Must Be Work-Derived	Progress comes from the underlying workload: completed/total units, durable checkpoints, output timestamps or explicit stage transitions. Observer count, poll count, process count, elapsed time and number of waiters are never evidence of workload progress.
F-337	Expected-Duration Watchdog	Track rolling normal duration by operation class. When runtime materially exceeds the established envelope (default investigation threshold 3x rolling normal where enough history exists), verify authoritative job state before continuing to wait, restart or escalate.
F-338	Non-Destructive Hang Diagnosis	Before killing/restarting a long-running operation, determine whether it is alive, advancing, completed-but-unobserved, blocked on a child/dependency, or duplicated. Preserve valid output and evidence already produced.
F-339	No Blind Restart	A slow or apparently stuck operation is not rerun until the prior operation's authoritative state is known. Restart/retry must not duplicate paid calls, mutate shared state twice, or discard a valid completed result.
F-340	Waiter Resource Bound	Waiting/monitoring has bounded wall time, process count and resource use. On bound breach, stop spawning observers and escalate to authoritative-state diagnosis; do not let a monitoring defect consume hours unattended.
F-341	Result Propagation Contract	The producer writes a terminal result exactly once and all consumers read that result. Result propagation is tested across shell exit, agent interruption, deploy/restart and delayed observers.
F-342	Observer/Work Separation	Monitoring code and workload code are separately identifiable in process/job metadata. A query for the workload cannot be satisfied by the observer itself, its parent shell, or another monitoring process.
F-343	No False Healthy/Busy State	Health dashboards must distinguish 'work executing', 'work completed', 'observer waiting', 'blocked', and 'unknown'. A waiter being alive cannot make the system healthy, busy, progressing or non-idle.
F-344	Operational Time Waste Accounting	Record avoidable wall-clock delay caused by orchestration/observer defects separately from compute/API spend. Large time loss without business progress is a reliability incident even when monetary spend is zero.
F-345	Test-Suite Run Record	Each full acceptance suite persists start time, end time, commit/SHA, total tests, pass/fail/skip counts, duration and log/result location. The latest validated suite is queryable without inspecting shell processes.
F-346	Suite Concurrency Guard	Only one full suite for the same commit/environment may run unless deliberate parallel execution is explicitly requested. Additional requests attach to/read the existing run.
F-347	Targeted Validation Before Full Rerun	After diagnosing a hang/observer defect, validate the smallest affected test/module first. Run the full suite only when needed to prove integration, preventing long redundant validation loops.
F-348	Monitor Regression Tests	Automated tests must prove: observers cannot match themselves; duplicate waiters do not multiply; completed jobs are observed as complete; stale observers cannot resurrect running state; and rising observer counts cannot be interpreted as progress.

ID	Requirement	Executable definition
F-349	Business-Progress Preservation	Monitoring, testing and orchestration may not monopolize autonomous work when independent zero-cost launch-critical work can safely proceed. Concurrency is allowed only where shared-state integrity remains protected.
F-350	Reliability Incident Escalation	Any observer defect that creates false progress, hides completion, duplicates work, or causes material launch delay is logged as a reliability/trust incident with root cause, prevention test and durable fix before closure.

39. REQUIRED OBSERVATION / WAITER ACCEPTANCE TESTS

- A completed suite whose observer misses the terminal event **MUST** still be discoverable as completed from durable run state.
- A waiter searching for the workload **MUST NOT** match itself, another waiter, or a monitoring shell as the target workload.
- Starting four observations of one suite **MUST** create one underlying suite and one authoritative state, not four independent waiter chains.
- An increasing number of observers/pollers **MUST NOT** increase reported workload progress.
- If a suite exceeds the configured duration anomaly threshold, the system **MUST** inspect authoritative state before spawning another waiter or rerunning the suite.
- A completed suite **MUST** expose commit/SHA, start/end timestamps, duration and pass/fail totals without parsing a live process list.
- A restart/deploy after durable suite completion **MUST NOT** cause the same suite to rerun solely because an observer lost state.
- A hung workload and a completed-but-unobserved workload **MUST** produce distinguishable diagnoses.
- Terminating redundant observers **MUST NOT** terminate the underlying workload.
- An observer defect that delays launch work materially **MUST** surface as a reliability incident even when API spend is CA\$0.

Observed build lesson incorporated: on 2026-09-23, the suite itself completed successfully with 3,127 passing and 0 failing, but four shell waiters continued for roughly 100 minutes because their pgrep signature matched their own command lines. The apparent rise in process count was observer multiplication, not test progress. This class of defect is now prohibited by F-331 through F-350.

Final-upgrade registry now runs F-001 through F-350.

Version note: v0.10 appends durable job-observation, waiter-safety and progress-truth requirements to v0.9. All earlier requirements remain in force unless explicitly superseded.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.11 Addendum - Baby & Children's Catalogue Priority

Owner direction: baby and children's crochet is a major Brambleloop catalogue opportunity and must receive explicit market-research and product-selection attention. The trigger evidence supplied by the owner is an established Etsy crochet-pattern shop with approximately 18,000 sales over seven years and a catalogue visibly concentrated in baby/children's cardigans, jackets, tops, dresses and related garments. This evidence establishes the category as a priority for investigation; it does not hard-code individual products or quotas.

Governing rule: PRIORITIZE THE CATEGORY, LET EVIDENCE SELECT THE PRODUCTS.

40. BABY & CHILDREN'S CROCHET CATALOGUE

ID	Requirement	Executable definition
F-351	Priority Product Family	Baby and children's crochet patterns are a high-priority Brambleloop product family for market research, launch planning, catalogue expansion and seasonal planning.
F-352	Evidence-Led Selection	Priority status does not mean blind production. Product selection must still use Brambleloop's demand, competition, pricing, differentiation, seasonality and launch-readiness evidence.
F-353	Core Category Coverage	Research newborn/baby cardigans, toddler/children's cardigans, sweaters, jackets, dresses, tops/vests, hooded pieces, matching sets, bonnets/hats, decorative collars, blankets, seasonal/holiday garments, unisex designs and sibling/multi-size products.
F-354	Multi-Size Value	Explicitly evaluate products that include multiple age/size ranges in one purchase. Measure customer value, search demand, grading complexity, support burden and conversion potential rather than automatically splitting every size into separate listings.
F-355	Beginner-Friendly Opportunity	Measure demand for easy/beginner-friendly children's construction and identify designs that can remain visually distinctive while reducing instruction complexity and failure risk.
F-356	Gift Intent	Model baby-shower, newborn-gift, birthday, holiday and handmade-gift purchase intent when selecting and merchandising children's products.
F-357	Seasonality	Track seasonal demand for children's garments and accessories early enough that research, pattern certification, imagery, listing preparation and launch occur before the demand window.
F-358	Gender-Neutral & Style Segments	Research gender-neutral as well as style-specific demand. Do not infer a fixed palette or aesthetic; use marketplace evidence and Brambleloop's creative differentiation system.
F-359	Sibling / Family Variants	Evaluate sibling sets, coordinated products and reusable design families where genuine customer value exists, while preventing low-value catalogue duplication.
F-360	Sizing Is Safety-Critical Product Truth	Children's sizing, gauge, finished measurements, ease and age/size labeling must be validated with the same or stronger Product Truth discipline as adult garments. Smaller garments do not justify weaker validation.
F-361	No Lower Quality Floor	Children's products must clear the same instruction clarity, stitch/count consistency, construction plausibility, image truth, download integrity and customer-support readiness floors as the rest of the catalogue.
F-362	Age-Range Clarity	Listings and PDFs must state supported size/age ranges and finished measurements clearly. Avoid implying that age alone guarantees fit where measurements are the authoritative basis.
F-363	Physical Plausibility	Wearables for babies/children require explicit checks for proportion, openings, closures, seams, trim and construction details that could make the finished object impractical or inconsistent with the certified pattern.
F-364	Visual Merchandising	Children's listing imagery and storefront presentation should optimize emotional/gifting appeal while remaining truthful to the certified product. Product Truth cannot be traded for cuteness or conversion.
F-365	Store Architecture	Brambleloop's storefront taxonomy must support a coherent Baby & Kids department/collection when catalogue depth justifies it, including useful navigation by garment/product family and size/age range where supported.
F-366	Launch Mix Evaluation	Once the Product Truth/image architecture is resolved, the launch-catalogue optimizer must explicitly consider baby/children's products alongside adult garments, home/decor, seasonal and accessories rather than treating kids as a later afterthought.
F-367	Competitive Learning	Successful children's crochet shops and listings may be studied for demand signals, product families, merchandising, pricing, bundling, sizing strategies and customer expectations. Brambleloop must use that evidence to make its own product decisions.
F-368	Differentiation Requirement	Do not fill the catalogue with undifferentiated near-duplicates merely because the category is popular. Each selected product must justify its place through demand, differentiation, bundle value, seasonality or strategic catalogue coverage.
F-369	Portfolio Measurement	Track children's-category impressions, clicks, favourites, conversion, revenue, refunds/support burden, seasonality and repeat-customer behaviour separately enough to learn whether priority should increase or decrease after launch.

ID	Requirement	Executable definition
F-370	Autonomous Rebalancing	After sufficient live evidence exists, Brambleloop may increase or reduce the share of baby/children's products based on measured performance, while preserving owner-set brand and quality constraints.

41. IMPLEMENTATION / ACCEPTANCE

- Market research explicitly covers the children's subcategories above and records demand/competition evidence before product selection.
- Launch-catalogue scoring includes baby/children's candidates once the current Product Truth visual blocker is resolved.
- Multi-size candidates are evaluated as bundled customer value rather than automatically duplicated into one listing per size.
- Children's wearables cannot be certified with unresolved sizing, gauge, finished-measurement or construction contradictions.
- Storefront information architecture can expose a Baby & Kids collection when enough certified products exist to make it useful.
- Post-launch learning can reweight the category from measured sales/conversion/support evidence rather than permanently freezing today's priority.

Final-upgrade registry now runs F-001 through F-370.

Version note: v0.11 appends baby/children's catalogue-priority requirements to v0.10. All earlier requirements remain in force unless explicitly superseded.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.12 Addendum - Continuous Autonomous Improvement & 24/7 Learning

Owner direction: Brambleloop is not a static Etsy automation. It must remain cloud-operational with the HQ PC powered off and continuously improve the business from durable evidence, while preserving hard product-truth, quality, safety, spend and owner-authority boundaries. Continuous improvement means disciplined observe-test-prove-deploy-monitor learning, not uncontrolled self-modification.

Governing loop: OBSERVE → DIAGNOSE → PROPOSE → TEST → COMPARE → PROVE → DEPLOY → MONITOR → LEARN → REPEAT.

42. CONTINUOUS AUTONOMOUS IMPROVEMENT

ID	Requirement	Executable definition
F-371	24/7 Cloud Operation	Production workers, scheduler/cadences, durable queues, observation, incident detection, spend governance and authorized business operations continue in cloud infrastructure while the owner PC is completely off.
F-372	Continuous Improvement Loop	Brambleloop continuously searches for evidence-backed improvements across products, patterns, CIR/digital twins, PDFs, imagery, storefront, Etsy search, conversion, pricing, ads, support, costs, seasonality and internal reliability.
F-373	Evidence Before Change	No autonomous improvement is deployed merely because an agent proposes it. Every material change requires evidence of the problem/opportunity, a defined hypothesis and measurable acceptance criteria.
F-374	Isolated Experimentation	Potential improvements are tested in shadow/staging/isolated work before production adoption. Experiments cannot silently mutate the production truth they are intended to evaluate.
F-375	Champion-Challenger	Where measurable, compare the proposed version against the current production champion. The challenger replaces the champion only after satisfying all applicable hard floors and demonstrating the intended improvement.
F-376	Hard Floors Cannot Be Optimized Away	Pattern Truth, Product Truth, identity/morphology, photographic realism, IP/policy, customer experience, financial controls and other hard gates cannot be averaged away by higher conversion, lower cost or another favorable metric.
F-377	Durable Learning	Validated findings become durable structured knowledge, provenance, regression fixtures, tests or capability changes so Brambleloop does not repeatedly pay to rediscover the same lesson.
F-378	Failure Memory	Known failed methods/providers/prompts/architectures are fingerprinted with the evidence that rejected them. They are not retried without a materially changed premise or new evidence.
F-379	Self-Improvement Is Bounded	Brambleloop may autonomously improve within owner-authorized boundaries but may not grant itself new authority, raise spending ceilings, bypass owner gates, publish outside the active phase, or weaken controls to make progress.
F-380	Code Change Governance	Autonomous code changes require tests, version control, deploy evidence, rollback capability and post-deploy observation appropriate to their blast radius. Production self-modification is never an unversioned live edit.
F-381	Rollback First-Class	Every material autonomous deployment preserves a known-good predecessor and an executable rollback/recovery path. Detected regression triggers containment and rollback according to severity.
F-382	Maturity Ladder	Track implemented → tested → deployed → exercised → evidenced → production-observed as distinct states. An improvement cannot be described as production-proven at an earlier maturity level.
F-383	Business Outcome Learning	The system learns from impressions, clicks, favourites, carts, sales, refunds, support, reviews, contribution margin and repeat behavior where available, while distinguishing correlation from demonstrated causal experiment evidence.
F-384	Product Portfolio Learning	Continuously evaluate which product families, skill levels, sizes, seasons, price points and bundles deserve more or less catalogue investment using measured market and customer evidence.
F-385	Pattern-System Learning	Defects, tester evidence, support issues, benchmark teardowns and real-product reconstruction findings feed improvements to pattern generation, CIR semantics, grading, charts, instructions and certification.
F-386	Visual-System Learning	Listing imagery learns from Product Truth failures, realism failures, click-through/conversion evidence and approved benchmark photography without allowing conversion optimization to alter the certified product.
F-387	Etsy Search Learning	Search terms, Search Visibility guidance, impressions, clicks and conversions continuously inform titles, tags, attributes, sections and merchandising through controlled/versioned changes.
F-388	Ads Learning	Paid-media allocation learns from attributable performance and contribution economics inside owner-approved ceilings; it cannot raise its own ad budget or use paid traffic to conceal weak organic/listing quality.
F-389	Cost Learning	Provider/model routing, context size, caching, deterministic substitutions and cadence frequency continuously improve cost efficiency, but cheaper execution must prove equivalent required quality before adoption.
F-390	Seasonal Learning	Seasonal performance and lead-time evidence update future research, design, certification, merchandising and advertising schedules so the system improves timing each cycle.

ID	Requirement	Executable definition
F-391	Customer Feedback Learning	Support questions, complaints, refunds, reviews and recurring confusion are converted into structured product/process evidence. Customer feedback may trigger fixes but does not automatically override validated product truth.
F-392	Incident Learning	Every meaningful reliability, spend, evidence or customer-impact incident produces root cause, durable prevention, regression coverage and a check that the same failure family cannot silently recur.
F-393	No Busywork Autonomy	Continuous operation does not mean constant paid inference or constant code churn. When no evidence-backed improvement is warranted, Brambleloop may remain stable rather than manufacturing work.
F-394	Improvement Value Ranking	Prioritize autonomous improvement by expected customer/business impact, risk reduction, confidence, cost and reversibility. Launch/revenue/customer blockers outrank elegance and theoretical optimization.
F-395	Owner Decision Compression	Brambleloop should resolve routine engineering/operational choices itself within policy and surface only genuinely consequential owner decisions with concise evidence, alternatives, cost and consequence of waiting.
F-396	PC-Off Improvement Continuity	Cloud-resident learning/observation continues with HQ offline. HQ-only heavy development or archives may pause, but production-required observation, evidence capture and bounded improvement orchestration cannot require the owner PC.
F-397	HQ Acceleration Without Dependency	When HQ is online, Brambleloop may exploit its storage/compute/development environment to accelerate analysis and builds; any artifact required by production must be promoted to durable cloud infrastructure before production depends on it.
F-398	Improvement Audit Trail	Every adopted autonomous improvement records trigger/evidence, old version, new version, tests, quality-floor verdicts, spend, deployment, observed outcome and rollback reference.
F-399	Learning Health Metrics	Expose useful improvement health: opportunities detected, experiments run, adoption/rejection rate, regressions caught, repeated-failure avoidance, cost per adopted improvement, and business impact where measurable.
F-400	Never-Finished Company, Finite Launch	Brambleloop is designed to keep improving indefinitely after launch, but continuous improvement must never make launch completion impossible. Separate launch blockers from post-launch required work and optimization; the existence of another possible improvement is not itself a launch blocker.

43. CONTINUOUS-IMPROVEMENT ACCEPTANCE TESTS

- With HQ PC powered off, cloud production continues applicable observation, scheduling, incident detection, spend governance and evidence capture.
- A proposed autonomous change that improves conversion but fails Product Truth is rejected.
- A cheaper model/provider cannot replace a quality-proven route without equivalent-floor evidence.
- A known failed method is not retried without a changed premise/new evidence.
- A material code change cannot reach production without versioning, validation and rollback evidence appropriate to blast radius.
- A completed experiment's durable lesson survives deploy/restart and prevents repeated paid rediscovery.
- When no justified improvement exists, the system can remain stable without creating artificial work or paid inference.
- Post-launch optimization opportunities do not reopen launch readiness unless they reveal a genuine launch-critical defect.

Final-upgrade registry now runs F-001 through F-400.

Version note: v0.12 appends continuous autonomous improvement, bounded self-upgrade and 24/7 learning requirements to v0.11. All earlier requirements remain in force unless explicitly superseded.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.13 Addendum - Autonomous Research Agent & Research-Before-Invention

Owner direction: Brambleloop must have a first-class Research Agent. When the system encounters a material unknown, stale assumption, capability gap, technical blocker, market question, integration uncertainty, policy ambiguity, or evidence conflict, it should not default to guessing or unnecessarily inventing from first principles. It should recognize the uncertainty, research it, test applicability, preserve provenance, and convert useful findings into durable system knowledge.

Governing loop: UNKNOWN -> RESEARCH -> UNDERSTAND -> COMPARE -> TEST -> PROVE -> ADOPT -> RECORD -> REUSE.

Never: UNKNOWN -> GUESS -> BUILD -> SPEND -> DISCOVER THE ESTABLISHED ANSWER LATER.

44. AUTONOMOUS RESEARCH AGENT

ID	Requirement	Executable definition
F-401	First-Class Research Agent	Operate a dedicated research capability that can be invoked by other agents and can autonomously recognize research-worthy uncertainty within its authority.
F-402	Research-Before-Invention	For material unknowns, search internal evidence first and then appropriate current external sources before designing novel machinery from first principles.
F-403	Unknown Detection	Agents must explicitly identify uncertainty, missing evidence, stale knowledge, conflicting evidence, unfamiliar failure modes and capability gaps rather than silently filling them with plausible assumptions.
F-404	Confidence Trigger	Material decisions made from weak, stale, indirect or conflicting evidence trigger research before implementation when research can materially reduce uncertainty.
F-405	Internal Evidence First	Search BUILD_STATE, DECISION_LOG, experiments, incidents, durable lessons, benchmarks, source data, prior failures, tests and current production evidence before paying to rediscover an answer externally.
F-406	External Research Escalation	When internal evidence is insufficient, use current external research appropriate to the question rather than treating the model's pretrained knowledge as current authority.
F-407	Source Hierarchy	Prefer primary and authoritative sources: official documentation/APIs/policies, original academic papers, standards, first-party marketplace material and original repositories. Use secondary/community sources for practical evidence and discovery, clearly distinguished from primary authority.
F-408	Multi-Source Verification	Consequential findings should be cross-checked against multiple independent sources where practical. A single search result is not automatically established truth.
F-409	Fact / Claim / Inference Separation	Research artifacts distinguish directly observed facts, source claims, Brambleloop inference, hypothesis and unresolved uncertainty. Do not flatten these into one confidence class.
F-410	Recency Awareness	Time-sensitive technical, marketplace, pricing, policy, platform and API claims must be checked for freshness. Record retrieval date/version and avoid treating historical behavior as current without evidence.
F-411	Applicability Test	Finding an established method does not authorize adoption. Evaluate whether its assumptions, topology, scale, data, runtime and failure modes transfer to Brambleloop's actual problem.
F-412	Research-to-Experiment Bridge	Translate promising research into the smallest bounded experiment that can falsify or support its applicability before a large build or spend.
F-413	Existing Work Before Reinvention	Prefer adapting proven methods, libraries, algorithms, renderers, standards and open implementations when they satisfy requirements, rather than rebuilding mature infrastructure unnecessarily.
F-414	Custom Invention Permitted	Novel Brambleloop-specific engineering remains allowed when research shows existing methods are inadequate, incompatible, unavailable, unsafe, uneconomic or insufficiently truthful.
F-415	Licensing / IP Review	Before adopting external code, models, datasets, assets or methods with implementation implications, record applicable license, attribution, commercial-use constraints and material IP restrictions.
F-416	Security / Supply-Chain Review	External packages, repositories, binaries, services and models require appropriate provenance/security review before entering production or trusted build infrastructure.
F-417	Cost & Infrastructure Review	Research adoption must record material compute, API, storage, GPU/CPU, deployment, maintenance and per-unit operating implications before architecture is committed.
F-418	Research Spend Governance	Research uses existing owner-approved spend ceilings. The Research Agent cannot raise its own budget or turn open-ended curiosity into uncontrolled paid browsing/inference.
F-419	Research Stop Conditions	Each substantial research task defines the question, evidence needed, time/spend bound and stop condition so research cannot become endless exploration.
F-420	No Research Theatre	Do not collect sources merely to appear rigorous. Research must change confidence, narrow alternatives, establish a constraint, produce a testable hypothesis or justify stopping.
F-421	Search Strategy	Use iterative search: broad discovery -> terminology extraction -> authoritative source retrieval -> contradiction/alternative search -> implementation evidence -> synthesis.

ID	Requirement	Executable definition
F-422	Terminology Discovery	When the system does not know the right vocabulary, use exploratory research to discover field-specific terminology before concluding that relevant work does not exist.
F-423	Cross-Domain Transfer	Search adjacent fields when underlying mechanics transfer - e.g., knitting/textile simulation for yarn geometry - while explicitly preserving domain differences such as crochet topology.
F-424	Research Reproducibility	Record queries/terms, source identifiers/URLs where permitted, versions/dates, key evidence, extracted constraints and resulting decisions sufficiently for later audit/reproduction.
F-425	Citation / Provenance	Material research conclusions retain source provenance. Future agents must be able to tell why the system believes a claim and where the supporting evidence came from.
F-426	Contradiction Handling	Conflicting credible sources trigger explicit reconciliation or an unresolved state. Do not choose the convenient source silently.
F-427	Negative Evidence	Record well-designed failed experiments and authoritative negative findings as durable knowledge; absence of success is not proof of impossibility unless the evidence supports that scope.
F-428	Scope Discipline	A failed implementation rejects only the tested method/class at the supported scope. Do not generalize a glyph-render failure into 'deterministic rendering is impossible' without evidence.
F-429	Research Agent Can Challenge Architecture	When credible evidence contradicts a current assumption, the Research Agent may flag architecture for review rather than forcing evidence to fit the existing design.
F-430	Research Agent Cannot Override Hard Floors	External research cannot be used to rationalize weakening Product Truth, quality, identity, spend, policy, security or owner-authority gates.
F-431	Durable Knowledge Promotion	Validated research findings become structured durable knowledge, regression tests, capability metadata, architecture notes or decision records so later agents reuse them.
F-432	Deduplication / Research Memory	Before starting research, check whether the same question or materially equivalent premise has already been answered. Reopen only with new evidence, changed assumptions or expired/stale findings.
F-433	Research Freshness	Durable research carries freshness/expiry semantics appropriate to the domain; evergreen mathematics differs from Etsy policy, API behavior, pricing or marketplace demand.
F-434	Automatic Re-Research	Where a decision depends on volatile external facts, schedule or trigger re-research when the evidence becomes stale, a dependency changes, an incident occurs or contradictory live evidence appears.
F-435	Research Outputs Are Machine-Usable	Produce structured conclusions including question, answer/status, confidence, evidence, alternatives, constraints, applicability, unresolved issues and recommended bounded next test.
F-436	Research Agent Tool Choice	Use the lowest-cost adequate research method: internal retrieval, official docs, web search, academic literature, repository inspection, marketplace observation, controlled browser work or bounded experiment.
F-437	Primary Documentation for Integrations	For APIs/platform behavior, consult current official documentation and live capability evidence before coding around remembered behavior.
F-438	Academic / Technical Research	For hard engineering/scientific gaps, actively search original papers, technical reports, reference implementations and established tools before inventing bespoke algorithms.
F-439	Marketplace Research	For Etsy/customer/product questions, combine marketplace evidence with first-party platform guidance and Brambleloop's own observed performance; do not equate search-result popularity with causal demand.
F-440	Competitive Learning Without Blind Copying	Research competitor catalogues, merchandising, pricing, product families and customer expectations as evidence, while requiring Brambleloop-specific differentiation and independent Product Truth.
F-441	Research-Driven Dependency Selection	New dependencies/providers are chosen from measured fit, license, reliability, maintenance, cost and reversibility - not because they are fashionable or first in search results.
F-442	Benchmark Acquisition Logic	When a real external benchmark can answer a material unknown more cheaply/reliably than continued synthetic testing, surface the acquisition to the owner with expected information value before purchase.
F-443	Real-Object Benchmarking	Use real products/patterns/objects where appropriate to validate architecture against external reality; preserve what the source can and cannot establish.
F-444	Research Failure Escalation	If research cannot resolve a material unknown, report what was searched, what remains unknown, why it matters and the smallest owner/external action capable of resolving it.
F-445	No Fabricated Citations or Capabilities	Never invent a source, paper, repository, API feature, benchmark result or external capability. Unknown remains unknown until evidenced.
F-446	Research Parallelism	Research may run alongside unrelated executable work when safe, so a blocked research question does not freeze the whole business.
F-447	Research Agent Authority Boundary	The Research Agent gathers, evaluates and recommends. Adoption follows the authority model of the affected subsystem; research itself does not grant deployment/publishing/spend authority.
F-448	Learning From Owner Discovery	When the owner supplies useful external evidence or terminology, ingest it as research evidence, verify where appropriate, and update durable knowledge rather than treating it as a one-off chat hint.
F-449	Research Quality Metrics	Track research questions resolved, reuse rate, stale findings refreshed, duplicate research avoided, experiments prevented, spend avoided, source quality and downstream decisions materially improved.

ID	Requirement	Executable definition
F-450	Research as Core Autonomy	A mature Brambleloop should not require the owner to Google routine unknowns. Detecting 'I do not know enough yet' and autonomously researching the answer is a core 24/7 self-improvement capability.

45. RESEARCH AGENT OPERATING CONTRACT

- Trigger research automatically when a material unknown would otherwise cause guessing, unnecessary invention, ambiguous testing, wasted spend or a potentially stale decision.
- Begin by asking: What exactly is unknown? What decision depends on it? What evidence would change the decision?
- Search Brambleloop's durable internal evidence before external research.
- When external research is needed, discover the correct terminology first, then prioritize primary/current sources and actively look for credible alternatives or contradictions.
- Return a structured research packet: question; status (answered/partial/unresolved); evidence; source class; confidence; applicability; constraints; alternatives; license/cost/security implications; recommended smallest test; stop condition.
- A research result does not become production truth until its applicability is demonstrated at the maturity level required by the affected subsystem.
- Promote validated findings into durable knowledge/tests so another agent does not repeat the same research without changed premises.

46. RESEARCH AGENT TRIGGERS

Trigger	Research response
Technical blocker	Unknown algorithm, renderer, data structure, numerical method, infrastructure or failure mode.
External integration	API/platform behavior uncertain, changed, undocumented or contradicted by live evidence.
Marketplace/business	Demand, pricing, competition, SEO, seasonality or customer behavior requires current evidence.
Policy/compliance	Marketplace/platform/legal/policy requirement is material and potentially current or jurisdiction-dependent.
Capability gap	The system is about to build substantial custom machinery because it does not know whether an established solution exists.
Evidence conflict	Two credible observations/sources disagree or production contradicts stored assumptions.
Stale knowledge	A decision depends on evidence beyond its freshness window.
Repeated failure	A method has failed enough times that continuing without external research would be re-asking a settled question.
Owner-supplied clue	The owner supplies a paper, screenshot, listing, example, product or terminology that may materially change the problem representation.

47. ACCEPTANCE TESTS

- Given an unfamiliar rendering failure, the system researches established textile/yarn methods before authorizing a large custom renderer build.
- Given a current API error that contradicts stored knowledge, the system checks current official documentation/live capability evidence before coding a workaround.
- Given a prior answered research question with unchanged premises, the system reuses durable findings rather than paying to research it again.
- Given a failed implementation, the system records the supported scope of failure and does not generalize it into an unsupported impossibility claim.
- Given a promising paper or open-source method, the system checks applicability, licensing, security, infrastructure and cost before adoption.
- Given conflicting credible sources, the system reports unresolved/reconciled status rather than silently selecting one.
- Given owner-provided evidence, the system incorporates and verifies it where appropriate, and preserves the resulting lesson durably.
- The Research Agent cannot publish, raise budgets, bypass quality floors or deploy consequential changes merely because research recommends them.

Final-upgrade registry now runs F-001 through F-450.

Version note: v0.13 appends the Autonomous Research Agent and Research-Before-Invention architecture to v0.12. It is designed as a permanent cross-cutting capability used by Build 2, post-launch continuous improvement, technical engineering, marketplace intelligence and

future agents. All earlier requirements remain in force unless explicitly superseded.

BRAMBLELOOP FINAL MASTER UPGRADE

v0.14 Addendum - Parallel Autonomous Organization & Dependency-Aware Execution

Owner direction: Brambleloop must operate like a real multi-department company, not one sequential agent changing hats. When one department enters a long-running build, render, research task, external wait, or difficult blocker, every independent READY workstream must continue. Parallelism is dependency-aware and conflict-safe: it accelerates independent work without allowing simultaneous agents to corrupt shared truth, deploy over one another, or multiply spend.

Governing rule: NO INDEPENDENT READY WORK SITS IDLE SOLELY BECAUSE ANOTHER DEPARTMENT IS BUSY.

48. PARALLEL AUTONOMOUS ORGANIZATION

ID	Requirement	Executable definition
F-451	Master Orchestrator	Maintain a master orchestrator with visibility across requirements, departments, dependencies, owner gates, external blockers, spend, deployments and active work.
F-452	Dependency DAG	Represent substantial build/operational work as a dependency graph. A task waits only for dependencies it genuinely requires; unrelated work remains dispatchable.
F-453	Work-Conserving Scheduler	Whenever safe independent READY work exists and capacity is available, dispatch it. A long render, research run, test suite or blocked integration cannot idle unrelated departments.
F-454	Department Agents	Use focused agents/workers for distinct domains such as Research, Pattern/CIR Engineering, Visuals, QA, Etsy/Commerce, SEO/Marketing, Catalogue, Analytics, Customer Experience, Finance/Cost, Reliability and Integration.
F-455	Context Isolation	Department agents receive the context/evidence relevant to their work rather than carrying the entire company's unrelated history, reducing distraction, token cost and cross-domain assumption leakage.
F-456	Shared Truth Contracts	Cross-department contracts - CIR schemas, product IDs, listing schemas, evidence formats, APIs and quality floors - are versioned and authoritative. Agents do not silently redefine shared contracts locally.
F-457	Ownership Boundaries	Assign clear code/data/module ownership for active tasks. Two agents may not concurrently make incompatible edits to the same shared surface without explicit coordination.
F-458	Isolated Workspaces	Concurrent engineering uses isolated branches/worktrees or equivalent safe workspaces so one agent's incomplete changes do not contaminate another agent's validation.
F-459	Merge Queue	Completed department changes enter a controlled integration/merge queue. Shared-branch races and last-writer-wins development are prohibited.
F-460	Integration Agent	A dedicated integration responsibility resolves cross-department conflicts, validates shared contracts and assembles compatible changes before production deployment.
F-461	Deploy Serialization	Production deployments are serialized/controlled even when development is parallel. Agents cannot independently race deployments or overwrite a newer verified production state.
F-462	Lease Scope	Use the narrowest safe lock/lease scope. A deployment or shared-schema lock must not unnecessarily freeze unrelated research, drafting, analysis or isolated development.
F-463	Long-Running Task Handoff	Renders, suites, research jobs and external waits persist durable job state so the initiating agent can yield capacity while another observer/job system tracks completion.
F-464	No Waiter Occupancy	An agent does not consume a scarce execution slot merely waiting on a durable long-running operation when it can safely perform or dispatch other work.
F-465	Critical-Path Awareness	Continuously calculate launch/business critical paths and prioritize work that shortens them, while still using spare capacity for independent non-critical but valuable work.
F-466	Blocker Fan-Out	When a blocker is discovered, identify every requirement actually dependent on it and only those requirements. Do not allow a local blocker to become a global stop.
F-467	Unblock Fan-Out	When a gate opens or dependency completes, immediately recompute downstream readiness and dispatch newly READY work without waiting for a manual heartbeat.
F-468	Parallel Build-2 Completion	During pre-launch Build 2, independent remaining requirements should execute concurrently by department where safe, rather than sequentially waiting for the current headline blocker.
F-469	Parallel Post-Launch Operation	After launch, Research, product creation, pattern engineering, visual production, commerce, marketing, analytics, support and reliability continue concurrently within resource/authority limits.
F-470	Resource-Aware Concurrency	Concurrency respects CPU, memory, storage, provider rate limits, database capacity, Railway capacity and other shared resources. Parallelism cannot destabilize production.
F-471	Spend-Aware Concurrency	Maintain aggregate spend reservations across concurrent agents. Five agents cannot each interpret the same remaining budget as independently available.
F-472	Provider Concurrency Governance	Centralize provider/API rate-limit and spend coordination so concurrent departments cannot create retry storms, duplicate experiments or surprise billing.
F-473	Priority Preemption	Higher-value launch/customer/reliability work may preempt lower-value optimization when capacity is constrained, while preserving durable state for later resumption.

ID	Requirement	Executable definition
F-474	Safe Pause/Resume	Department work can pause and resume from durable checkpoints without losing evidence, repeating paid work or requiring the original agent session to remain alive.
F-475	Cross-Agent Evidence Reuse	Evidence produced by one department is discoverable and reusable by others through durable provenance rather than being trapped in one agent's context.
F-476	No Duplicate Department Work	Before dispatch, check active/completed jobs and evidence fingerprints so multiple agents do not unknowingly solve the same question in parallel.
F-477	Independent QA	Where practical, the agent that builds a consequential artifact is not the sole authority that certifies it. QA consumes the artifact/evidence independently.
F-478	Parallel Research	Research Agent may investigate blockers/opportunities while engineering continues on independent work; research does not require the whole company to wait.
F-479	Parallel Etsy Preparation	Storefront, SEO, listing schemas, policies, merchandising, analytics preparation and other non-image-dependent Etsy work may proceed while visual R&D; remains blocked, subject to Shadow Mode.
F-480	Parallel Catalogue Work	Research/design/certification work on products that do not depend on an unresolved downstream presentation step can continue, while publication remains gated by complete launch floors.
F-481	Parallel Reliability Work	Reliability, observability, cost governance and regression work may proceed independently when it does not alter shared contracts under active modification.
F-482	Contract Change Barrier	A department changing a shared schema/API/CIR contract declares the change before merge; dependent agents either pin the old version or explicitly migrate. Silent contract drift is prohibited.
F-483	Integration Regression Gate	Parallel speed never bypasses integrated validation. Merged changes run the appropriate targeted and full regression suites before production maturity is claimed.
F-484	Conflict Detection	Detect overlapping file/module/schema ownership before work begins where possible, not only at merge time.
F-485	Deadlock Detection	The orchestrator detects circular dependencies, stale locks and mutually waiting departments; it surfaces/fixes orchestration deadlocks instead of reporting the company as merely busy.
F-486	Starvation Prevention	Long-running high-profile work cannot permanently starve smaller READY tasks. Scheduler fairness is balanced against critical-path priority.
F-487	Department Health	Expose each department's active task, blocker, READY backlog, last durable progress, spend, output/evidence and whether capacity is idle.
F-488	Company-Wide Progress	Expose business progress across departments: launch blockers closed, listable products, storefront readiness, integration readiness and critical-path movement - not only commits/tests from the busiest department.
F-489	Parallelism Efficiency	Measure wall-clock compression, idle-capacity time, duplicate-work avoidance, conflict rate, merge delay and useful throughput attributable to parallel execution.
F-490	Concurrency Backpressure	If conflicts, regressions, resource saturation or spend anomalies rise, reduce concurrency automatically rather than preserving agent count for appearance.
F-491	Failure Isolation	A failed department task does not roll back unrelated validated work. Contain failures to their dependency branch and preserve successful independent outputs.
F-492	Department-Specific Stop Conditions	Each workstream has its own success/stop/escalation criteria; one department reaching an owner gate does not create an owner gate for every department.
F-493	Owner Decision Batching	Where possible, consolidate genuine owner decisions from multiple departments into concise evidence-backed batches rather than interrupting the owner repeatedly.
F-494	Autonomous Dispatch	Within existing authority, the orchestrator may dispatch newly READY department work without waiting for the owner to say 'continue' after every completed task.
F-495	24/7 Parallel Continuity	Cloud departments continue independent authorized work while HQ is off and while other departments are blocked, sleeping, rendering or waiting on external systems.
F-496	HQ Acceleration Pool	When HQ compute is available, assign suitable heavy isolated workloads to it without making cloud production or scheduling dependent on HQ remaining online.
F-497	Build vs Operate Separation	Development agents and production-operation agents have distinct authority and deployment paths. Building in parallel cannot destabilize the live business.
F-498	Finite Parallel Launch	Parallelism is used to shorten the finite launch path, not to create infinite simultaneous pre-launch optimization. Launch blockers remain explicitly bounded.
F-499	Post-Launch Department Autonomy	Once live, each department may continuously improve within its approved authority while the orchestrator coordinates shared constraints and integration.
F-500	Parallel Company Standard	Brambleloop's mature operating model is a dependency-aware parallel autonomous organization: many departments advancing simultaneously, one coherent source of truth, controlled integration, and no unnecessary global waiting.

49. PARALLEL EXECUTION OPERATING CONTRACT

- Recompute the dependency DAG whenever a requirement closes, a gate opens, a blocker appears, or a shared contract changes.

- Dispatch all safe independent READY work up to resource/spend/concurrency limits; never serialize merely because one agent is the conversational foreground.
- Use isolated workspaces and ownership declarations for concurrent code changes; serialize integration/deployment, not all development.
- A long-running render/test/research operation persists durable state and releases scarce agent capacity rather than occupying the company with waiting.
- Before merge, verify shared-contract compatibility and run targeted regression; before production maturity, run required integrated/full validation.
- Central spend/resource coordination applies across every concurrent department.
- When concurrency causes more conflicts/regressions than useful throughput, automatically reduce it and record why.

50. IMMEDIATE BUILD-2 APPLICATION

- Map the remaining Build-2 requirements to departments and explicit dependencies rather than treating the current visual blocker as the global build.
- Keep the Visual/Product-Truth workstream independent while Etsy/storefront, SEO/research, catalogue, PDF/pattern presentation, analytics preparation, reliability, cost governance and other genuinely independent work advance concurrently.
- Do not publish or bypass Shadow Mode merely because upstream departments finish early; publication still waits on all launch-critical dependencies.
- Use the parallel system to compress calendar time to launch, not to inflate activity metrics.

51. ACCEPTANCE TESTS

- Given a two-hour visual render and five independent READY tasks, at least one independent department advances rather than all execution slots waiting on the render.
- Two agents attempting incompatible edits to the same shared module are prevented/isolated before production merge.
- Concurrent agents cannot each spend the full remaining global API budget.
- A completed long-running job wakes/recomputes dependent readiness without requiring a manual owner heartbeat.
- A failed Visual task does not discard validated Etsy/SEO/reliability work completed concurrently.
- A shared CIR schema change forces explicit version/migration handling for dependent workstreams.
- Parallel development may produce many commits, but production deploys remain ordered and production-observed.
- The operator dashboard can show which departments are active, blocked, idle, READY and on the launch critical path.

Final-upgrade registry now runs F-001 through F-500.

Version note: v0.14 appends dependency-aware parallel autonomous organization requirements to v0.13. This architecture is intended for immediate Build-2 calendar compression and permanent post-launch operation. All earlier requirements remain in force unless explicitly superseded.

v0.15 Addendum - Continuous Autonomous Business & Etsy Improvement

52. CONTINUOUS AUTONOMOUS BUSINESS IMPROVEMENT

Brambleloop is not complete merely because it can create and publish products. After launch, the system must continuously collect real operating evidence and use it to improve the business. Launch establishes the production feedback loop; real marketplace operation becomes part of the evidence system used to improve Brambleloop over time.

Non-delay rule. Continuous improvement must never become continuous launch delay. Once the defined launch-quality floor and all launch-critical gates are genuinely satisfied, non-blocking improvements move to post-launch operation. The company must sell, measure, learn and improve while operating rather than indefinitely expanding the pre-launch build.

53. ETSY STOREFRONT AS A CONTINUOUSLY OPTIMIZED SYSTEM

The Etsy storefront is an owned operating surface, not a static export target. Brambleloop must continuously evaluate listing titles, tags, descriptions, attributes, SEO, thumbnails, image order, presentation, pricing, promotions, product/category selection, shop organization, branding, merchandising, customer questions, reviews, refunds/cancellations, and listing state using real evidence.

Where Etsy supports event delivery, webhook/event signals such as order paid, cancelled, shipped, delivered and inventory/out-of-stock events should feed Brambleloop's event layer. These signals complement - not replace - listing/search/conversion analytics and marketplace research.

54. CLOSED-LOOP LEARNING ARCHITECTURE

The required operating loop is: Etsy and other approved evidence sources -> durable event/data layer -> specialized departments -> research/diagnosis -> bounded experiment -> validation/gates -> controlled deployment -> measured outcome -> durable learning. The system must not equate 'self-improving' with uncontrolled self-modification.

Every autonomous change must preserve attribution: what changed, why it changed, what evidence justified it, expected effect, measured result, whether it was retained or rolled back, and which requirement/department owned the decision. Failed experiments remain durable knowledge so later agents do not repeatedly rediscover the same failure.

55. POST-LAUNCH DEPARTMENT RESPONSIBILITIES

Department	Continuous post-launch responsibility
Visual	Improve image realism, presentation quality and conversion-relevant image performance without weakening Product Truth.
Marketplace / SEO	Improve discoverability, titles, tags, attributes, search positioning and listing presentation from marketplace evidence.
Catalogue / Product	Learn which products, categories, sizes, styles and seasonal opportunities deserve expansion, retirement or testing.
Commerce	Improve pricing, promotion, listing state, conversion path and Etsy integration while preserving financial and policy gates.
Customer Experience	Learn from questions, reviews, refunds, cancellations and support friction; feed recurring issues back to product and deliverable teams.
Deliverable QA	Use real customer outcomes to improve pattern clarity, terminology, charts, instructions, packaging and regression coverage.
Reliability / Cost	Continuously improve uptime, spend efficiency, observability, reservations, retries, failure handling and resource governance.

Department	Continuous post-launch responsibility
Coordinator	Keep independent departments productively filled in parallel, recompute dependencies, integrate safely and prevent one blocked lane from idling the company.

56. AUTONOMY, SAFETY AND OWNER-RESERVED BOUNDARIES

Agents may research, diagnose, propose, test and run bounded experiments automatically within their granted authority. Product Truth, customer safety, legal/policy compliance, material financial exposure, destructive actions, publication/activation and other owner-reserved boundaries remain gated according to the Master. Evidence-driven improvement may never weaken an invariant merely to improve a metric.

57. ETSY EVENT / WEBHOOK OPERATING REQUIREMENTS

Etsy webhook integration is a post-authentication Commerce capability. It must be implemented with endpoint verification, authenticated/validated event handling where supported, idempotency, duplicate-event protection, durable event storage, retry handling, observability, replay/reconciliation strategy, and tests before production subscription. OAuth/shop authorization and the controlled draft-image-update-readback-delete shadow round trip remain prerequisites; webhook work must not distract from or bypass those gates.

58. LEARNING QUALITY STANDARD

Brambleloop should become materially better with accumulated operating history. At 3, 6 and 12 months, the system should have stronger evidence for what to make, how to present it, what to charge, what customers value, what causes support friction, and which experiments actually improve outcomes. Improvement is measured by validated business/customer outcomes and preserved quality floors - not by commit count, agent activity or code volume.

59. ACCEPTANCE TESTS

- A paid-order event can enter the durable event layer once, be processed idempotently, update the appropriate business evidence, and not create duplicate effects when replayed.
- An underperforming listing can be detected from evidence, assigned to the correct department, changed through a bounded experiment, measured, retained or rolled back, and recorded with full attribution.
- A customer-support pattern or review issue can create a durable improvement candidate without silently rewriting Product Truth.
- Independent post-launch departments continue useful work in parallel when another lane is blocked, rendering, waiting on an external system or awaiting an owner decision.
- A non-launch-critical improvement cannot keep launch in Shadow once every defined launch-quality floor and launch-critical dependency is satisfied.
- A proposed optimization that violates a safety, policy, Product Truth, spend or owner-reserved gate is refused even if its conversion metric is better.
- Failed experiments remain queryable and prevent automatic repetition unless new evidence explicitly justifies reopening them.

Registry additions

F-501	Etsy Event Feedback	Supported Etsy events feed a durable, idempotent business-event layer after authenticated integration is proven.
F-502	Storefront Continuous Optimization	Continuously evaluate and improve Etsy listing/storefront presentation from real evidence.
F-503	Evidence-Driven Experiments	Storefront/business changes use bounded experiments with explicit hypotheses and outcome measurement.
F-504	Experiment Attribution	Record what changed, why, evidence, expected effect, measured effect, owner/department and retain/rollback decision.

F-505	Failure Memory	Persist failed experiments and causal findings so agents do not repeatedly rediscover known failures.
F-506	Customer Feedback Loop	Questions, reviews, refunds, cancellations and support friction become structured improvement evidence.
F-507	Sales Feedback Loop	Sales and conversion evidence inform catalogue, pricing, merchandising and product-family decisions.
F-508	SEO Learning Loop	Search/listing evidence informs titles, tags, attributes and discoverability experiments.
F-509	Image Performance Loop	Listing-image ordering/presentation may improve from measured performance without weakening Product Truth.
F-510	Pricing Learning Loop	Pricing/promotion experiments remain bounded by margin, fee, spend and owner-policy floors.
F-511	Post-Launch Department Continuity	Every department has an ongoing evidence-backed improvement role after launch.
F-512	Webhook Reliability	Webhook/event handling includes validation, idempotency, duplicate protection, retries, storage and observability.
F-513	Controlled Self-Improvement	Autonomous improvement operates through research, experiments, gates and controlled deployment - not unrestricted self-modification.
F-514	Outcome-Based Improvement	Measure improvement by customer/business outcomes and preserved quality floors, not activity volume.
F-515	Launch Non-Delay Rule	Once launch-critical quality floors pass, non-blocking improvements cannot indefinitely delay launch.
F-516	3/6/12-Month Learning	Accumulated operating evidence must materially strengthen decisions over 3, 6 and 12 month horizons.
F-517	Cross-Department Evidence Routing	Route each signal to the department capable of acting on it while preserving a common source of truth.
F-518	Safe Rollback	Every autonomous production experiment has a defined rollback/stop condition where technically applicable.
F-519	Owner-Reserved Boundaries	Continuous improvement cannot bypass publication, safety, legal/policy, Product Truth, destructive-action or material-spend gates.
F-520	Sell-Learn-Improve Standard	Brambleloop's mature mode is to sell, measure, learn and improve continuously while operating.

Final-upgrade registry now runs F-001 through F-520.

Version note: v0.15 appends continuous autonomous business and Etsy improvement requirements to v0.14. All earlier requirements remain in force unless explicitly superseded.

BRAMBLELOOP FINAL MASTER UPGRADE v0.16

53. ETSY SHOP OPERATING SYSTEM - DEEP AUDIT & CONTINUOUS STORE IMPROVEMENT

This section strengthens rather than replaces Build 2 and the existing F-001+ / F-501-F-513 Etsy requirements. The master already covers query matching vs ranking, listing conversion, Search Visibility, policy freshness, event ingestion, marketplace performance, storefront improvement, pricing learning, customer evidence, bounded experiments and long-horizon learning. This increment closes the broader Shop Manager operating loop: shop completeness, branding, Marketplace Insights, Stats, ads/promotions, service standards, policy/finance health, OAuth/API health, remote verification and permanent self-improvement.

Deep-audit findings

- Etsy is an operating surface, not merely a publishing endpoint. Shop Manager contains first-party evidence that internal guesses cannot replace.
- Search Visibility is a first-party defect queue. Brambleloop should ingest warnings, fix them, and confirm Etsy clears them.
- Marketplace Insights is first-party demand intelligence. Etsy Plus currently provides unlimited searches, so BrambleloopStudio should use it deliberately for product and keyword research.
- Etsy Plus is paid infrastructure: listing credits, Ads credits and customization should be measured and used only when they create value.
- Shop trust and customer-service quality affect conversion and search readiness; logo/icon, About, policies, reviews, response and case health need explicit gates.
- HTTP success is not business success. Important writes require remote readback and field-by-field verification.
- Self-improvement includes the Etsy page itself: merchandising, branding, SEO, economics, service and policy health should continuously improve from measured evidence.

New executable requirements - additive to all earlier requirements

ID	Requirement	Executable definition
F-514	Etsy Surface Inventory	Maintain a machine-readable inventory of every Shop Manager surface Brambleloop can observe or act on: Listings, Search Visibility, Stats, Marketplace Insights, Marketing/Ads, Sales & Discounts, Messages, Orders, Customer Service Stats, Policy Violations, Finances, Apps, shop customization, policies, subscription and API/OAuth. Mark each API, browser-only, owner-only, unsupported or unknown.
F-515	Shop Completeness Gate	Before first publication and continuously thereafter, verify shop icon/logo, banner or chosen intentional absence, About/story, policies, branding consistency, support expectations and required seller information. Missing trust surfaces cannot silently pass launch readiness.
F-516	Search Visibility Ingestion	Capture Etsy Search Visibility findings as first-party evidence with timestamp, affected listing/shop area, recommended action, severity, owner and status. Do not substitute an internal SEO score for Etsy's own warning.
F-517	Search Visibility Closure Proof	After a corrective change, re-check Etsy Search Visibility and require the finding to clear or materially change before marking it resolved. Record time-to-clear and confirmation evidence.
F-518	Marketplace Insights Harvester	Use Marketplace Insights to collect first-party demand, related-query and competition evidence for approved product/category research. Store query, timestamp, market window, searches, listing competition, related terms and provenance.
F-519	Marketplace Insights Research Budget	Exploit Etsy Plus unlimited Marketplace Insights deliberately but deduplicate research. Prioritize launch categories, rising terms, seasonal demand, weak-competition opportunities and questions generated by Catalogue/SEO.
F-520	Demand-to-Catalogue Loop	Route first-party demand evidence into Catalogue. Candidate products require a traceable demand hypothesis, competition context, Product Truth feasibility and margin estimate before promotion to build.
F-521	Stats Truth Layer	Ingest Shop Stats and listing-level traffic evidence where obtainable: visits/views, search terms, traffic sources, favorites, orders, revenue and listing performance. Preserve zero-data as zero-data; never infer performance from absence.
F-522	Organic-vs-Paid Separation	Keep Etsy Ads traffic and organic Etsy search performance analytically separate. Ads must never be credited as organic ranking improvement.
F-523	Listing Funnel Model	Track visibility -> visits/clicks -> favorites -> orders -> revenue/refunds at listing and product-family level. Diagnose whether the constraint is matching, click-through, conversion, price, trust, imagery or product-market fit.
F-524	First-Image Program	Treat the first image as a conversion/search asset. Maintain thumbnail readability, crop safety, mobile legibility, truthful representation and controlled hero-image experiments with rollback.
F-525	Gallery Sequence Optimization	Optimize image order by buyer question sequence: hero, finished-result proof, scale/context, detail/texture, options, what's included, difficulty/requirements and trust/support as applicable. Measure changes rather than rotating by taste.
F-526	Shop Home Merchandising	Continuously optimize shop sections, featured listings, shop-home layout and seasonal merchandising. Use Etsy Plus customization only when it improves navigation, trust or conversion.
F-527	Brand Asset Health	Maintain logo/icon, banner, color, typography and listing-image consistency across Etsy. Detect stale, missing, low-resolution or off-brand assets and route fixes to Brand/Visual.

ID	Requirement	Executable definition
F-528	Etsy Plus Value Ledger	Track value captured from Etsy Plus: listing credits, Ads credits, Marketplace Insights, customization and other included benefits. Compare realized value against subscription cost.
F-529	Ads Credit Experimentation	Use included Etsy Ads credit for bounded experiments only after organic listing readiness passes. Define hypothesis, listings, spend cap, attribution window and stop condition.
F-530	Paid Acquisition Guard	Spend beyond included credits requires budget authority, expected unit economics and reconciliation. Track CAC/ROAS/contribution margin where measurable and stop experiments that breach approved economics.
F-531	Promotion Governance	Sales, coupons and discounts require a reason, margin floor, target/event, start/end, attribution and postmortem. Prevent permanent discounting from becoming the unmeasured default price.
F-532	Customer-Service Standards Watch	Continuously monitor message response, review quality, case rate and other Etsy service signals available to the shop. Escalate approaching-risk conditions before they can reduce trust/search visibility.
F-533	Message Intelligence	Structure recurring pre-sale and post-sale questions into evidence. Feed ambiguity back to listing copy, images, FAQs, PDFs, product design and support macros. Never relabel a real defect as customer confusion.
F-534	Review Intelligence	Classify reviews by product, expectation, difficulty, accuracy, imagery, instructions, value and support. Themes become measurable product/listing evidence.
F-535	Refund/Cancellation Root Cause	Every refund, cancellation, case or complaint receives a reason code and owner. Repeated causes trigger listing, deliverable, pricing, policy or product investigation.
F-536	Policy-Violation Watch	Monitor Etsy policy-violation surfaces and policy freshness. A new warning or policy change can invalidate publication readiness immediately.
F-537	Shop Policy Consistency	Ensure listing promises, PDF/customer-use terms, shop policies, FAQ/support language and internal commerce rules do not contradict one another. One canonical source owns each policy fact.
F-538	Finance Reconciliation	Reconcile Etsy orders, fees, taxes/remittances, refunds, credits, ads/subscription charges and payouts against Brambleloop's ledger. Unknown fee behavior remains explicit until proven by a real settlement.
F-539	Profit-by-Listing Truth	Compute contribution margin by listing/product family from real Etsy fees, discounts, ad spend, model/API cost and support/refund burden. Revenue alone cannot select winners.
F-540	OAuth Credential Health	Continuously monitor stored/openable Etsy credential status, granted scopes, expiry and refresh health without exposing values. Refresh-token rotation must be proven in production and persisted atomically.
F-541	OAuth Failure Recovery	Handle invalid_grant, revoked access, expired/reused refresh token, changed encryption key and scope drift with fail-closed owner actions.
F-542	Remote Write Readback	Every Etsy mutation that can affect customers must be followed by authoritative remote readback where the API permits. A 2xx response is insufficient proof that the intended business field changed.
F-543	Draft-to-Live Activation Gate	A listing may activate only when remote draft state, image presence, digital file where required, taxonomy/properties, price/inventory, disclosures, Product Truth, safety and owner publication authority all pass.
F-544	Live Listing Drift Detector	Compare intended listing state with Etsy's live remote state. Detect manual edits, missing images/files, changed tags/attributes, price drift, state changes and taxonomy drift; reconcile deliberately rather than overwriting blindly.
F-545	Listing Lifecycle Audit Trail	Persist create/update/image/file/activate/deactivate/delete actions, actor, evidence, remote IDs, before/after state and rollback/cleanup result. Test listings must be unmistakable and independently confirmed deleted.
F-546	Etsy Feature Discovery	Continuously audit new/changed Shop Manager capabilities, API fields, Seller Handbook guidance and Etsy Plus features. Classify each as useful, irrelevant, owner-only, unsupported or research-needed before adoption.
F-547	Browser-Only Work Queue	For valuable Etsy capabilities unavailable through API, maintain an explicit browser/manual queue with exact owner action, estimated minutes, risk and evidence required. Do not pretend unsupported automation exists.
F-548	Storefront Experiment Engine	Run bounded experiments on titles, thumbnails, gallery order, descriptions, tags, pricing and merchandising only when enough traffic exists to learn. Protect against simultaneous changes that destroy attribution.
F-549	Etsy Executive Scorecard	Maintain one evidence-backed shop scorecard: live listings, Search Visibility findings, traffic, conversion, revenue, margin, reviews, response/case health, refunds, ad efficiency, policy status, OAuth/API health and unresolved owner actions.
F-550	Continuous Etsy Improvement Cadence	Operate a permanent Etsy loop: daily health/exceptions, weekly listing/search/market review, monthly economics/storefront review, seasonal trend planning and quarterly deep audit. Re-prioritize from evidence; do not wait for the owner to ask.

54. ETSY ACCEPTANCE TESTS & OPERATING CADENCE

- An empty Search Visibility ingest cannot mean 'no issues' unless Etsy was actually read successfully.
- A Shop Manager warning remains OPEN until Etsy confirms closure or confirmation is explicitly impossible.
- Marketplace Insights evidence records timestamp/window and never turns demand into permission for prohibited content.
- Organic and Etsy Ads traffic cannot be blended into one ranking-success metric.
- A 2xx Etsy response cannot certify a write unless relevant remote state is read back and matches.
- A draft cannot activate without remotely verified image(s) and all required file/taxonomy/property/disclosure gates.
- A test listing is not cleaned up until Etsy independently confirms deletion.
- OAuth refresh rotation must survive process/container replacement and never exist only in memory.
- Manual Etsy edits create drift evidence; automation does not blindly overwrite them.
- Etsy Plus value is measured from actual benefit usage, not assumed.
- Customer questions/reviews/refunds route back to the owning product/listing/deliverable department.
- Policy or customer-service risk can block publication even when SEO and product files are green.
- Quarterly deep audit compares current Etsy guidance/features/API/shop surfaces against this master and adds requirements when material gaps appear.

Permanent cadence

Continuous/event-driven	OAuth/API health, supported events, policy/violation alerts, publication/destructive safeguards.
Daily	exceptions, credential health, live-listing drift, urgent service issues, spend anomalies.
Weekly	Search Visibility, Stats/query evidence, Marketplace Insights, funnel review, questions/reviews/refunds, experiment decisions.
Monthly	profit-by-listing, Etsy Plus value, ads/promotions economics, shop-home merchandising, brand assets, catalogue allocation.
Seasonal	trend/occasion planning from Etsy first-party demand plus external research, with Product Truth and policy gates.
Quarterly	full Etsy capability/policy/API/storefront deep audit and master-file delta review.

Current primary-source audit basis (September 2026)

- Etsy Seller Handbook: How Etsy Search Works (updated guidance: query matching + ranking; shop/listing/customer-service quality).
- Etsy Seller Handbook: Checklist - Optimize Your Shop for Etsy Search; Search Visibility guidance.
- Etsy Help: Marketplace Insights (real Etsy search data; recent searches/listing competition; Etsy Plus unlimited searches).
- Etsy Help: Etsy Plus and Advanced Shop Customization (listing credits, Ads credit, banners/featured listings/color customization).
- Etsy Seller Handbook: Shop Stats and current title/search guidance.

Version note: v0.16 appends F-514 through F-550. All earlier requirements remain in force unless explicitly superseded. The purpose is not to create endless pre-launch scope; launch-critical gates remain separate from post-launch optimization. The Etsy operating system must continue improving after launch without silently expanding publication authority.

BRAMBLELOOP FINAL MASTER UPGRADE v0.17

55. COMPLETE ETSY SHOP MANAGER SURFACE AUDIT

This increment was triggered by direct inspection of the current BrambleloopStudio Shop Manager navigation. It extends v0.16 from a broad Etsy operating system to an explicit surface-by-surface contract. The left navigation currently exposes Dashboard, Listings, Messages, Orders, Etsy Search Visibility, Stats (including Shop Traffic and Marketplace Insights), Customer Service Stats, Policy Violations, Marketing (Etsy Ads, Offsite Ads, Sales and Discounts, Social Media, Share & Save), Finances (Payment Account, Monthly Statements, Payment Settings, accounting/tax surfaces), Apps, Help, Settings (Your Shop, Info & Appearance, Options, Shared Access, Delivery Settings, Policy Settings, Partners You Work With) and sales-channel surfaces. Nothing visible is assumed irrelevant merely because Brambleloop sells digital products.

Operating rule

Every Etsy surface must be classified as **OBSERVE, ANALYZE, ACT, OWNER-ONLY, NOT-APPLICABLE** or **UNSUPPORTED**. 'Not built' and 'not applicable' are different states. A surface may be useful even when it cannot be automated: Brambleloop must still know it exists, know what evidence it contains, and maintain the exact manual/owner action required.

New executable requirements - additive to F-001 through F-550

ID	Requirement	Executable definition
F-551	Dashboard Task Harvester	Ingest Shop Manager dashboard top tasks, Shop Advisor warnings, recent activity, open orders, messages needing response, renewals/expiry and shop-specific reminders. Every actionable item becomes a durable task with owner, deadline, evidence and closure proof.
F-552	Dashboard KPI Snapshot	Persist dashboard views/visits/orders/revenue and relevant date range/channel context. Dashboard numbers are quick signals, not substitutes for detailed Stats.
F-553	Listing Estate Manager	Inventory every draft, active, inactive, sold-out and expired listing; track renewal state, fees, remote IDs, product family, assets, digital files, taxonomy, properties and intended state.
F-554	Bulk Listing Operations Guard	Support safe bulk edits/renewals only with preview, affected-count proof, invariant checks and rollback plan. Never bulk-edit Product Truth, safety or legal fields from a heuristic.
F-555	Renewal & Expiry Strategy	Measure manual/automatic renewal economics and performance. Renew because evidence supports the listing, not merely because expiry is approaching.
F-556	Message SLA Engine	Track first-message response clocks and unresolved buyer threads. Etsy's customer-service standards and stronger internal targets must be visible before breach.
F-557	Message Triage & Escalation	Classify pre-sale, order, download, pattern-help, refund, safety, policy and abuse/spam messages; route to the correct department and escalate risk without auto-sending unsupported claims.
F-558	Order Operations Ledger	For every order, reconcile payment, item/version, digital delivery state, buyer issue, refund/cancellation/case state and downstream accounting evidence.
F-559	Digital Delivery Verification	For digital products, verify the exact customer files attached to each live listing, version/hash, language variants and post-purchase availability. Prevent stale or mismatched PDFs.
F-560	Search Visibility Three-Area Model	Model Etsy Search Visibility exactly across Listings, Shop and Customer Service. Keep each area independently green/amber/red with first-party findings and timestamps.
F-561	Search Finding Aging	Track how long each Etsy Search Visibility issue remains open; re-check after fixes and alert when Etsy has not cleared an expected issue within its normal refresh window.
F-562	Stats Traffic-Source Model	Track Etsy Search, Etsy app/other Etsy pages, direct traffic, social, Etsy Ads, Offsite Ads and other available sources separately; preserve Etsy's attribution definitions.
F-563	Search-Term Performance	Capture Etsy search terms that actually delivered visits/orders and map them to intended query families, listing fields and Marketplace Insights demand evidence.
F-564	Listing-Level Stats	Maintain listing views, favorites, orders, revenue and conversion evidence by timeframe/status. Detect high-view/low-order and low-view/high-conversion listings as different problems.
F-565	Marketplace Insights Opportunity Score	For researched terms, combine buyer searches, competing listings, trend direction, Product Truth feasibility, policy eligibility, expected margin and catalogue fit. Keep raw Etsy evidence separate from the derived score.
F-566	Customer Service Daily Watch	Once Etsy exposes service stats, ingest daily progress and monthly overview for response rate, on-time shipping/tracking where applicable, review rating and case rate.
F-567	Star Seller Separation	Track Etsy minimum customer-service standards separately from Star Seller targets. Never report meeting the minimum as earning or approaching Star Seller without the required evidence.
F-568	Policy Violation Incident System	Every Policy Violations item becomes a severity-ranked incident with source text, affected listing/action, deadline, remediation owner, owner/legal escalation and closure proof.
F-569	Etsy Ads Campaign Manager	Track advertised listings, budget, clicks, visits, orders, revenue and spend. Require listing readiness before paid traffic and measure incremental economics rather than vanity clicks.
F-570	Offsite Ads Economics	Track Offsite Ads attribution, fee rate, 30-day attribution behavior and the shop's 365-day revenue threshold/participation status. Recompute contribution margin on attributed orders.
F-571	Sales & Discounts Engine	Manage coupon/sale experiments with margin floor, audience, timing, inventory/listing eligibility, attribution and postmortem. Expired promotions must not remain assumed active.

ID	Requirement	Executable definition
F-572	Social Media Distribution	Create an evidence-backed distribution queue for eligible Etsy social updates, listings, milestones, promotions and reviews. Track destination, asset, link, date and resulting traffic.
F-573	Share & Save Optimization	Where enrolled/eligible, generate and use Etsy Share & Save trackable links for external traffic, track eligible orders/refunds and compare fee savings against ordinary external links.
F-574	Marketing Channel Attribution	Compare organic Etsy search, Etsy Ads, Offsite Ads, social and Share & Save at listing/product-family level. One order cannot be double-credited to multiple channels internally.
F-575	Payment Account Reconciliation	Reconcile Etsy Payment Account entries to orders, fees, refunds, credits, ads, subscription charges and payouts. Unexplained balance movements create finance incidents.
F-576	Monthly Statement Archive	Archive monthly Etsy statements and extract revenue/fee/tax/refund/payout summaries into durable finance evidence. Preserve originals for audit.
F-577	Payment Settings Guard	Treat bank/payment-setting changes as owner-reserved high-risk actions. Monitor drift but never autonomously change payout/bank details.
F-578	Tax Information Registry	Track required Etsy legal/tax information, filing/export availability and jurisdictional thresholds as facts with source dates. Software monitors and reports; it does not invent a tax position.
F-579	Accounting Integration Decision	Evaluate QuickBooks Online, Xero and other finance integrations against Brambleloop's ledger needs, duplication risk, cost and reconciliation quality before connecting one.
F-580	Apps Inventory & Permissions	Inventory every connected Etsy app, granted permissions/scopes, purpose, owner, last use, data access, cost and revocation path. Unknown or unused integrations are security debt.
F-581	Third-Party App Evaluation	Evaluate Etsy apps such as SEO/research, fulfillment and finance tools by capability overlap, data quality, cost, permissions, vendor risk and whether Brambleloop can build/measure the function better.
F-582	No Redundant Tool Spend	Do not subscribe to eRank/EverBee/other tools merely because they are popular. Require a capability gap, evidence advantage and expected ROI beyond Etsy Marketplace Insights + Brambleloop's own research.
F-583	Shared Access Governance	If Shared Access is used, inventory team members, access dates and activity; enforce least privilege within Etsy's available model. Owner remains accountable for team actions.
F-584	Shop Appearance Optimizer	Continuously evaluate logo, banner mode, featured layout, color theme, shop title, announcement, About/story and receipt branding for trust, readability, mobile presentation and conversion.
F-585	Shop Options Registry	Audit every Settings > Options control, its current value, business consequence, owner authority and desired state. Detect drift after Etsy UI or policy changes.
F-586	Delivery Settings Applicability	Audit delivery/shipping settings even for a digital-first shop; mark physical-only controls explicitly not applicable rather than silently ignoring them.
F-587	Policy Settings Canonicalization	Map Etsy return/exchange, cancellation, privacy and fixed policy settings to Brambleloop's canonical policy facts; block contradictions across Etsy, listing copy, PDFs and support.
F-588	Partners You Work With	Maintain accurate production-partner disclosures when applicable. Refuse fabricated or stale partner data and re-audit when fulfillment/manufacturing changes.
F-589	Help & Education Change Feed	Treat relevant Etsy Help/Seller Handbook updates as dated external evidence. Material changes open review tasks; educational content does not silently overwrite Product Truth or owner policy.
F-590	Sales Channel Inventory	Inventory every enabled Etsy sales channel/integration and its effect on listings, inventory, payments, customer data and attribution. New channels require explicit readiness.
F-591	Pattern / External Channel Boundary	If Etsy surfaces Pattern or other owned-site channels, evaluate them separately from Etsy marketplace sales. Do not assume Etsy marketplace SEO, fees, policies or analytics transfer unchanged.
F-592	Shop Security Posture	Continuously verify OAuth credential health, connected apps, shared access, secret rotation history, callback configuration and suspicious authorization changes without exposing credentials.
F-593	Owner-Only Action Queue	Maintain exact, minimal instructions for Etsy actions automation cannot safely perform: bank/tax/legal settings, browser-only disclosures, access invitations and other owner-reserved actions.
F-594	First-Sale Readiness Gate	Before first publication, require remote Etsy write/read/delete proof, exact customer file verification, shop completeness, policies, disclosures, search readiness, support readiness and rollback/incident plan.
F-595	First-10-Sales Learning Protocol	Treat the first ten sales as a protected learning cohort: inspect search source, conversion path, questions, download/support friction, reviews/refunds and economics after each sale without destabilizing listings with constant edits.
F-596	First-100-Sales Quality Program	Aggregate early evidence before scaling catalogue breadth. Repeated confusion or defect signals outrank growth experiments; protect review quality and customer trust.
F-597	Storefront Change Budget	Limit simultaneous storefront/listing experiments so causal attribution remains possible. Emergency policy/safety fixes bypass the experiment budget but are separately tagged.
F-598	Cross-Surface Consistency Audit	Weekly compare Dashboard, Listings, Search Visibility, Stats, Marketing, Finances, Apps and Settings for contradictions: active counts, fees, traffic, state, policies, credentials and warnings.
F-599	Etsy UI Drift Detector	Quarterly and after material Etsy announcements, compare the observed Shop Manager navigation/surfaces with the maintained surface inventory. New, renamed or removed tabs create review tasks.
F-600	Complete Etsy Autopilot Contract	Brambleloop's Etsy department is never 'done'. It must observe, diagnose, prioritize, research, test, prove, deploy, verify remotely, monitor and learn across every supported shop surface while preserving owner authority, Product Truth, safety, policy and spend gates.

56. CURRENT ETSY FACTS THAT MUST BE ENCODED

- Etsy's Dashboard surfaces top tasks such as orders needing shipping, messages needing response and listings needing renewal, plus Stats, Shop Advisor and recent activity.
- Search Visibility explicitly evaluates three areas: listings, shop and customer service; Etsy says improvements typically disappear from that page within about 24 hours after correction.
- Marketplace Insights uses real Etsy search data, including searches and competing listings over the last 30 days. Etsy Plus provides unlimited keyword searches.
- Etsy Stats exposes visits, orders, conversion rate, revenue, traffic sources, Etsy-search terms and listing-level performance. Etsy says Stats is the accurate place to assess search performance rather than manually searching for your own listings.
- Customer service standards include message response, on-time shipping/tracking where applicable, review rating and case rate; Etsy says these can affect search visibility and selling privileges.
- Etsy Plus includes 15 listing/renewal credits per cycle, USD \$5 Etsy Ads credit, expanded shop customization and unlimited Marketplace Insights; unused monthly credits do not roll over.
- Share & Save uses trackable external links and currently provides fee refunds on eligible attributed orders; attribution and promotion terms must be read from Etsy rather than hard-coded forever.
- Offsite Ads have a 30-day attribution window after a click; participation/fee treatment depends on the shop's 365-day Etsy revenue threshold.
- Shared Access permits team members to manage orders, listings and messages but restricts finance, marketing, stats, settings and destructive/refund actions; the shop owner remains responsible.
- Etsy shop policies include returns/exchanges, cancellations, privacy and fixed policy surfaces; some policy editing is web-only.
- Shop appearance includes logo, banners, featured layout and Etsy Plus customization. A deliberate no-banner configuration is valid and should not be treated as missing by default.

57. POST-AUTHENTICATION EXECUTION ORDER

- **1. Finish the controlled real-Etsy shadow-write.** Draft only: create -> image -> digital file if prepared -> update -> remote readback -> taxonomy/property proof -> delete -> independent deletion proof. No activation.
- **2. Prove OAuth refresh rotation in production.** The current credential is authenticated; durable rotation across refresh/process replacement remains a separate proof.
- **3. Snapshot every Shop Manager surface.** Record current state, warnings, values, owner-only controls and automation support. This becomes the baseline for drift detection.
- **4. Close first-sale blockers only.** Shop identity/policies/customer files/search readiness/support/remote transport. Do not hold launch for post-launch optimization that can safely learn from real traffic.
- **5. Start the permanent cadence after launch.** Daily exceptions, weekly Etsy optimization, monthly economics, seasonal planning, quarterly full surface audit.
- **6. Resume Visual in parallel.** Etsy work does not supersede the crochet/image realism programme; Milestone D remains independent and must continue without serializing the rest of the company.

Version note

v0.17 appends F-551 through F-600. Together with v0.16, the Final Master now explicitly treats the entire Etsy shop as a continuously operated and continuously improving business system, not merely an API destination. These requirements are additive; no earlier Product Truth, safety, policy, spend, publication or owner-authority gate is weakened.

BRAMBLELOOP FINAL MASTER UPGRADE v0.18

58. AI COMPANY OPERATING SYSTEM - PERMANENT TEAMS, ACCOUNTANT & COMMAND CENTER

Brambleloop is specified as an AI-operated company with permanent specialist departments working in parallel. It must keep running, learning, adapting and improving after launch. Finance/Accounting is now a first-class department with an Accountant/Controller agent, and the owner receives a clean Command Center rather than having to interpret developer logs.

New executable requirements - additive to F-001 through F-600

ID	Requirement	Executable definition
F-601	Permanent AI Company	Brambleloop is a 24/7 multi-department AI company. Executive, Product/Catalogue, Visual/Creative, Etsy/Marketplace, Marketing/Growth, Customer Experience, Finance/Accounting, Analytics/Intelligence, Compliance/Policy, Reliability/Engineering, QA/Product Truth and Market Radar/Research each own a mandate, backlog, KPIs, evidence and improvement loop.
F-602	Executive COO	Coordinate company goals, critical path, department priorities, file/authority ownership, budgets, merge/deploy queue and owner decisions. Refill independent lanes immediately instead of waiting behind long jobs.
F-603	Parallel Work Contract	Independent departments work concurrently. A long render, suite or Etsy task cannot stall unrelated Finance, Marketing, Catalogue, CX or QA work. Serialize only genuine shared-state dependencies.
F-604	Controlled Self-Improvement	Every team follows Observe -> Diagnose -> Research -> Hypothesis -> Test -> Independent Verify -> Adopt -> Monitor -> Learn. Constant improvement never means uncontrolled rewriting or weakening gates.
F-605	Company Learning Memory	Persist experiments, failures, reversals, evidence, confidence and withdrawn claims so agents compound knowledge and do not repeatedly rediscover the same lesson.
F-606	Constraint Finder	Continuously identify the dominant growth/quality constraint - impressions, CTR, conversion, price, imagery, product-market fit, trust, support, refunds, margin, policy, quality, uptime or capacity - and route effort there.
F-607	Finance & Accounting Department	Create a permanent Finance/Accounting team led by an Accountant/Controller agent. It owns books, reconciliation, profitability, cash flow, expenses, tax/reserve monitoring, financial controls and owner reporting.
F-608	Independent Accountant	The Accountant independently reconciles Etsy, internal orders, Payment Account/monthly statements and future accounting integrations. Disagreement becomes a reconciliation incident; missing data is UNKNOWN, never silently zero.
F-609	Auditable Ledger	Maintain an auditable ledger for sales, discounts, refunds, Etsy fees, ads, subscriptions, AI/model spend, infrastructure, software, professional costs, taxes/reserves, payouts, owner funding and adjustments. Every row has source, external ID, time, currency, classification and reconciliation state.
F-610	Financial Truth & Close	Track gross/net sales, refunds, fees, expenses, cash and liabilities. Monthly close imports statements, reconciles balances, investigates variances, locks evidence and issues monthly/YTD P&L, cash-flow and reconciliation reports.
F-611	Profit Intelligence	Compute contribution profit/margin by company, channel, product family, listing and campaign using real revenue, refunds, Etsy fees, ads, variable AI/API cost and attributable support burden where measurable.
F-612	Tax & GST/HST Monitoring	Track tax evidence, marketplace-remitted taxes, configurable tax reserves and Canadian GST/HST threshold progress. Agents alert and prepare evidence but never autonomously choose registration, filing or legal positions.
F-613	Marketing & Growth Department	Permanent team for Etsy organic growth, Etsy Ads, Offsite Ads, Share & Save, social/external traffic, seasonal campaigns and controlled acquisition experiments.
F-614	Etsy Plus Credit Wallet	Track Etsy Plus benefits separately from cash. Official current entitlement is USD \$5 Etsy Ads credit per monthly subscription cycle plus 15 listing/renewal credits; the Canadian shop currently displays about CA\$7.10. Credit expires and does not roll over.
F-615	Free-Credit-First Rule	When an approved Etsy Ads experiment exists, prefer expiring included credit before incremental owner-funded spend where Etsy billing permits. Never advertise merely to avoid wasting credit.
F-616	Paid Ads Budget & Economics	Owner-funded ads are a separate budget. Every campaign records free credit, cash spend, clicks, CPC, visits, orders, revenue, contribution margin, ROAS, hypothesis, attribution window and stop condition.
F-617	Customer Experience Department	Own messages, support, reviews, refunds/cancellations and buyer friction. Repeated questions and negative signals route upstream to Listing, Product, Visual or QA. Early customer trust outranks growth acceleration.
F-618	Analytics & Intelligence	Own metric definitions, attribution, experiment registry, anomaly detection and forecasting. Revenue, profit, conversion, CAC, ROAS, refund rate and other KPIs have one canonical definition across departments.
F-619	Brambleloop Command Center	Build a clean owner-facing web app with Today, Month, Finance, Products, Listings, Marketing, Customers, Experiments, Departments, Agent Activity, Owner Actions and System Health. It is the executive interface; the ledger/evidence stores remain sources of truth.
F-620	Command Center Finance	Show MTD/YTD sales, refunds, Etsy fees, free ad credit, cash ad spend, AI/infrastructure/software expenses, contribution profit, margin, tax reserve, payout/cash position, unreconciled items and monthly-close status.
F-621	Command Center Departments	Show each department's active agents/jobs, objective, blockers, completed work, queue, spend and KPI trend. Green requires measured evidence, not merely absence of an error.
F-622	Command Center Marketing & Customers	Show organic vs paid traffic, ad credit/cash spend, CPC/CTR/conversion/ROAS, campaigns, Share & Save/Offsite Ads, orders, support queue, review themes, refunds and unresolved customer/product feedback.
F-623	Owner Actions & Evidence Drilldown	One prioritized owner-action queue with exact steps, minutes, urgency, cost and consequence. Every KPI/status can drill to source, timestamp, external ID where safe, transformation and reconciliation state.
F-624	Executive Cadence	Daily executive brief; weekly company review; monthly business review after financial close; seasonal product/marketing planning; quarterly full company/Etsy capability audit. Each review produces decisions, not ceremonial reporting.

ID	Requirement	Executable definition
F-625	90-Day Growth & Forecasting	Maintain base/upside/downside 90-day revenue/profit scenarios using traffic, conversion, AOV, catalogue, repeat behavior, ads and margin. Compare actuals weekly; forecasts are hypotheses, not promises.
F-626	Scale Gate	Do not scale paid acquisition or catalogue volume faster than quality, support, review/refund health, margin and reliability can sustain.
F-627	Agent Performance & Upgrades	Measure agents by verified business/quality outcomes, not commits/tool calls. Periodically improve prompts, models, tools and workflows from measured failures; changes are versioned, tested and reversible.
F-628	24/7 Runtime & Continuity	Monitoring, research, accounting imports, event handling and safe recurring operations continue without the owner's PC on. Jobs, ledger, experiments, evidence, credentials and critical artifacts survive restarts.
F-629	Department Spend Governance	Each department has ceilings and attribution; the company-wide ceiling dominates. Concurrent reservations prevent overshoot. Promotional/cloud credits are tracked separately from real cash exposure.
F-630	Greatness Without Finish-Line Drift	Permanent teams continuously improve products, marketing, finances, customer value and themselves after launch. Continuous improvement is mandatory; endless pre-launch expansion is prohibited.

59. ACCOUNTANT / CFO OPERATING PACK

- Owner view: gross sales -> discounts -> refunds -> net sales -> Etsy fees -> advertising -> AI/model -> infrastructure -> software/professional costs -> contribution profit -> tax reserve -> operating cash.
- Required drilldowns: company -> channel -> product family -> listing -> campaign -> order -> source transaction.
- Daily accountant loop: import evidence, reconcile orders/fees/ads/payouts, classify exceptions, update cash/profit and alert anomalies.
- Monthly accountant loop: close the month, archive Etsy statements, reconcile material balances, issue P&L;cash/reconciliation pack and unresolved-item schedule.
- Accounting principle: the Command Center is the clean owner interface; the auditable ledger underneath is the financial source of truth.
- Tax principle: monitor thresholds, reserves and evidence; never silently choose a registration, filing or legal position.

60. PERMANENT DEPARTMENT CADENCE

Executive / COO	continuous	critical path, lane refill, owner actions, budgets
Finance / Accountant	daily + monthly close	ledger, reconciliation, cash, profit, tax reserve
Etsy / Marketplace	daily + weekly	shop/API, listings, search, Stats, Marketplace Insights
Marketing / Growth	daily + weekly	organic/paid channels, ad credit, campaigns, experiments
Product / Catalogue	weekly + seasonal	demand, portfolio, build/expand/retire decisions
Visual / Creative	continuous	image realism, galleries, brand, creative variants
Customer Experience	continuous	messages, reviews, refunds, support friction
Analytics / Intelligence	daily + weekly	metric truth, attribution, forecasts, anomalies
Compliance / Policy	event + quarterly	policy/safety/privacy/licence/tax escalation
Reliability / Engineering	continuous	uptime, jobs, secrets, deploys, spend, recovery
QA / Product Truth	per release	independent adversarial verification
Market Radar / Research	daily/weekly/seasonal	demand, market changes, trends, opportunities

61. ETSY PLUS AD CREDIT RULE

The owner's Etsy Plus page currently displays about CA\$7.10 of Etsy Ads credit per monthly cycle. Etsy's current official documentation defines the entitlement as **USD \$5 per monthly subscription cycle**, plus 15 listing/renewal credits. The local Canadian display can vary with currency conversion. Credits expire and do not roll over. Brambleloop must record official entitlement, local displayed value, cycle dates, credit consumed, credit expired and owner-funded spend as separate fields.

Free credit is an input to a good experiment, not permission to advertise a weak listing. Marketing still requires an approved listing, hypothesis, measurement plan and stop condition. Cash spend remains separately governed.

62. VERSION / LAUNCH BOUNDARY

v0.18 appends F-601 through F-630. These are permanent company-operating requirements, not a command to delay the first sale until every department and dashboard is mature. After launch, the teams are expected to keep growing Brambleloop, improving themselves, learning from customers/Etsy and compounding capability without weakening Product Truth, safety, policy, finance or owner authority.

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63. BRAMBLELOOP STUDIO — UNIFIED AUTONOMOUS COMPANY OPERATING SYSTEM

Brambleloop Studio is the single premium operating environment for the company. Desktop/web is headquarters; Galaxy tablet is portable headquarters; Galaxy S25 Ultra is owner command. The cloud-hosted company continues operating when every owner device is off. Studio is not merely a dashboard: it is the visible operating environment for permanent departments, products, evidence, finance, experiments, queues and owner authority.

F-651–F-660 · PLATFORM & 24/7 EXECUTION

- F-651 Unified Studio — one Brambleloop Studio app contains Command Center, Products, Visual, Etsy, Marketing, Finance, Customers, QA, Research, Reliability, Agents and Owner Inbox.
- F-652 Desktop Headquarters — desktop/web is the primary deep-work experience; premium, spacious, image-rich and executive-readable rather than a developer admin panel.
- F-653 Galaxy Access — Galaxy tablet provides near-full portable-HQ functionality; Galaxy S25 Ultra provides owner-command views optimized for touch and small screens.
- F-654 Cloud-Always-On Core — owner PC/browser/phone/tablet may all be off without pausing the company. Production queues, events, schedules, accounting, monitoring and safe autonomous work remain live.
- F-655 24/7 Valuable Work — no valuable READY work sits untouched because another lane is busy. Idle compute is acceptable; idle opportunity is not.
- F-656 RUN/GROW/IMPROVE — every department maintains operations, growth and self-improvement queues. Empty RUN pulls GROW, then IMPROVE; otherwise the department enters WATCHING.
- F-657 Event Wakeups — orders, messages, reviews, refunds, Etsy events, policy changes, incidents, spend anomalies, research signals, schedules and owner decisions wake the appropriate team.
- F-658 Durable Coordinator — company DAG, priorities, ownership, authority and completion evidence persist across deploys/restarts and automatically recompute READY work.
- F-659 Durable Workers — persistent queue, leases/locks, heartbeats, bounded retries, dead-letter queue, idempotency, checkpoints and crash recovery are required.
- F-660 Elastic Capacity — cloud sessions are optional burst capacity, never a structural dependency; normal multi-agent operation continues when promotional/paid cloud credit is zero.

F-661–F-670 · COMMAND CENTER & OWNER EXPERIENCE

- F-661 Executive Command Center — Today/Month/90-day objective, revenue/orders/profit/spend, conversion, live listings, customer risk, incidents, active departments, completed work and owner decisions.
- F-662 Live Company Timeline — human-readable causal activity stream across departments instead of raw tool-call noise.
- F-663 Owner Inbox — one prioritized queue containing only genuine human decisions, with evidence, exact steps, minutes, cost, risk and consequence of deferral.
- F-664 Ask Brambleloop — universal company command/search bar answering what changed, why profit moved, what blocks launch, what agents are doing, opportunities, spend and evidence.
- F-665 Evidence WHY — every important KPI/status/gate/recommendation exposes source, timestamp, transformation, confidence, reconciliation/calibration state and safe external identifiers.
- F-666 Overnight Brief — 'While you were away' shows autonomous hours, tasks/experiments, products advanced/rejected, image candidates, Etsy findings, compute/ad spend, sales/profit, incidents and decisions.
- F-667 Department Tabs — each permanent department has its own workspace with objective, KPIs, agents/jobs, queues, evidence, spend, experiments, blockers, learning and next actions.
- F-668 Agents Workspace — org chart with director/worker status, objective, current work, heartbeat, queue, evidence, cost, completions, failures and learning.
- F-669 Progressive Authority — actions are classified OBSERVE/RESEARCH/BUILD/TEST/DRAFT/DEPLOY/SPEND/PUBLISH/LEGAL-TAX/CUSTOMER-REMEDY/CREDENTIALS; no agent silently expands authority.
- F-670 Premium Responsive Design — desktop first; excellent Galaxy tablet and S25 experiences; coherent design system, dark/light support, strong typography, clean cards/charts and drill-down diagnostics.

F-671–F-680 · PRODUCTS & VISUAL STUDIO

- F-671 Product Digital Dossier — one record spans opportunity, research, Product Truth/CIR, materials/sizing, 3D, imagery, PDFs, QA, safety, Etsy, marketing, customers, finance, experiments and history.
- F-672 Product Lifecycle Board — DISCOVERED → RESEARCHING → SELECTED → PATTERN → QA → VISUAL → LISTING READY → LIVE → GROWING/MAINTAIN/RETIRED, with explicit rejection/rollback states.
- F-673 Opportunity Cards — agents surface demand, competition, seasonality, economics, confidence, policy feasibility and recommended next experiment.
- F-674 Visual Pipeline — PATTERN/CIR → structurally faithful finished-product 3D → conditioning assets → downstream AI photoreal image → Product Truth validation → approved listing/ad asset.
- F-675 3D Structural Truth — deterministic 3D preserves silhouette, dimensions, construction, stitch structure, shaping, colour placement and believable macro drape; it need not itself be the final photograph unless evidence requires it.
- F-676 AI Photography — downstream generation supplies realistic yarn/material appearance, folds/contact, lighting, environment, model/display context and camera realism without redesigning the product.
- F-677 Final Image Gate — beautiful images are rejected if they change certified neckline, sleeves, silhouette, stitch family, colour placement, dimensions, construction or other Product Truth.
- F-678 Benchmark Pattern — the purchased benchmark pattern plus its real listing photographs become the recurring end-to-end test: pattern → CIR → full-product 3D → AI ad image → Product Truth/real-outcome comparison.
- F-679 Conditioning Research — compare RGB, depth, normals, edges, segmentation/material masks, multi-view and hybrids to determine the minimum deterministic fidelity needed for faithful photoreal generation.
- F-680 Visual Evidence Panel — show real benchmark reference, Brambleloop 3D, conditioning passes, generated candidates, rejected candidates/reasons and final approved asset together.

F-681–F-690 · FINANCE, MARKETING & ETSY

- F-681 Finance Workspace — Finance/Accountant is a major Studio tab, not a separate owner app: sales, refunds, Etsy fees, ads/credits, AI/infrastructure/software expense, profit, margin, tax reserve, cash/payouts, forecasts and unreconciled items.
- F-682 Accountant Activity — show last reconciliation, discrepancies, close status, tax/threshold alerts and evidence; never silently force mismatches to zero.
- F-683 Financial Drilldown — company → channel → product family → listing → campaign → order → source transaction with provenance and reconciliation state.
- F-684 Marketing Laboratory — organic/paid growth, Share & Save, Offsite Ads, social, seasonal campaigns and experiments with hypothesis, baseline, budget, attribution and stop/scale rules.
- F-685 Etsy Ads Eligibility — persist `etsy_ads_status`, `eligible_from/next_check`, `evidence_source` and `last_verified_at`. Current 2026-09-25 owner evidence says 9 days remain; do not hard-code a derived date as authoritative.
- F-686 Ads Recheck — automatically re-verify Etsy eligibility at/near the expected unlock; countdown reaching zero alone cannot change state to ELIGIBLE.
- F-687 Pre-Eligibility Preparation — research keywords, choose candidate listings, capture organic baselines, prepare creative and pre-register economics before Ads unlock; do not attempt activation while ineligible.
- F-688 Ads Credit Accounting — Etsy Plus included Ads credit is separate from owner cash; record official entitlement, locally displayed CAD value, cycle, used/expired credit and incremental cash spend.
- F-689 Customer Workspace — messages, orders, reviews, refunds/cancellations, recurring confusion, case/policy risk and upstream fixes in one CX workspace.
- F-690 Cross-Team Work Creation — departments create structured tasks for one another so customer/search/finance/quality evidence automatically becomes corrective or growth work.

F-691–F-700 · COMPANY BRAIN & CONSTITUTION

- F-691 90-Day Goal — make CA\$5,000/month visible and decompose it into traffic, conversion, AOV, catalogue/product quality, launch cadence, organic/paid acquisition, retention and margin assumptions with actual-vs-plan tracking.
- F-692 Constraint Routing — executive coordinator continually identifies the dominant business constraint and directs capacity toward it rather than optimizing every metric equally.
- F-693 Agent Learning Ledger — hypothesis → evidence → experiment → result → proposed workflow/behavior change → validation → promoted/rejected; improvements are measurable, versioned and reversible.

- F-694 Cross-Department Evidence Bus — validated evidence moves between teams with provenance; no department traps knowledge in its own chat or private state.
- F-695 Spend Governance — model/API/infrastructure/cloud/ad spend uses company and department ceilings, reservations, provenance and ROI/learning value; cloud burst capacity cannot become uncontrolled baseline cost.
- F-696 Security & Privacy — secrets, raw tokens, sensitive customer data and destructive controls are never exposed merely because Studio is convenient; privileged actions are separated from read-only views.
- F-697 Business Continuity — database, queue, ledger, experiments, credentials, artifacts and job state survive restarts; recovery and backups are tested.
- F-698 No Launch Paralysis — mature Studio/department capabilities are long-term operating requirements, not automatic first-sale blockers. First-sale gates remain deliberately small, evidence-based and customer-protective.
- F-699 Continuous Greatness — teams continuously grow, adapt, upgrade prompts/tools/models/workflows and business strategy after launch, while preserving Product Truth, safety, policy, finance and owner authority.
- F-700 Brambleloop Constitution — sell truthfully; Product Truth over speed; never manufacture evidence; UNKNOWN is valid; customer harm outranks revenue; no silent authority expansion; every dollar has provenance; consequential decisions are auditable; experiments are falsifiable; failures remain knowledge; never unknowingly repeat known failures; improve continuously without preventing shipping; idle compute is acceptable, idle opportunity is not.

64. 24/7 ACCEPTANCE TESTS

- Turning off every owner device does not stop scheduled/event-driven work or lose queue state.
- A process/container restart recovers coordinator, queues, ledger, experiments and long-job checkpoints without duplicate side effects.
- A long Visual job does not prevent unrelated Finance, Marketing, Etsy, Catalogue, CX or QA work from advancing.
- Departments with no valuable RUN/GROW/IMPROVE work enter WATCHING without token burn and wake on relevant events/schedules.
- Completed work is consumed from authoritative terminal/evidence state and dependent work becomes READY without owner heartbeat.
- Retries cannot duplicate Etsy writes, ledger rows, customer notifications or remedies.
- Cloud promotional credit reaching zero does not disable normal multi-agent operation.
- Phone, tablet and desktop read the same durable company state; no local device is a business dependency.
- Every green owner-facing status is backed by actual observed evidence rather than empty-data success.

v0.19 appends F-651 through F-700. All prior requirements remain in force. These define the mature Studio/company OS while preserving the rule that continuous improvement must not become an endlessly retreating pre-launch finish line.

BRAMBLELOOP FINAL MASTER UPGRADE v0.20

67. AUTONOMY, CONTINUOUS LEARNING & MULTI-MODEL TEAMWORK

This section strengthens the permanent company contract. Autonomy and continuous measurable improvement are non-negotiable. Early owner approval is an authority boundary around consequential actions, not a manual operating dependency. The long-run target is a cloud-resident company that continuously operates, learns and improves while preserving Product Truth, evidence, financial integrity and explicit authority.

F-701 · Autonomy Is Core — Brambleloop autonomy is a launch architecture requirement, not a future convenience. The company must continue initiating safe useful work without owner prompting, including research, build, test, QA, finance, marketing preparation, monitoring and improvement.

F-702 · Authority May Be Gated; Initiative May Not — Human approval may gate a consequential external action; it must never gate initiative. While publication, spend, legal/tax, credential or other owner-gated actions await approval, every department continues all other safe unblocked work.

F-703 · Earned Autonomy Ladder — Begin with conservative authority for irreversible/external actions such as first Etsy publications. Expand authority only through explicit owner policy plus measured safe history. Agents may never self-grant broader authority.

F-704 · Autonomous Approval Packages — When owner approval is required, Studio presents a complete approval packet: proposed action, product/listing, QA/Product Truth evidence, policy state, expected economics, risk, rollback/remedy, and Approve/Modify/Decline. Owner approval authorizes the action; it does not wake the company.

F-705 · Permanent Learning Loop — Every department continuously runs OBSERVE -> detect opportunity/failure -> HYPOTHESIS -> EXPERIMENT -> MEASURE -> ADVERSARIAL VALIDATION -> PROMOTE/REJECT -> MONITOR.

F-706 · Learning Ledger — Persist successful and failed experiments, evidence, conditions, cost, result and decision. Failed approaches remain institutional knowledge so Brambleloop does not unknowingly repeat known failures.

F-707 · Evidence-Backed Self-Improvement — Agents may improve prompts, models, tools, workflows, product selection, patterns, Visual methods, Etsy/SEO, marketing, CX, finance, reliability and strategy only through versioned, testable, reversible changes with post-promotion monitoring.

F-708 · Protected Truth and Authority — Continuous improvement may not silently alter Product Truth, safety/evidence thresholds, publication authority, spend authority, credentials, legal/tax positions or customer-remedy authority. Those remain governed by their existing gates and owner/professional policies.

F-709 · Regression Reversal — Every promoted improvement must define monitoring and a rollback/reversion path. If real outcomes regress materially, Brambleloop reverts or suspends the change and records the failure.

F-710 · Outcome-Based Model Routing — Claude, GPT/ChatGPT Work/Codex, Fable and future specialist models are bounded workers/capabilities under Brambleloop, not competing company coordinators. Route tasks by demonstrated quality, cost, latency, context, tool access and risk.

F-711 · Single Integration Authority — There is one authoritative integration path. External AI workers may propose, implement or review isolated work but cannot independently redefine company state, merge conflicting work or deploy outside granted authority.

F-712 · Model-to-Model Task Contract — Cross-model tasks include exact base SHA/version, objective, owned files/data boundary, authority, budget, dependencies, expected outputs, required tests/evidence and objective completion sentinel.

F-713 · Isolation Before Collaboration — Claude/GPT/Fable specialist work occurs in isolated branches/checkouts/sandboxes or other explicit boundaries. No worker edits another active worker's owned surface without coordinator-approved handoff.

F-714 · Independent Review — High-risk customer-facing, financial, marketplace, security, infrastructure and Product Truth work may receive independent adversarial review by a different model/agent before promotion.

F-715 · Evidence Resolves Disagreement — Model prestige never decides a technical disagreement. Resolve competing proposals using reproducible tests, primary evidence, Product Truth, real marketplace/customer outcomes and measured economics.

F-716 · Collaboration Workspace — Brambleloop Studio exposes proposer, implementer, reviewer, disagreement, evidence, tests, cost, accepted/rejected approach and final integration result for important multi-agent/model work.

F-717 · Collaboration Is Not a Launch Blocker — Multi-model teamwork is a compounding capability, not an excuse to postpone first sale. Claude-only or another safe single-worker path remains valid whenever collaboration infrastructure would delay customer-ready launch work.

F-718 · No Chat as Source of Truth — Institutional knowledge must live in durable project/company objects: repository, database, evidence records, experiment ledger, decisions, Product Truth, tasks and artifacts. No critical state may exist only in a Claude, GPT or Fable conversation.

F-719 · 24/7 Useful-Work SLOs — Measure ready-work age, queue starvation, blocked-on-owner time, autonomous task completion, WATCHING time, restart recovery, failed/retried work, learning experiments completed, promoted/reverted improvements and valuable opportunities left untouched.

F-720 · Ready-Work Age Alarm — If valuable READY work exceeds its allowed age while capacity exists, the coordinator treats that as an operational defect, investigates why it was not claimed, and records corrective evidence.

F-721 · Owner Block Does Not Cascade — A task blocked on owner approval blocks only dependent work. Independent departments and unrelated DAG branches continue automatically.

F-722 · Continuous Department Refill — On terminal completion/failure/refusal, the coordinator immediately recomputes department and company queues and refills capacity with the highest-value safe READY work; no manual 'continue' message is required.

F-723 · Learning Value Accounting — Track cost and learning value of experiments and self-improvement work. Continuous improvement is not unlimited token burn; expensive research must earn evidence, risk reduction, capability or expected business value.

F-724 · Real-Outcome Feedback — After launch, sales, traffic, conversion, favourites, refunds, reviews, support questions, ad results, fees and contribution profit become first-class learning evidence. Models must distinguish pre-launch assumptions from observed marketplace outcomes.

F-725 · Customer Feedback Closes the Loop — Repeated customer friction automatically creates structured upstream investigation work for the responsible Product/Pattern/Visual/Commerce/QA team and later measures whether the correction reduced recurrence.

F-726 · Maturity: Launch — Launch maturity means high internal autonomy with conservative external authority: Brambleloop researches/builds/tests/prepares autonomously while the owner may still approve first publications, material spend and other high-consequence actions.

F-727 · Maturity: Early Revenue — As real evidence accumulates, Brambleloop optimizes from its own marketplace/customer/financial outcomes and proposes specific authority expansions backed by a safe-history record.

F-728 · Maturity: Compounding Company — Long-run maturity means accumulated experiments, failures, customer interactions and business outcomes materially improve product selection, pattern quality, imagery, marketing, accounting, CX, reliability and operating strategy while remaining auditable.

F-729 · Autonomy Health in Command Center — Command Center shows autonomous operating hours, workers/agents active or WATCHING, ready/blocked queues, oldest ready work, owner-blocked work, restarts recovered, experiments, improvements promoted/reverted and whether any department is unintentionally idle.

F-730 · No Downtime Means No Lost Opportunity — The operating definition of 'no downtime' is that useful, safe, unblocked opportunity is continuously discovered, queued and acted on. It does not require wasteful model calls when there is genuinely nothing valuable to do.

68. ACCEPTANCE TESTS FOR AUTONOMY & LEARNING

- Leave a publication approval pending for eight hours: unrelated safe work continues and the owner-blocked task alone remains blocked.
- Complete a worker task: dependent READY work is automatically discovered/claimed without an owner heartbeat or 'continue' prompt.
- Restart the runtime during queued work: durable state recovers without duplicate external effects or lost completed evidence.
- Present the same failed improvement months later: the learning ledger identifies the prior experiment and prevents an unknowing repeat unless new evidence justifies retest.
- Promote an improvement, then feed a measurable regression: monitoring detects it and the configured revert/suspend path executes or escalates.
- Give Claude and GPT conflicting solutions to a bounded test problem: neither wins by identity; the integration decision cites reproducible evidence.
- Exhaust optional cloud-session credit: normal Brambleloop autonomous operation remains functional.
- Keep owner devices powered off: queues, schedules, monitoring, finance, research and safe autonomous work continue.
- Attempt self-expansion of publish/spend/legal/credential authority: the request is refused and surfaced through the appropriate owner/governance path.
- Command Center can distinguish productive ACTIVE work, legitimate WATCHING, owner-blocked work, starved READY work and failed/retrying work.

v0.20 appends F-701 through F-730. All v0.19 and earlier requirements remain in force. This section does not declare multi-model collaboration implemented or production-proven; it defines the required operating architecture and rollout constraints.

BRAMBLELOOP FINAL MASTER UPGRADE v0.21

69. CANONICAL BRAMBLELOOP MODEL - PERMANENT BRAND IDENTITY

This section removes ambiguity from the Visual and listing-image contract. The established Brambleloop human model is a permanent brand asset. When a customer-facing image contains a human model, the system must use the exact canonical Brambleloop model identity. A random, stock, substitute, newly invented, or silently drifted person is not acceptable.

F-731 · Canonical Model Is Mandatory - Every customer-facing on-model image uses the canonical Brambleloop model. This applies across products, listings, hero/model shots, lifestyle imagery, ads, social creative and future channels. Product-only images may omit a person; if a person is present, the canonical model is mandatory.

F-732 · Exact Identity, Not a Similar Person - The requirement is identity continuity, not demographic similarity or aesthetic resemblance. Face, identity-defining features and overall recognizable personhood must remain the same across generations, providers, poses, scenes and time.

F-733 · Canonical Asset Registry - Maintain versioned canonical model reference assets, identity specification, hashes, provenance, approved views and provider-specific conditioning packages. Agents must resolve the current approved canonical version from this registry rather than from chat memory.

F-734 · Dual Truth Contract - Every on-model generation has two independent truth obligations: PRODUCT IDENTITY must remain faithful to Product Truth/CIR, and MODEL IDENTITY must remain faithful to the canonical Brambleloop model. Failure of either rejects the image.

F-735 · Model Identity Gate - On-model imagery cannot become approved customer-facing creative until model-identity validation and Product Truth validation both PASS. UNKNOWN blocks approval. A beautiful image with identity drift is a failed image.

F-736 · No Silent Substitution - If the selected provider cannot preserve the canonical model reliably, Brambleloop changes provider/mode/workflow, rejects and retries, or escalates. It never solves provider limitations by substituting a different person.

F-737 · Canonical Model Improvement Loop - Visual/Brand agents continuously improve the canonical model system: realism, skin/hair detail, lighting response, pose range, expression consistency, anatomy, garment interaction, camera realism, cross-view consistency and provider conditioning. Improvements must make the same model more convincing - not evolve her into a different person.

F-738 · Identity-Preserving Promotion - A proposed model upgrade is promoted only when evidence shows equal-or-better identity retention plus improved realism/utility. Previous approved assets remain recoverable; regressions trigger rollback.

F-739 · Cross-Provider Identity Benchmark - Continuously benchmark available image providers/modes for canonical-model identity retention, Product Truth retention, realism, yield, latency and cost. Route on-model work to the best demonstrated configuration rather than permanently hard-coding one provider.

F-740 · Persistent Character Continuity - The canonical model remains consistent across the catalogue so customers repeatedly encounter the same recognizable Brambleloop model. Wardrobe, pose, hair styling, expression, scene and camera may vary when appropriate; identity may not.

F-741 · Product Before Pose - For apparel, product fidelity has priority over pose novelty. The system chooses poses and framing that preserve and reveal the actual garment rather than allowing a dramatic pose to hide or distort construction, fit, proportions, colour placement or features.

F-742 · Exact Product on Exact Model - The on-model stage must use the certified/accepted product representation from the same listing pipeline. It may not regenerate a loosely similar garment around the canonical model. The required chain is Product Truth -> certified product reference -> photoreal product -> canonical model wearing/using that product -> independent validation.

F-743 · No Model Drift Through Learning - Continuous self-improvement may alter prompts, references, adapters, conditioning, validation, provider routing and rendering techniques, but cannot silently redefine canonical identity. Identity changes require an explicit owner-approved canonical-model decision.

F-744 · Model Evidence Ledger - Persist every material model experiment: canonical version, provider/model/mode, references, prompts/settings, output hashes, identity score/evidence, Product Truth result, realism result, cost, acceptance/rejection and reason. Failed identity approaches remain institutional knowledge.

F-745 · Autonomous Model R&D; - When safe capacity exists, Visual/Brand agents continuously search for measurable ways to make the canonical model more realistic and more consistent. They may run bounded experiments under spend/governance policy without waiting for an owner prompt, while customer-facing promotion remains gated by evidence and authority.

F-746 · Listing Set Consistency - All on-model images within a listing must depict the same canonical person and the same accepted product. Cross-image inconsistencies in face, body identity, garment construction, colour or features reject the affected image/set.

F-747 · Brand Asset Portability - Canonical identity must not depend on one vendor. Preserve provider-neutral references/specification and sufficient evidence so Brambleloop can migrate to a better model/provider without losing the established person.

F-748 · Studio Visibility - Brambleloop Studio exposes canonical model version, current best provider/mode, recent identity/realism yield, rejected drift examples, active improvement experiments and whether each listing image passed both Model Identity and Product Truth.

70. CANONICAL MODEL ACCEPTANCE TESTS

- Generate the same accepted garment in multiple poses/scenes: every on-model image resolves to the same canonical identity and the same product.
- Attempt a provider/model change: the candidate is rejected if realism improves but canonical identity drifts.
- Attempt a visually similar substitute person: identity gate rejects it even if Product Truth passes.
- Improve skin, hair, lighting or anatomy realism: promotion requires evidence that identity retention did not regress.
- Generate a listing set across hero, model and lifestyle contexts: identity and product consistency hold across the set.
- Remove all owner devices: bounded canonical-model improvement experiments may continue under normal autonomy/spend rules without changing approved identity.
- Restart the runtime: canonical references, hashes, current version, experiments and validation state recover from durable storage.
- Provider outage or degradation: routing can move to another demonstrated provider without silently changing the model identity.

Permanent rule: Brambleloop does not have a rotating cast of generated models. It has one canonical Brambleloop model identity for on-model customer imagery unless the owner explicitly authorizes a future canonical-identity change. Agents continuously make that same model more realistic, consistent and capable.

v0.21 appends F-731 through F-748. All v0.20 and earlier requirements remain in force.

BRAMBLELOOP FINAL MASTER UPGRADE v0.22

71. PURCHASED-PATTERN AUDIT: MINI STAR STITCH CARDIGAN

This upgrade records the system-level lessons from a full audit of the purchased Mini Star Stitch Cardigan by MJ's Off The Hook Designs. It does not copy the purchased pattern into Brambleloop's product catalogue. The purchase is treated as private benchmark evidence for architecture, quality, merchandising, validation and workflow design.

Product architecture	10 child sizes (0-3 months through 12), oversized fit, DK #3 yarn, 5.0/5.5 mm hooks, optional hood, 4-6 buttons.
Construction	Worked flat in one piece from the bottom up. Back band/panel first; sleeve chains extend from both sides; neckline is marked; fronts are completed separately; hood is added; garment is folded and seamed; cuffs and ribbed button band finish the product.
Fabric identity	Star stitch is not merely a visual texture. It has explicit loop/eye/leg/base relationships, alternates with HDC return rows, and has a measurable gauge of 18 stitches (9 stars) x 10 rows per 4 inches. Ribbing has its own SC-BLO gauge.
Sizing truth	The source supplies chest, finished measurements, positive ease guidance, yardage, schematic dimensions, sleeve/body/neck/hood measurements and size-specific stitch counts.
Customer support	The product combines a polished PDF, schematic, size chart, yarn substitution guidance, project tips, a step-by-step video for the smallest size, a Learn Hub/community support funnel and gallery imagery.
Commercial system	The same design ecosystem is distributed across paid PDF storefronts plus a free blog/tutorial funnel and is cross-sold into yarn, kits/collections and related designs.

72. NEW PERMANENT REQUIREMENTS - PRODUCT TRUTH AND STITCH IDENTITY

F-749 - Purchased Pattern = Benchmark Evidence, Not Product Source - A purchased competitor pattern may be used privately to test ingestion, validation, rendering, merchandising quality and system capability. It is not authority to publish a copied, translated, lightly modified, reformatted or derivative competing pattern.

F-750 - Instruction-to-Topology Compiler - Product Truth must represent construction order and relationships, not only final silhouette. It must distinguish worked-flat, in-the-round, top-down, bottom-up, side-to-side, built-in sleeves, separate sleeves, joins, seams, shaping, bands, hoods, pockets and other construction semantics.

F-751 - Stitch Semantics Are First-Class - Named stitches and stitch patterns must compile to explicit structural semantics where commercially material. A texture label alone is insufficient when the product's appearance depends on loop placement, eyes, legs, posts, BLO/FLO use, clusters, stars, cables, mosaic placement or similar structure.

F-752 - Stitch-Pattern Identity Gate - A generated listing image fails when the visible fabric is a different stitch pattern even if silhouette, colour and garment features are correct. The gate must distinguish the intended stitch family/pattern from plausible-looking generic crochet.

F-753 - Gauge Is Product Truth - Store stitch gauge, row gauge, hook/yarn context and separate ribbing/border gauges. Validate apparent stitch scale in generated imagery. A photoreal image with materially wrong stitch scale is not certified.

F-754 - Multi-Gauge Regions - A product may contain multiple gauge regimes. Body fabric, ribbing, edging, cuffs, collars, bands and decorative sections must carry their own gauge/stitch identities when the source specifies them.

F-755 - All-Size Parse Before Single-Size Render - When a pattern contains graded sizing, parse and consistency-check the complete size family before freezing a benchmark/render size. This catches indexing, parenthetical and grading errors that a one-size parser can hide.

F-756 - Schematic Cross-Check - When schematics/measurement tables exist, reconcile them against stitch counts, gauge and written construction. Disagreement becomes an explicit UNKNOWN/CONFLICT; do not silently choose whichever source makes the render easier.

F-757 - Optional Feature State - Optional hood, pockets, buttons, sleeves, colourways or other variants are explicit configuration state. The listing image manifest must state which variant is being represented.

F-758 - Reference Defect Feedback Loop - If a generator faithfully reproduces a defect in Brambleloop's deterministic reference, classify the defect as upstream reference error, fix the reference, add a regression test and regenerate. Do not blame or tune the image model around bad Product Truth geometry.

F-759 - Distinctive-Stitch Benchmark Set - Maintain adversarial positive/negative examples for each supported distinctive stitch family. The validator must reject visually plausible but wrong alternatives before that stitch family is promoted to production support.

F-760 - Commercial Certification Requires Dual Pass - A listing image must independently pass Product Truth and photographic realism. Neither can compensate for the other; UNKNOWN blocks approval.

73. NEW PERMANENT REQUIREMENTS - PATTERN PRODUCT QUALITY

F-761 - Pattern Package Completeness - Brambleloop patterns should ship as a coherent product package: clear title/skill level, materials, yarn/hook data, gauge, abbreviations, special stitches, size chart, finished measurements/ease, construction overview, schematic where useful, row/round instructions, finishing, care/use notes where appropriate and support path.

F-762 - Size Architecture - Graded patterns must maintain a machine-readable size matrix connecting measurements, stitch counts, row counts, yardage and schematic dimensions. Agents test monotonicity and expected grading relationships before release.

F-763 - Ease and Fit Are Explicit - Garment patterns state intended fit and ease and separate body measurements from finished garment measurements. Listing copy and imagery must not imply a different fit.

F-764 - Yarn Substitution Intelligence - Where useful, include substitution guidance based on yarn weight plus measurable attributes such as yards/metres per mass, fibre behaviour and gauge requirements, while clearly requiring the maker to obtain gauge.

F-765 - Construction Overview Before Instructions - Every nontrivial garment gets a concise construction explanation before row-level instructions so the customer understands how the pieces/sections become the finished item.

F-766 - Project Tips from Failure Modes - Testing and support failures become concise project tips at the exact points customers are likely to struggle: tension, markers, joins, symmetry, stitch counts, sizing, seaming and finishing.

F-767 - Tutorial Linkage - When a video/tutorial exists, the PDF and listing identify exactly which size/version it demonstrates. Video support supplements the written pattern; it does not become the only source of required instructions.

F-768 - Pattern QA Must Recompute - Release QA independently recomputes counts, repeats, grading, dimensions and transitions instead of merely checking that the PDF matches the drafting database.

F-769 - Visual Gallery as Validation - Finished-object gallery/reference imagery is useful as independent evidence that instructions produce plausible finished results, but it does not override written Product Truth or replace physical testing.

F-770 - Supportability Is a Product Requirement - The product records what support exists, where it is offered and what modifications are outside support. Recurring support questions feed the Research/Pattern agents and future revisions.

74. NEW PERMANENT REQUIREMENTS - COMMERCIAL FUNNEL AND MERCHANDISING

F-771 - Free-to-Paid Content Ladder - Where strategically appropriate, Brambleloop may use a free searchable article/tutorial as discovery and sell a cleaner premium PDF/package. The free and paid experiences must be intentionally differentiated rather than accidental duplicates.

F-772 - Multi-Channel Product Distribution - A pattern can be distributed through multiple appropriate storefronts/content channels from one canonical product record. Pricing, version, files, description claims and update state stay synchronized.

F-773 - Video as Acquisition + Support - Tutorial video is both customer support and discovery content. Agents track which patterns justify video based on demand, complexity, search opportunity, conversion and support reduction.

F-774 - Collection Flywheel - Successful stitch systems, yarn palettes or design languages may become coherent collections across garments, accessories and home products. Reuse is at the design-system level, not by copying another creator's protected expression.

F-775 - Cross-Sell Graph - Every product records legitimate related products, bundles, collections, tutorials, materials and next-project recommendations. Agents optimize attach rate without misleading customers.

F-776 - Ownable Listing Assets - Brambleloop generates and owns/controls its listing copy, diagrams and imagery. Third-party seller photographs may be private benchmark evidence only unless separately licensed for commercial use.

F-777 - Evidence-Backed Listing Claims - Claims such as beginner/intermediate, oversized, anti-pilling, machine washable, quick, reversible or size coverage must trace to Brambleloop's own product evidence or properly licensed factual source.

F-778 - Review-to-Product Learning - Reviews and support signals are mined for clarity, fit, errors, difficulty, missing explanation and desired variants. Agents convert recurring evidence into bounded product improvements and regression tests.

F-779 - Listing Conversion Is Not Product Truth - SEO/CTR/conversion optimization may change titles, thumbnails, ordering and copy, but may not distort construction, fit, included sizes, stitch appearance, difficulty, materials or deliverables.

F-780 - Benchmark Competitors by System, Not Clone - Competitor audits may identify useful business patterns - package completeness, tutorial support, collections, distribution, imagery standards, support loops - but Brambleloop must create independently authored designs, instructions, assets and brand expression.

75. IP / LICENSING FIREWALL FOR PURCHASED PATTERNS

The audited PDF is explicit: it is licensed for personal use, may not be copied/shared/reproduced/redistributed/translated/alterd/published without permission, permits small-scale sale of finished items with designer credit, prohibits use of the designer's photographs/images or pattern text to sell those finished items, and prohibits mass production/wholesale. Brambleloop therefore needs a hard boundary between private benchmark analysis and publishable Brambleloop products.

F-781 - Competitor File Quarantine - Purchased pattern files, extracted text, diagrams and seller imagery remain in a restricted benchmark corpus and are never directly fed into a publishable-pattern drafting path.

F-782 - No Derivative Rewrite Pipeline - Do not take a purchased pattern and produce a Brambleloop pattern by paraphrasing, translating, reformatting, resizing, recolouring, renaming or making superficial construction changes.

F-783 - Independent Design Provenance - Every Brambleloop pattern needs provenance showing how its design brief, dimensions, stitch system, construction choices, grading and instructions were independently developed or derived from Brambleloop-owned/licensed sources.

F-784 - Similarity/Contamination Review - Before release, compare a Brambleloop candidate against benchmark sources used during research for suspicious overlap in wording, unusual numeric sequences, construction sequence, distinctive combinations and visual presentation. Escalate material similarity.

F-785 - Benchmark Images Never Become Listing Inputs - Competitor/seller photos can be used for private evaluation, never as generation references for Brambleloop customer-facing imagery unless explicit commercial rights permit it.

F-786 - Licence Terms Are Machine-Readable - For every purchased benchmark, store licence/copyright terms, allowed uses, prohibited uses, attribution requirements and source provenance so agents cannot accidentally cross the boundary later.

F-787 - Public/Free Version Does Not Erase Rights - If substantially similar instructions are also publicly viewable on a creator's blog/video, that does not authorize Brambleloop to republish the creator's expression. Public material may inform general research; publishable output still requires independent authorship.

F-788 - Finished-Item Permission != Pattern Permission - Permission to sell finished physical items does not imply permission to sell, adapt or redistribute the pattern itself. Agents treat these rights as separate.

76. BENCHMARK ACCEPTANCE TESTS ADDED

- Parse all ten sizes and prove the size-index mapping remains correct through every instruction section.
- Recompute star counts and HDC return-row counts and verify the 2:1 stitch/star relationship where specified.
- Render a correct star-stitch control and adversarial generic-crochet controls; the identity gate must reject the wrong textures.
- Perturb apparent gauge while preserving silhouette; the gauge gate must reject the image.
- Swap built-in sleeves for separately attached sleeves in the structural representation; construction truth must detect the mismatch.
- Toggle hood/buttons as explicit variants; the listing manifest and generated imagery must agree.
- Inject a deterministic reference defect; the pipeline must classify and fix the upstream reference rather than tune around it.
- Attempt to move seller imagery into a listing-generation input; the licensing firewall must block it.
- Attempt to create a publishable Brambleloop pattern by paraphrasing the purchased instructions; the provenance/firewall must reject the path.
- Restart agents/runtime; benchmark licence metadata, source provenance, tests and learned failures must recover durably.

77. AUDIT CONCLUSION

What Brambleloop should copy: the operational quality system - complete size architecture, explicit gauge, construction overview, schematic, special-stitch definitions, tutorial support, gallery evidence, multi-channel distribution, collection strategy, related-product flywheel and strong customer support.

What Brambleloop must not copy: this creator's protected pattern text, photographs, layout, exact expressive instructions or a derivative rewrite of the purchased design. The benchmark should make Brambleloop better at creating and validating its own products, not better at cloning hers.

Visual consequence: Bench1 proved full-product hero generation can pass a strict gate. This audit raises the next bar: distinctive stitch identity and gauge must be first-class Product Truth. A beautiful image of the wrong fabric is a failed listing image.

Business consequence: the strongest lesson is not one cardigan. It is the integrated product engine: searchable free content + premium PDF + video + multiple storefronts + related collection + materials ecosystem + community/support + consistent imagery. Brambleloop Studio should orchestrate that flywheel autonomously while preserving independent authorship and evidence.

v0.22 appends F-749 through F-788. All v0.21 and earlier requirements remain in force.

BRAMBLELOOP FINAL MASTER UPGRADE v0.23

78. SUPERSEDING CLARIFICATION - COMPETITOR PATTERN R&D;

This section supersedes any interpretation of v0.22 that would prevent Brambleloop from aggressively studying purchased competitor patterns. Brambleloop may fully audit purchased patterns for private R&D;: parse them, reconstruct their functional construction, test stitch systems, compare sizing and grading, benchmark imagery, identify weaknesses, and use the resulting knowledge to improve Brambleloop's own design and validation systems.

However, the publishable-output boundary remains: Brambleloop must not sell or publish a competitor's protected pattern text, images, or a lightly disguised derivative created primarily by paraphrasing or making superficial tweaks. Purchased patterns are permitted as deep engineering benchmarks and inspiration; Brambleloop's sellable pattern must be independently authored and contain meaningful original design decisions.

79. REVISED REQUIREMENTS (SUPERSEDE F-781 THROUGH F-788 WHERE CONFLICTING)

F-789 - Full Competitor Reverse-Engineering Allowed for Private R&D; - Agents may ingest the complete purchased pattern, reconstruct its functional garment architecture, stitch logic, grading, dimensions, workflow, customer experience, merchandising system and weaknesses inside the private benchmark environment.

F-790 - Improve the Idea, Not the Protected Expression - Agents may use learned functional concepts and general crochet techniques to design a better Brambleloop product. Improvements may include different fit, proportions, stitch system, shaping, construction, sizing, features, finishing, materials, grading, usability, support, imagery and packaging. The final sellable work must be Brambleloop-authored rather than a rewritten copy.

F-791 - Meaningful Redesign Gate - Before a competitor-inspired design can enter the publishable pipeline, require documented original design decisions that materially distinguish the Brambleloop product. Merely changing colour, yarn, wording, size labels, button count, small dimensions or a few rows is not sufficient.

F-792 - Independent Instruction Draft - Publishable instructions are generated from Brambleloop's own Product Truth/CIR and design record, not by paraphrasing the competitor's row-by-row prose. The competitor file is excluded from the final instruction-writing context once the independent design specification is frozen.

F-793 - Functional Technique Reuse - Standard crochet methods, stitch names, general construction techniques, gauge concepts, sizing practices and other functional methods may be used when appropriate. Brambleloop should combine them into its own independently designed product rather than preserve a competitor's distinctive overall expression by default.

F-794 - Design-Difference Ledger - For competitor-informed products, store a machine-readable ledger showing the benchmark source, what was learned, what Brambleloop changed, why each change improves the product, and which aspects were independently designed.

F-795 - Similarity Review Before Sale - Before publication, an independent agent compares the Brambleloop candidate with benchmark materials for excessive similarity in wording, unusual numeric sequences, distinctive feature combinations, construction sequence and presentation. Material overlap is escalated for redesign rather than hidden.

F-796 - Competitor Weakness Mining - Agents should explicitly identify opportunities to outperform benchmark products: clearer instructions, better grading, more sizes, easier construction, better fit, stronger schematics, better tutorials, more useful yarn substitution, better accessibility, superior imagery, improved support and fewer customer failure points.

F-797 - No Artificially Weak Benchmarking - Do not avoid understanding a competitor's pattern because it is commercially successful. Deep private analysis is encouraged. The restriction applies to what Brambleloop publishes, not to how thoroughly the agents learn.

F-798 - Originality Is Part of Product Truth - A product is not release-ready solely because it is technically correct and photorealistic. Its provenance must also show that the sellable design and instructions are Brambleloop's own work or otherwise properly licensed.

80. PRACTICAL RULE FOR THE AGENTS

Allowed private workflow: buy competitor pattern -> audit every page -> reconstruct how it works -> identify what is strong/weak -> extract general engineering lessons -> create a materially redesigned Brambleloop specification -> remove the competitor text from the drafting context -> generate Brambleloop's own instructions from its own specification -> similarity/provenance review -> test -> publish if cleared.

Not an acceptable publishable workflow: buy competitor pattern -> preserve essentially the same product -> tweak a few counts/features -> paraphrase the instructions -> sell it as Brambleloop.

This distinction lets the autonomous company learn aggressively from the market while keeping the final catalogue independently authored and defensible. All non-conflicting requirements in v0.22 and earlier remain in force.

BRAMBLELOOP FINAL MASTER UPGRADE v0.24

81. BRAMBLELOOP LEARN - AUTONOMOUS EDUCATION, CONTENT & AUDIENCE ENGINE

This section converts the audited education/content architecture into a permanent Brambleloop department. The objective is not to copy another site's prose or visual design. It is to build Brambleloop's own durable learning library that teaches crochet, answers recurring customer problems, creates search discovery, supports pattern customers and compounds into email, video, social and product demand.

Reference architecture audited September 2026:

MJ's Off The Hook Designs - Homepage: https://www.mjsoffthehookdesigns.com/
MJ's Off The Hook Designs - Learn to Crochet: https://www.mjsoffthehookdesigns.com/learn-to-crochet/

The reference homepage currently connects patterns, seasonal categories, garments/home decor, a Learn-to-Crochet entry point, a yarn collection, a free gauge lead magnet, YouTube tutorials, social channels and a community/support group. These links are retained in the Master as future audit references, not as content to reproduce.

82. PERMANENT REQUIREMENTS F-799 THROUGH F-830

F-799 - Learn Department - BrambleloopStudio has a first-class Learn/Content department with its own queue, metrics, experiments, assets, calendar, knowledge graph and durable state. It operates 24/7 under the same RUN/GROW/IMPROVE contract as Product, Visual, Marketing and Finance.

F-800 - Structured Curriculum - Maintain an original beginner-to-advanced curriculum rather than disconnected posts. Core tracks should cover foundations, tools/yarn, tension, core stitches, pattern reading, gauge/fit, finishing, garment construction, troubleshooting and advanced stitch systems as evidence and demand justify.

F-801 - Pattern-to-Lesson Trigger - When a new Brambleloop pattern introduces a stitch, technique, construction method, gauge issue or finishing method not adequately covered in Learn, automatically create a content-gap task.

F-802 - Support-to-Lesson Trigger - Cluster customer questions, refunds, support tickets, failed validations and review complaints. Repeated confusion automatically proposes a tutorial/FAQ/diagram/video/pattern clarification experiment.

F-803 - Search-to-Lesson Trigger - SEO/Search agents identify high-intent educational queries and compare them against the Learn knowledge graph. Missing high-value topics become prioritized content opportunities.

F-804 - One Truth, Many Formats - Each approved lesson has one canonical factual/teaching specification that can produce an article, concise FAQ, printable guide, diagram set, short video script, long video script, email, social carousel, short-form clips and pattern-specific help without factual drift.

F-805 - Education Product Truth - Tutorial instructions are tested like patterns. Counts, stitch semantics, hook/yarn assumptions, diagrams, terminology and demonstrations must reconcile with the canonical lesson specification.

F-806 - Technique Visual Truth - Educational stitch images/diagrams must show the correct loops, insertion points, yarn overs, eyes, posts and orientation. Generic crochet imagery cannot illustrate a different stitch merely because it looks attractive.

F-807 - Difficulty Ladder - Lessons and linked patterns carry evidence-backed prerequisite relationships. The system can recommend 'learn these two skills first' rather than relying only on vague beginner/intermediate labels.

F-808 - Contextual Help Inside Patterns - Pattern PDFs/apps/listings can link directly to Brambleloop-owned lessons for stitches, gauge, seaming, blocking and other techniques. Help appears where the maker needs it without bloating every pattern.

F-809 - Gauge Learning System - Gauge receives a dedicated educational system: stitch vs row gauge, swatching, hook adjustment, yarn substitution, tension consistency, blocking and garment-fit consequences. Visual examples and calculators must agree with Product Truth.

F-810 - Free Lead Magnets - Agents may build original high-value downloadable guides/checklists/calculators to earn email subscribers. Every lead magnet has a measurable acquisition goal, provenance, version and downstream attribution.

F-811 - Free-to-Paid Ladder - Educational/free content may lead naturally to premium Brambleloop patterns, bundles or tools when relevant. The free content must remain genuinely useful and must not become deceptive teaser content.

F-812 - Video Coverage Strategy - Track which lessons/patterns have video support, search demand, support burden and expected ROI. Create long-form tutorials only when evidence justifies them; derive Shorts/Reels/TikToks/Pins from canonical assets where

useful.

F-813 - Pattern Video Contract - When a video supports only a specific size/version, state that explicitly. The written pattern remains complete enough to stand on its own unless the product explicitly promises otherwise.

F-814 - Content Constellation - A product launch can automatically generate a constellation: listing, hero/detail imagery, stitch lesson, gauge help, construction article, video plan, email, social derivatives, Pinterest assets, FAQs, internal links and support entries - but only where each asset has real value.

F-815 - Knowledge Graph - Persist relationships among stitches, techniques, yarn weights, hooks, gauge, construction methods, products, lessons, videos, FAQs, support issues and search queries so agents can reuse knowledge without duplicating content.

F-816 - Internal Linking Engine - Automatically propose and test useful links among lessons, patterns and support pages. Links are based on reader intent and prerequisite/product relevance, not keyword stuffing.

F-817 - Education SEO - Learn pages have independent search briefs, intent classification, title/meta/schema strategy, FAQ opportunities and update cadence. Search optimization may not distort crochet truth.

F-818 - Content Freshness - Agents monitor lessons for stale links, changed terminology, broken media, obsolete products, new evidence and support-driven gaps. Updates are versioned and regression-checked.

F-819 - Email Learning Funnel - Track subscriber source, lead magnet, lesson engagement and downstream product conversion. Email sequences teach first and sell contextually; unsubscribes and fatigue constrain frequency.

F-820 - Community Feedback Loop - Community/project-photo/support signals feed the learning ledger. Common successes/failures can improve tutorials, patterns and FAQs while respecting privacy and content rights.

F-821 - Original Education Assets - Brambleloop creates its own tutorial prose, diagrams, photos, renders and videos or uses properly licensed assets. Competitor education pages may be benchmarked privately but are not copied into Brambleloop Learn.

F-822 - Education Quality Judge - Before publication, lessons are reviewed for correctness, completeness, sequencing, clarity, accessibility, visual accuracy and whether a learner could actually execute the skill.

F-823 - Beginner Failure Simulation - For foundational lessons, adversarial QA checks likely beginner mistakes: wrong loop, wrong stitch count, skipped turning chain, excessive tension, wrong hook, twisted foundation, missed gauge, reversed RS/WS and similar errors.

F-824 - Accessibility - Educational pages and downloadable guides use readable hierarchy, alt text, captions/transcripts where appropriate, sufficient contrast, mobile-friendly layouts and clear terminology.

F-825 - Content Economics - Track creation cost, maintenance cost, search traffic, email acquisition, assisted conversions, support deflection and attributable sales. High-cost/low-value formats are redesigned or retired.

F-826 - No Content Mill - Autonomy does not mean publishing filler. If an agent cannot identify a useful learner problem, search opportunity, support need or product connection, it should WATCH rather than generate low-value pages.

F-827 - Studio Learn Dashboard - Show curriculum coverage, content gaps, drafts, QA state, traffic, rankings, email captures, video queue, assisted sales, support-deflection signals, update needs and experiments in BrambleloopStudio.

F-828 - Cross-Department Ownership - Research finds gaps; Pattern supplies technical truth; Visual supplies truthful instructional assets; Marketing/SEO packages discovery; CX supplies confusion signals; Finance measures economics; Learn coordinates the educational product.

F-829 - Continuous Teaching Improvement - Every lesson participates in OBSERVE -> HYPOTHESIS -> EXPERIMENT -> MEASURE -> VALIDATE -> PROMOTE/REJECT. Improve explanations, visuals, sequencing and formats while preserving the underlying technical truth.

F-830 - Education Compounds - A solved customer question becomes reusable company knowledge. The target is that each new product and each support interaction makes future products easier to discover, understand, make and support.

83. REFERENCE-ARCHITECTURE OBSERVATIONS

Discovery	Homepage routes visitors into seasonal patterns, garments, home decor, blankets, Learn, stitch/gauge/yarn topics and search.
Lead capture	A free gauge guide is used as an email acquisition asset, with gauge meaning, stitch/row gauge, common fixes, consistency, yarn adjustment and blocking topics.
Video	Homepage states that almost every pattern offered on the site can be found on its YouTube channel, making video both support and acquisition.
Community	A Facebook community group is positioned for project photos and pattern-support questions.
Product ecosystem	A named yarn collection links curated colours, yarn sales/kits and multiple patterns, showing how content, materials and designs reinforce each other.
Learn entry	The Learn section explicitly promises step-by-step basics through pattern reading, providing a clear learner journey instead of only a category archive.

84. LAUNCH VS MATURE SCOPE

Launch-critical: Learn architecture, canonical knowledge graph, content-gap detection, pattern/support links, provenance, correctness QA and the ability to queue valuable education work. Brambleloop does not need hundreds of lessons before first sale.

Post-launch autonomous expansion: curriculum depth, search-led articles, videos, lead magnets, email sequences, social derivatives, community feedback loops and conversion optimization grow continuously from real demand. This prevents the education engine from becoming another moving pre-launch finish line.

v0.24 appends F-799 through F-830. All non-conflicting requirements in v0.23 and earlier remain in force. The two audited URLs above are retained as explicit future reference sources.

BRAMBLELOOP FINAL MASTER - AUDITED v1.0

85. FINAL AUDIT AFTER BUILD 2

This addendum audits v0.24 against the evidence produced by Build 2 certification, independent Codex review, Visual V1/V2 research and runtime-independence testing. Earlier requirements remain in force except where this addendum explicitly strengthens their interpretation.

Primary audit finding: code, tests, handlers, readings and declared cadences must never be mistaken for an implemented business capability when the real runtime does not reach, consume and act on them. The Final Build uses end-to-end proof as its definition of done.

Baseline	Audited state
Build 2	claude/visual-investigation @ 019ebf0
Tests	5,242 passing; 0 failing suites; 285 suites
Closure	218 COMPLETE+PROVEN; 0 OPEN; 58 owner-gated; 37 data-gated; 7 external-blocked
Independent audit	0 INVALID after repair/adjudication
Runtime independence	4/4 including SIGKILL/restart without loss or duplication
Production	fcb982d still deployed/integrated baseline; Build 2 candidate not yet merged/deployed
Visual V1	NOT LOCKED
Visual V2	codex/visual-v2-rnd @ 035ff0c; deterministic render + protected composition (B+C) recommended

86. RUNTIME-PROOF DEFINITION OF DONE

F-831 - Capability Proof Chain - COMPLETE+PROVEN requires the applicable chain: requirement -> producer -> durable evidence/state -> runtime consumer -> decision -> protected action/effect -> observable result -> independent proof.

F-832 - Existence Is Not Execution - A module, function, handler, cadence, row, dashboard card or unit test cannot by itself prove a business capability operates.

F-833 - Live-Root Reachability - Runtime requirements must be reachable from real scheduler/API/worker/event roots; arbitrary callable paths are insufficient.

F-834 - Consumption Required - Producing a reading is insufficient when the requirement exists to influence a downstream decision/action; prove the consumer.

F-835 - Protected-Action Revalidation - Revalidate gates at consequential execution when authority/evidence can change after planning.

F-836 - No Fixture-Only Proof - Injected test evidence with no production producer cannot prove end-to-end capability.

F-837 - Proxy Honesty - Heuristics/proxies stay labelled proxies and cannot silently satisfy direct-evidence requirements.

F-838 - Closure Integrity - Only an explicit valid park/gate removes executable work from OPEN. Defaults, stale labels or broad gate tuples cannot hide reopened work.

F-839 - Final-Head Re-Audit - Material integration requires a new proof/reachability audit on the final effective source tree.

F-840 - Green Suite Necessary, Not Sufficient - A green full suite is mandatory but cannot replace requirement-level runtime proof.

87. FINAL BUILD EXECUTION CONTRACT

F-841 - Build 2 Freeze - Treat 019ebf0 as the certified engineering baseline until an explicitly reviewed successor is created.

F-842 - Final Build Purpose - Final Build integrates the certified foundation, satisfies the Final Master, executes owner-gated activation, launch rehearsal, deployment, production observation and customer-ready polish.

F-843 - Final Master Closure Matrix - Map every applicable F-requirement to implementation, maturity, evidence, gate and owner action.

F-844 - Explicit Maturity - Keep implemented, tested, integrated, deployed, exercised, production-observed and commercially evidenced distinct.

F-845 - Release Candidate Freeze - Freeze a launch candidate SHA before adversarial certification; defects are recorded against that exact candidate.

F-846 - Audit-Repair Loop - Freeze -> audit -> defect -> repair -> regression -> full suite -> independent re-audit. Reporting never substitutes for repair.

F-847 - No Final-Build Drift - New ideas are launch-critical, post-launch scheduled or rejected; nice-to-have expansion cannot indefinitely move launch.

F-848 - Shadow Rehearsal - Before activation, exercise opportunity -> design -> pattern -> QA -> Visual -> listing -> pricing -> approval -> publish refusal/simulation -> order/support/accounting through real runtime paths.

F-849 - Claude/Chat Independence - Launch certification proves persistent runtime execution does not depend on an active Claude/Codex/Fable/ChatGPT development conversation.

F-850 - Restart Recovery - Restart isolated runtime components during queued/in-flight work and prove durable recovery, fencing/idempotence and no unauthorized duplicate effect.

88. VISUAL V2 - PRODUCT TRUTH OWNS THE PRODUCT

Visual V1 remains honestly unresolved: deterministic Product Truth and validation work, but generative providers repeatedly redraw fine crochet structure. The audited Visual V2 direction moves product authority into deterministic rendering and bounds generative AI to presentation.

F-851 - B+C Default - Visual V2 defaults to deterministic/procedural product rendering plus protected photoreal presentation unless later evidence disproves it.

F-852 - No Product Redraw by Default - Do not ask generative AI to recreate structure Brambleloop already knows when that can change stitch count, gauge, cable pitch, silhouette or features.

F-853 - Authoritative Product Region - Certified product geometry/pixels remain authoritative through presentation wherever technically feasible.

F-854 - Construction Fidelity - Deterministic rendering must encode commercially material stitch/fabric/construction truth, not merely a schematic symbol.

F-855 - Bounded Material Enhancement - AI may enhance fibre/fuzz/micro-material only when macro geometry/stitch boundaries remain protected and revalidated.

F-856 - Structural vs Photographic Gates - Structural Product Truth and photographic realism remain independent hard gates; both must pass.

F-857 - Blind Original Graduation - Product-only Visual V2 graduates on a Brambleloop-original design with no finished-product visual reference.

F-858 - On-Model After Product Lock - On-model graduation follows product-only lock and must preserve both exact product and exact canonical Brambleloop model.

F-859 - Visual Economics - Track yield, rejection reasons, latency, compute, spend and cost per certified listing set.

F-860 - Visual Does Not Stall Independent Work - Unresolved Visual R&D; cannot idle unrelated Final Build engineering, commerce, finance, QA or infrastructure.

89. MULTI-MODEL & SESSION RESILIENCE

F-861 - Repository Is Shared Memory - Critical state lives in Git/database/evidence artifacts, not conversation context.

F-862 - Durable Handoffs - Long AI-development tasks persist objective, branch/head, decisions, defects, tests, unfinished work and next actions.

F-863 - Context Is a Cache - Persist conclusions before context pressure and reread durable state after compaction/restart.

F-864 - Session-Limit Survival - Checkpoint partial work, patch series and assignments before session limits can kill subagents.

F-865 - Bounded Parallelism - Use disjoint ownership and one integrator; optimize for integrated tested work per resource window, not maximum agent count.

F-866 - Independent Model Review - A second model may adversarially review evidence without becoming certification authority.

F-867 - Worker Does Not Self-Certify - Implementation workers return tested candidates; high-risk completion is independently adjudicated.

F-868 - Empirical Model Routing - Route Claude/Opus, Fable, Codex/Astra and future models by demonstrated outcome, cost, latency, context and tool fit.

F-869 - Approval Friction Is Cost - Measure excessive development approval prompts as throughput loss; prefer bounded autonomous sandboxes while keeping production/spend/credential gates.

90. OWNER ACTIONS & LAUNCH

F-870 - Owner Action Packets - Every owner gate states exact action, why software cannot do it, time/cost, unlock and consequence of delay.

F-871 - Provider Funding Is Authority - Model/vision credit is owner financial authority; bounded experiment and max spend are prepared first.

F-872 - Etsy Scope Reauthorization - Required OAuth scopes such as transactions_r are explicitly reauthorized and proven by readback before trusted use.

F-873 - Browser/Vision Ruling - Ambiguous acceptance wording receives a durable owner ruling without fabricating past evidence.

F-874 - No Premature KYC - Do not force Etsy identity/banking/tax activation before the next safe launch step requires it.

F-875 - Ad Authority Explicit - Advertising remains disabled until owner spend/authority policy and prerequisites pass.

F-876 - Physical Evidence Specificity - Request physical/tester evidence only for the exact unresolved property it can prove.

F-877 - Bounded Visual Spend - Paid Visual experiments require hypothesis, provider/model, call count, max spend and pass/fail criteria.

F-878 - Launch Packet - Owner receives candidate SHA, tests/audit, Visual status, listings/products, policy state, owner actions, spend limits, economics, rollback and activation steps.

F-879 - Launch Starts Feedback - Launch occurs when customer-protective launch gates pass; mature/post-launch requirements remain scheduled and continuous improvement continues.

91. AUDIT FINDINGS / SUPERSESSIONS

Runtime proof: F-831 through F-840 supersede any weaker interpretation of completion elsewhere in the Master. Existing production-readback and maturity requirements remain valid.

Multi-model: F-710 through F-718 remain valid. F-861 through F-869 add the practical session/context/handoff rules learned during Build 2 certification.

Visual: Product Truth, dual-verdict and canonical-model requirements remain mandatory. F-851 through F-860 change the default implementation direction from generative product redraw toward deterministic product authority plus protected presentation.

Build 2 vs production: certification of 019ebf0 does not mean it is deployed. Final Build must integrate/deploy only through explicit authorization and production readback.

Launch scope: the Master remains the end-state specification. Launch-critical items are deliberately smaller than the mature-company backlog so continuous improvement does not become launch paralysis.

92. FINAL ACCEPTANCE TESTS ADDED

- Delete/disable a runtime consumer while leaving its library/tests intact: requirement proof must reopen.
- Seed a test-only fact with no production producer: end-to-end certification must refuse it.
- Close a gate at planning then revoke authority before execution: protected action must revalidate and refuse.
- Integrate a branch that silently drops a previously claimed repair: final-head audit must detect the missing behavior.
- Kill the runtime mid-job and restart with owner devices and AI development chats absent: durable work resumes without loss/duplication.
- Render a blind original product with exact repeated structure: structural gate measures truth independently of photographic realism.
- Attempt generative presentation that alters the authoritative product region: Product Truth rejects it.
- Run Final Build with one lane blocked on Visual/owner action: unrelated READY work continues.
- Present an owner gate: packet contains exact action, time/cost, unlock and consequence; no unrelated work waits.
- Freeze a launch candidate, repair a defect, and prove the successor candidate receives a fresh audit rather than inheriting the old verdict.

93. FINAL AUDIT CONCLUSION

Final Master v1.0 is now the canonical end-state specification: v0.24 plus this audited addendum and all non-conflicting earlier requirements. The major correction is epistemic: Brambleloop may only claim capabilities that are actually reachable, consumed and evidenced at the maturity level being claimed.

Final Build begins from certified Build 2 engineering at 019ebf0, while production remains on fcb982d until explicitly merged/deployed. Visual V1 remains NOT LOCKED; Visual V2 proceeds on deterministic product rendering plus protected presentation. Final Build must not hide either fact.

v1.0 appends F-831 through F-879. All non-conflicting F-001 through F-830 remain in force.